Companion volume · Phases 0–5

Catalogue of Proofs

Exact recovery of a linear differential operator from a single solution

Part I	**Part II**
the 31 load-bearing proofs	all the other results
Dim. A1–A31	Dim. B1–. . .

Every card carries its ledger identifier (for example C3-T1), its trust label, and the page of the statement in the companion paper *Exact Recovery of a Linear Differential Operator from a Single Solution, complete edition, Phases 0–5*.

October 2026

How to read this catalogue

This volume contains the proofs of the results stated in the paper *Exact Recovery of a Linear Differential Operator from a Single Solution* (complete edition, Phases 0–5). It is a catalogue: every result has one card, and every card has the same shape.

The shape of a card

Dim. A0 **ID-T0**

Name of the result [proved]

The full statement, with every hypothesis.

> **Idea.** One paragraph that says why the result is true, before any computation.

Proof.

Step 1 First claim

One claim per step, with its computation displayed on its own line.

Step 2 Second claim

...

Why it works. one sentence that names the mechanism of the proof.

■

Uses: the results this proof relies on, by catalogue number.

The two parts

Part I contains the 31 results on which everything else rests. They were selected from the project's dependency graph: a result is in Part I if many other results depend on it, or if it is a headline theorem of the paper. Their proofs are given in full, step by step.

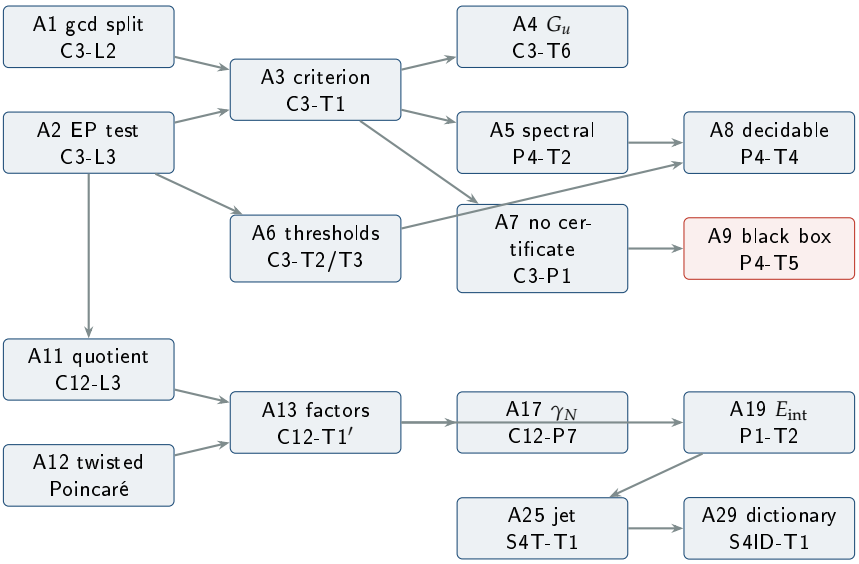
Part II contains all the other results of Phases 0–5, grouped by phase and ordered by identifier. Short and medium proofs are given in full. Proofs that take many pages in the research files (the Burgers asymptotics, the scattering identification, the Hirota computations) are given as a complete chain of steps with every key estimate displayed; routine computations are referred to the research file and marked *condensed*. Results that are cited, conditional, or only checked on examples are listed with their label and no proof.

Trust labels

Label	Meaning
[proved]	complete proof, passed an independent review
[proved mod. X]	complete except for a cited published theorem X, whose source was checked
[proof written, not audited]	complete proof, not independently reviewed
[from cited source]	a published theorem, used as a black box
[conditional]	depends on a lemma that is not proved
[checked on examples]	checked on examples only
[false]	a counterexample is proved (the card proves the counterexample)
[open]	unsolved

The reviewers were independent AI agents, not human referees. Identifiers such as C3-T1 are those of the project ledger `claims/LEDGER.md`; the research file that holds each original proof is named in the index at the end.

The backbone of Part I



The arrows mean "is used in the proof of". Blue: positive results; red: a negative (impossibility) result. The arrows come from the `depends_on` edges of the project graph.

Contents

How to read this catalogue i

I The load-bearing proofs 1

1 Algebra and identification 2

1.1 The algebraic toolkit . . . 2

1.2 The identification theorems . . . 7

1.3 Finite data . . . 11

2 Decomposition and regularity 19

2.1 Terms and factors . . . 19

2.2 The factor decomposition . . . 20

2.3 Regularity . . . 26

3 The interaction term 33

3.1 Exact form and cost . . . 34

3.2 Instability . . . 36

3.3 Perturbation and non-commuting terms . . . 40

4 Nonlinear equations 44

4.1 The interaction term . . . 44

4.2 After the shock *(condensed proofs)* . . . 48

4.3 Identification in a dictionary . . . 51

II The other results 56

5 Phases 0–1: the ODE prototype, identification, decidability 57

5.1 The one-variable prototype and the companion matrix . . . 57

5.2 Refinements of identification . . . 63

5.3 Decidability and variable coefficients . . . 68

5.4 Results used without proof . . . 71

6 Phase 2: decomposition and regularity **72**

6.1 Rectangles and convex sets . . . 72
6.2 Regions with holes . . . 78
6.3 Closedness of sums of kernels . . . 79
6.4 Complements: variable-coefficient identification . . . 83
6.5 Results used without proof . . . 84

7 Phase 3: the interaction term **85**

7.1 The class $\mathfrak{B}$ . . . 85
7.2 Factors and second-order evolutions in $\mathfrak{B}$ . . . 88
7.3 Non-commuting terms . . . 89
7.4 Boundaries . . . 90
7.5 Where the interaction is continuous . . . 92
7.6 Regularisation of the term split . . . 93
7.7 Results used without proof here . . . 96

8 Phase 4: nonlinear equations **98**

8.1 Tools before the shock . . . 98
8.2 After the shock . . . 100
8.3 Identification in dictionaries . . . 103
8.4 Nonlinear results whose proofs are not reproduced . . . 106

A Index of results **108**

Bibliography **111**

Part I

The load-bearing proofs

CHAPTER 1

Algebra and identification

In this chapter

Starting from	$\Omega \subset \mathbb{R}^2$ open and connected; the ring $\mathcal{R} = \mathbb{C}[\tau, \xi]$ acting on distributions by $\tau \mapsto \partial_t$, $\xi \mapsto \partial_x$. Classical facts used without proof: unique factorisation in $\mathcal{R}$ (Gauss), the Nullstellensatz, the Chinese remainder theorem, and the fact that distributions agreeing near every point agree.
Why	every identification argument reduces a question about solutions to a question about ideals and exponential-polynomials. This chapter proves those reductions in full, then the identification theorems built on them.
Role	Dim. A1–A9. They prove Chapters 3–5 of the paper: the gcd splitting, the EP test, the identification criterion, the intrinsic operator, the spectral form, the finite-data thresholds, and the decidability results.

Notation for the whole chapter

$V(I) \subset \mathbb{C}^2$ is the common zero set of an ideal I. For a point $z = (\tau_z, \xi_z) \in \mathbb{C}^2$,

$$\mathfrak{m}_z = (\tau - \tau_z,\ \xi - \xi_z)$$

is its maximal ideal, and $e^{\langle z, \cdot \rangle}$ denotes $e^{\tau_z t + \xi_z x}$. $\mathrm{Ann}(w) = \{Q \in \mathcal{R} :\ Q(\partial) w = 0\}$. $\mathrm{EP}(\Omega)$ is the set of restrictions to Ω of finite sums $\sum_k q_k\, e^{\langle z_k, \cdot \rangle}$ with polynomials q_k.

1.1 The algebraic toolkit

Dim. A1 **C3-L2**

Splitting off the gcd [proved]

Let $0 \neq I \subseteq \mathcal{R}$ be an ideal and $G = \gcd(I)$. Then:

(i) $I = G \cdot J$ for an ideal J with $\gcd(J) = 1$;

(ii) $V(J)$ is finite;

(iii) $(\sqrt{J})^N \subseteq J$ for some N, and $J = \mathcal{R}$ if $V(J) = \emptyset$.

Two facts are proved on the way:

- **(V)** two nonzero coprime polynomials have finitely many common zeros;
- **(F)** an irreducible polynomial has infinitely many zeros; hence if $V(\operatorname{Ann} w)$ is finite, some $h \in \operatorname{Ann}(w)$ is not divisible by a given irreducible P_i.

Stated in the paper as no. 3.2, p. 12

Idea. Over the field $\mathbb{C}(\xi)$ two coprime polynomials have a Bézout identity; clearing denominators gives a polynomial in ξ alone that vanishes at every common zero, and likewise in τ. After dividing an ideal by its gcd, a generic combination of its elements is coprime to any fixed element, and (V) applies.

Proof.

Step 1 **Fact (V): coprime polynomials meet in finitely many points**

Let $f, g \neq 0$ be coprime and $K = \mathbb{C}(\xi)$. A common factor of positive τ-degree in $K[\tau]$ would give, by the Gauss lemma, a common factor in $\mathbb{C}[\xi][\tau]$; so f, g are coprime in the principal ideal domain $K[\tau]$. Hence there are $a, b \in K[\tau]$ with

$$af + bg = 1.$$

Multiplying by a common denominator,

$$a'f + b'g = r(\xi), \qquad a', b' \in \mathcal{R}, \quad r \in \mathbb{C}[\xi] \setminus \{0\}.$$

At a common zero (τ_0, ξ_0) the left side vanishes, so $r(\xi_0) = 0$: finitely many ξ_0. Exchanging τ and ξ gives $s(\tau) \neq 0$ with $s(\tau_0) = 0$. Therefore

$$V(f, g) \subseteq \{r = 0\} \times \{s = 0\},$$

a finite set.

Step 2 **Fact (F): an irreducible polynomial has infinitely many zeros**

If $\deg_\tau P_i \geq 1$, then for all ξ_0 outside the finitely many zeros of the leading coefficient, $P_i(\cdot, \xi_0)$ has a root τ_0. This gives infinitely many points. If $\deg_\tau P_i = 0$, then P_i is an irreducible polynomial in ξ, so $P_i = \alpha(\xi - \xi_0)$ and $V(P_i) = \mathbb{C} \times \{\xi_0\}$.

For the consequence: if every $h \in \operatorname{Ann}(w)$ were divisible by P_i, then $V(P_i) \subseteq V(\operatorname{Ann} w)$, an infinite set inside a finite one.

Step 3 (i) Dividing by the gcd

Put $J = \{f/G : f \in I\}$. It is closed under sums and under multiplication by $\mathcal{R}$, so it is an ideal, and $I = G \cdot J$. If a prime π divided every element of J, then $G\pi$ would divide every element of I, against the definition of G. So $\gcd(J) = 1$.

Step 4 (ii) A generic combination is coprime to a fixed element

If $J = \mathcal{R}$ there is nothing to prove. Otherwise pick $0 \neq f \in J$ with distinct prime factors $\pi_1, \dots, \pi_s$. By Step 3, for each a there is $g_a \in J$ with $\pi_a \nmid g_a$. For $c \in \mathbb{C}^s$ put

$$g_c = \sum_{b=1}^{s} c_b \, g_b \in J.$$

For each a the set $H_a = \{c : \pi_a \mid g_c\}$ is a linear subspace, and it is proper because the a-th unit vector is not in it. A finite union of proper subspaces does not cover $\mathbb{C}^s$, so some c lies in no H_a. Then f and g_c are coprime, and by Step 1

$$V(J) \subseteq V(f, g_c) \quad \text{is finite.}$$

Step 5 (iii) Nullstellensatz

The polynomials vanishing on $V(J)$ form $\sqrt{J}$, which is finitely generated, so a power of it lies in J. If $V(J) = \emptyset$, then $\sqrt{J} = \mathcal{R}$ and $1 \in J$.

Why it works. dividing by the gcd leaves an ideal with no common curve, and two curves without common component meet in finitely many points.

■

Uses: Gauss lemma, Nullstellensatz.

Dim. A2 **C3-L3**

Exponential-polynomials are the finite-dimensional functions [proved]

For $w \in \mathcal{D}'(\Omega)$, Ω connected, the following are equivalent:

$$w \in \mathrm{EP}(\Omega) \iff \dim_{\mathbb{C}} \mathcal{R}w < \infty \iff V(\operatorname{Ann} w) \text{ is finite.}$$

Moreover, with distinct z_k:

(a) $Q(\partial)\big(q\, e^{\langle z_k, \cdot \rangle}\big) = \big(Q(z+\partial)q\big)\, e^{\langle z_k, \cdot \rangle}$ for every polynomial q;

(b) if $\sum_k q_k\, e^{\langle z_k, \cdot \rangle} = 0$ on a nonempty open set, every $q_k = 0$;

(c) if $w = \sum_k q_k\, e^{\langle z_k, \cdot \rangle}$ with $q_k \neq 0$, then $V(\operatorname{Ann} w) = \{z_k\}$.

Stated in the paper as no. 3.3, p. 13

Idea. Exponentials are independent, so an exponential-polynomial lives on finitely many points. Conversely, a finite-dimensional derivative-stable space gives two one-

variable ODEs, $p(\partial_t)w = 0$ and $q(\partial_x)w = 0$. On a rectangle they force w to be an exponential-polynomial, and connectedness glues the rectangles together.

Proof.

Step 1 (a) Derivatives of an exponential times a polynomial

For one derivative,

$$\partial_t\big(q\,e^{\langle z,\cdot\rangle}\big) = \big(\partial_t q + \tau_z q\big)\,e^{\langle z,\cdot\rangle},$$

and the same with ∂_x and ξ_z. Iterating and using linearity gives (a).

Step 2 (b) Exponentials are linearly independent

The sum is real-analytic on $\mathbb{R}^2$ and vanishes on an open set, so it vanishes everywhere. Suppose some $q_k \neq 0$, and let K' be the set of such k. Choose $v \in \mathbb{R}^2$ with $\langle z_k - z_l, v\rangle \neq 0$ for $k \neq l$, and y_0 with $\prod_{K'} q_k(y_0) \neq 0$. Restricting to the line $y_0 + sv$ gives

$$f(s) = \sum_{k\in K'} r_k(s)\,e^{\mu_k s} \equiv 0, \qquad \mu_k = \langle z_k, v\rangle \text{ distinct}, \quad r_k(0) \neq 0.$$

We remove one term at a time. Multiply by $e^{-\mu_1 s}$ and differentiate $\deg r_1 + 1$ times. The first term disappears. Each other term becomes

$$\Big(\big(D + \mu_k - \mu_1\big)^{\deg r_1 + 1} r_k\Big)\,e^{(\mu_k - \mu_1)s},$$

and since $\mu_k \neq \mu_1$ the operator $(D + c)^N$ with $c \neq 0$ preserves the degree, so the new coefficient is nonzero. Induction ends with $r(s)e^{\mu s} \equiv 0$, $r \neq 0$: impossible.

Step 3 (c) The zero set of the annihilator is the set of exponents

By (a), the derivatives of $q\,e^{\langle z,\cdot\rangle}$ stay in $\mathbb{C}[t,x]_{\leq \deg q}\cdot e^{\langle z,\cdot\rangle}$, which is finite-dimensional. If $|\beta| > \deg q$, the product $(\tau - \tau_z)^{\beta_1}(\xi - \xi_z)^{\beta_2}$ acts on q as ∂^β and kills it. So

$$\prod_k \mathfrak{m}_{z_k}^{\deg q_k + 1} \subseteq \operatorname{Ann}(w), \qquad V(\operatorname{Ann} w) \subseteq \{z_k\}.$$

Conversely, if $Q \in \operatorname{Ann}(w)$ then $Q(z_k + \partial)q_k = 0$ by (a) and (b). The top-degree part of $Q(z_k + \partial)q_k$ is $Q(z_k)$ times the top-degree part of q_k, so $Q(z_k) = 0$.

Step 4 Finite zero set implies finite dimension

If $V(\operatorname{Ann} w) = \varnothing$ then $\operatorname{Ann} w = \mathcal{R}$ and $w = 0$. Otherwise, by A1(iii),

$$\operatorname{Ann}(w) \supseteq \mathfrak{I}(V)^N,$$

where $\mathfrak{I}(V)$ is the ideal of the finite set $V = V(\operatorname{Ann} w)$. By the Chinese remainder theorem

$$\mathcal{R}/\mathfrak{I}(V)^N \cong \bigoplus_{z\in V} \mathcal{R}/\mathfrak{m}_z^N,$$

which is finite-dimensional. Hence $\mathcal{R}w \cong \mathcal{R}/\operatorname{Ann}(w)$ is finite-dimensional too.

Step 5 Finite dimension gives two ODEs

Let $W = \mathcal{R}w$, stable under ∂_t and ∂_x. Let p and q be the minimal polynomials of $\partial_t|_W$ and $\partial_x|_W$. Then

$$p(\partial_t)\, w = 0, \qquad q(\partial_x)\, w = 0.$$

Step 6 On a rectangle the ODE in t separates the variables

Let $Q_0 = I' \times J' \subset \Omega$ be an open rectangle. We claim that every $v \in \mathcal{D}'(Q_0)$ with $p(\partial_t)v = 0$ has the form

$$v = \sum_{j,l} c_{jl} \otimes t^l e^{\lambda_j t}, \qquad c_{jl} \in \mathcal{D}'(J'),$$

with λ_j the roots of p. Induction on $\deg p$. Write $p = (\tau - \lambda)\tilde{p}$. By induction $(\partial_t - \lambda)v = \sum \tilde{c}_{jl} \otimes t^l e^{\lambda_j t}$. Each summand has a particular solution $\tilde{c} \otimes g(t)$:

$$g = r(t)\, e^{\lambda_j t}\ (\deg r = l)\ \text{ if } \lambda_j \neq \lambda, \qquad g = \frac{t^{l+1} e^{\lambda t}}{l+1}\ \text{ if } \lambda_j = \lambda.$$

With v_p their sum, $\partial_t\big(e^{-\lambda t}(v - v_p)\big) = 0$.

A distribution V on $J' \times I'$ with $\partial_t V = 0$ is $c \otimes 1$. Indeed, fix $\chi \in C_c^\infty(I')$ with $\int \chi = 1$. For a test function φ put $\Phi(x) = \int \varphi(x,t)\,\mathrm{d}t$ and

$$\psi(x,t) = \int_{-\infty}^{t} \big(\varphi(x,s) - \chi(s)\Phi(x)\big)\,\mathrm{d}s.$$

Since I' is an interval, ψ has compact support, and $\varphi = \chi \otimes \Phi + \partial_t \psi$. Hence $V(\varphi) = V(\chi \otimes \Phi)$, which depends only on Φ. So $v = v_p + c \otimes e^{\lambda t}$, completing the induction.

Step 7 The ODE in x makes the coefficients exponential-polynomials

Applying $q(\partial_x)$:

$$\sum_{j,l} q(\partial_x) c_{jl} \otimes t^l e^{\lambda_j t} = 0.$$

The functions $g_a = t^l e^{\lambda_j t}$ are independent on I', so there are test functions ψ_b with $\int g_a \psi_b = \delta_{ab}$. Testing against $\varphi(x)\psi_b(t)$ gives $q(\partial_x)c_{jl} = 0$. The one-variable case of Step 6 gives $c_{jl} \in \mathrm{EP}(J')$. So w restricted to Q_0 is the restriction of an exponential-polynomial E_{Q_0} on $\mathbb{R}^2$.

Step 8 Connectedness glues the rectangles

Fix Q_0 and $E = E_{Q_0}$. Let $A = \{y \in \Omega :\ w = E \text{ near } y\}$. It is open and nonempty. It is closed in Ω: if $y \in \bar{A}$, take a rectangle $Q_y \ni y$. On the open set $Q_y \cap A$ we have $E_{Q_y} = w = E$, so by Step 2 the exponential-polynomial $E_{Q_y} - E$ vanishes identically, and $w = E$ near y. By connectedness $A = \Omega$, so $w = E|_\Omega \in \mathrm{EP}(\Omega)$.

Why it works. finite dimension turns differentiation into two commuting matrices; their eigenvalues are the exponents, and rectangles plus connectedness globalise the local form.

■

Uses: A1.

Why rectangles in Step 6

On a U-shaped region, $v = \phi(x)\mathbf{1}_{t>2}$ (with $\phi = 0$ for $x < 1$, $\phi = 1$ for $x > 2$) is smooth and satisfies $\partial_t v = 0$, but it is not of the form $c(x) \otimes 1$. So the local form is proved on rectangles and glued in Step 8. This v is not a counterexample to A2: its derivative space is infinite-dimensional.

1.2 The identification theorems

Dim. A3 **C3-T1**

Identification criterion

[proved]

Let Ω be connected, $u \in \mathcal{D}'(\Omega)$, $P(\partial)u = 0$ with $P = c\prod_i P_i^{e_i}$ nonconstant, and

$$w_i = \Big(\frac{P}{P_i}\Big)(\partial)u, \qquad w_i' = \Big(\frac{P}{P_i^{e_i}}\Big)(\partial)u.$$

(a) $\operatorname{Ann}(u) = (P) \iff w_i \notin \mathrm{EP}(\Omega)$ for every i.

(b) $V(\operatorname{Ann} u) = V(P) \iff w_i' \notin \mathrm{EP}(\Omega)$ for every i.

(c) If P is squarefree, the two conditions coincide.

(d) If P is monic in τ of degree m: $\operatorname{Ann}(u) = (P)$ iff the only $(b_0, \dots, b_{m-1}) \in \mathbb{C}[\xi]^m$ with $\sum_j b_j(\partial_x)\partial_t^j u = 0$ is zero.

(e) In general $\operatorname{Ann}(u) = G \cdot J$ with $G \mid P$, $V(J)$ finite, and $G \sim P \iff \operatorname{Ann}(u) = (P)$.

Stated in the paper as no. 4.2, p. 18

Idea. Write $\operatorname{Ann}(u) = G \cdot J$. If G misses a copy of some P_i, then w_i is killed by a power of the ideal of the finite set $V(J)$, so it is an exponential-polynomial. Conversely, an exponential-polynomial w_i is killed by some h coprime to P_i, and $h \cdot P/P_i$ is an annihilator of u that P does not divide.

Proof.

Let $I = \operatorname{Ann}(u)$. It contains P. By A1, $I = G \cdot J$ with $G = \gcd I$ dividing P, so

$$G = c' \prod_i P_i^{f_i}, \qquad 0 \le f_i \le e_i.$$

Step 1 **(e) $G \sim P$ exactly when $I = (P)$**

If $G \sim P$, then $I = PJ \subseteq (P) \subseteq I$. If $I = (P)$, then P divides every element of I and is itself in I, so it is the gcd.

Step 2 (a), first direction: if $I \neq (P)$, some w_i is an exponential-polynomial

If $I \neq (P)$, then $G \not\sim P$, so some $f_i < e_i$, and $P/P_i = G\,H$ for some $H \in \mathcal{R}$. By A1(iii), $J \supseteq \mathfrak{n}^N$ with $\mathfrak{n}$ the ideal of $V(J)$. For $h \in \mathfrak{n}^N$:

$$h \cdot \frac{P}{P_i} = G\,(Hh) \in G\,J = I, \qquad \text{so} \qquad h(\partial)\,w_i = 0.$$

Hence $\mathrm{Ann}(w_i) \supseteq \mathfrak{n}^N$, its zero set lies in the finite $V(J)$, and $w_i \in \mathrm{EP}$ by A2.

Step 3 (a), second direction: an exponential-polynomial w_i gives a new annihilator

Suppose $w_i \in \mathrm{EP}$. Then $V(\mathrm{Ann}\,w_i)$ is finite, so by A1(F) some $h \in \mathrm{Ann}(w_i)$ has $P_i \nmid h$. Put

$$Q = h \cdot \frac{P}{P_i}.$$

Then $Q(\partial)u = h(\partial)w_i = 0$, so $Q \in I$. If $Q = PA$, cancelling P/P_i in the domain $\mathcal{R}$ gives $h = P_i A$, a contradiction. So $Q \in I \setminus (P)$.

Step 4 (b) The variety sees only whether each $f_i \geq 1$

We have $V(I) = V(G) \cup V(J)$, with $V(J)$ finite. If $f_i \geq 1$, then $V(P_i) \subseteq V(G) \subseteq V(I)$. If $f_i = 0$, then $V(P_i) \cap V(I)$ lies in the finite set

$$\bigcup_{f_j \geq 1} \big(V(P_i) \cap V(P_j)\big) \;\cup\; V(J),$$

while $V(P_i)$ is infinite, so $V(P_i) \not\subseteq V(I)$. Hence $V(I) = V(P)$ iff all $f_i \geq 1$. Finally, $f_i = 0$ iff $w_i' \in \mathrm{EP}$: one direction is Step 2 with $P/P_i^{e_i}$ in place of P/P_i; for the other, an $h \in \mathrm{Ann}(w_i')$ with $P_i \nmid h$ gives $G \mid h\,P/P_i^{e_i}$, and P_i divides neither factor, so $f_i = 0$.

Step 5 (c) and (d)

For $e_i = 1$, $w_i = w_i'$, which is (c). For (d), divide by the monic P in $\mathbb{C}[\xi][\tau]$:

$$Q = AP + \sum_{j<m} b_j(\xi)\,\tau^j \quad \text{uniquely.}$$

Since $A(\partial)P(\partial)u = 0$, $Q \in I$ iff $\sum_j b_j(\partial_x)\partial_t^j u = 0$, and $Q \in (P)$ iff all $b_j = 0$.

Why it works. the gcd of the annihilator records the infinite part of u; any factor it misses leaves behind a function killed by a zero-dimensional ideal, that is, an exponential-polynomial.

■

Uses: A1, A2.

> **Example 1.1 The proof on a wave**
>
> Let $P = \tau^2 - \xi^2$ and $u = e^{x-t} + e^{-(x+t)^2}$. For the factor $P_i = \tau + \xi$:
>
> $$w_i = (\partial_t - \partial_x)u = -2e^{x-t} \in \mathrm{EP}.$$

Step 3 takes $h = \xi - 1$, which kills e^{x-t} and is not a multiple of $\tau + \xi$, and produces

$$Q = (\xi - 1)(\tau - \xi) \in \mathrm{Ann}(u) \setminus (P).$$

Dim. A4 **C3-T6**

The intrinsic operator [proved]

Let $0 \neq P \in \mathrm{Ann}(u)$, $P = c\prod_i P_i^{e_i}$, and

$$S_i = \Big\{k \in \{0, \dots, e_i\} : \Big(\frac{P}{P_i^k}\Big)(\partial)u \in \mathrm{EP}\Big\}.$$

Then $G_u = \gcd \mathrm{Ann}(u) \sim \prod_i P_i^{f_i}$ with $f_i = e_i - \max S_i$. Moreover $\mathrm{Ann}(u)$ is principal iff $G_u(\partial)u = 0$, and then $\mathrm{Ann}(u) = (G_u)$.

Stated in the paper as no. 4.3, p. 21

Idea. The multiplicity f_i of P_i in G_u is detected by stripping copies of P_i from P one at a time: as long as fewer than $e_i - f_i$ copies are removed the leftover is an exponential-polynomial, and as soon as more are removed it is not.

Proof.

Step 1 The key equivalence

By A1, $\mathrm{Ann}(u) = G_u J_u$ with $G_u \mid P$. We show, for $1 \le k \le e_i$,

$$f_i \le e_i - k \iff \Big(\frac{P}{P_i^k}\Big)(\partial)u \in \mathrm{EP}.$$

($\Rightarrow$) Then $G_u \mid P/P_i^k$; write $P/P_i^k = G_u H$. As in A3, Step 2, $h \cdot P/P_i^k \in \mathrm{Ann}(u)$ for every h in a power of the ideal of $V(J_u)$, so the function is an exponential-polynomial.

($\Leftarrow$) If it is an exponential-polynomial, pick h in its annihilator with $P_i \nmid h$ (A1(F)). Then G_u divides $h \cdot P/P_i^k$. Comparing multiplicities of P_i gives $f_i \le e_i - k$.

Step 2 Reading off f_i

The equivalence holds trivially for $k = 0$. So $S_i = \{0, 1, \dots, e_i - f_i\}$, and $\max S_i = e_i - f_i$. (The set is an interval because EP is stable under derivatives:

$$\Big(\frac{P}{P_i^{k-1}}\Big)(\partial)u = P_i(\partial)\Big[\Big(\frac{P}{P_i^k}\Big)(\partial)u\Big].)$$

Step 3 Principality

If $G_u(\partial)u = 0$, then $(G_u) \subseteq \mathrm{Ann}(u) = G_u J_u \subseteq (G_u)$. Conversely, if $\mathrm{Ann}(u) = (Q)$, then Q is the gcd: $Q \sim G_u$, and $G_u(\partial)u = 0$.

Why it works. each copy of P_i that u does not need leaves an exponential-polynomial behind; counting them gives the exponent.

■

Uses: A1, A2, A3.

Dim. A5 **P4-T2**

Spectral theorem for the cyclic module [proved]

Let $W = \mathcal{R}u \cong \mathcal{R}/I$ with $I = \mathrm{Ann}(u) = G \cdot J$ as in A1. A *joint eigenvector* for $z \in \mathbb{C}^2$ is $w \in W$, $w \neq 0$, with $\mathfrak{m}_z(\partial)w = 0$; $\sigma_p(u)$ is the set of z having one.

(i) Every joint eigenvector for z is $c\,e^{\langle z,\cdot\rangle}$ with $c \neq 0$, and $\sigma_p(u) = V(J)$.

(ii) $W \cap \mathrm{EP}(\Omega) = \mathcal{R}\,G(\partial)u \cong \mathcal{R}/J$, of finite dimension $\dim \mathcal{R}/J$.

(iii) $\mathrm{Ann}(u)$ is principal $\iff J = \mathcal{R} \iff \sigma_p(u) = \emptyset \iff W \cap \mathrm{EP} = 0$.

(iv) Hence, if $P(\partial)u = 0$: $\mathrm{Ann}(u) = (P) \iff \sigma_p(u) = \emptyset$ and $G \sim P$.

Stated in the paper as no. 4.3, p. 20

Idea. Translate statements about the quotient ring $\mathcal{R}/I$ (maximal ideals, socle) into statements about W. The only analytic input is the EP test A2.

Proof.

Two small facts are used.

- **(LG)** If $G \mid (\tau - \tau_z)Q$ and $G \mid (\xi - \xi_z)Q$, then $G \mid Q$: in a unique factorisation domain the gcd of the two products is Q.
- **(LS)** If $M \neq 0$ is a module with $\mathfrak{m}_z^N M = 0$, some $v \neq 0$ in M has $\mathfrak{m}_z v = 0$: take $y \neq 0$ and k maximal with $\mathfrak{m}_z^k y \neq 0$.

Step 1 Joint eigenvectors are single exponentials

If $w = Q(\partial)u \neq 0$ and $\mathfrak{m}_z^N(\partial)w = 0$, then $V(\mathrm{Ann}\, w) \subseteq \{z\}$, so $w \in \mathrm{EP}$ and by A2 $w = q\,e^{\langle z,\cdot\rangle}$. The eigenvector condition gives $\partial_t q = \partial_x q = 0$: q is a constant.

Step 2 $\sigma_p(u) \subseteq V(J)$

Let $Q(\partial)u$ be a joint eigenvector at z. Then $(\tau - \tau_z)Q$ and $(\xi - \xi_z)Q$ lie in $I \subseteq (G)$, so $G \mid Q$ by (LG). Write $Q = GQ'$. Cancelling G:

$$(\tau - \tau_z)Q' \in J, \qquad (\xi - \xi_z)Q' \in J, \qquad Q' \notin J.$$

So $\mathfrak{m}_z \subseteq (J : Q') \neq \mathcal{R}$. As $\mathfrak{m}_z$ is maximal, $(J : Q') = \mathfrak{m}_z \supseteq J$, and $z \in V(J)$.

Step 3 $V(J) \subseteq \sigma_p(u)$

By A1(iii), $J \supseteq \prod_{z' \in V(J)} \mathfrak{m}_{z'}^N$, and by the Chinese remainder theorem

$$\mathcal{R}/J \cong \prod_{z' \in V(J)} \mathcal{R}/\left(J + \mathfrak{m}_{z'}^N\right).$$

The factor at z is nonzero and killed by $\mathfrak{m}_z^N$. By (LS) it contains $v \neq 0$ with $\mathfrak{m}_z v = 0$. Through the isomorphism of Step 4 this gives a joint eigenvector $(vG)(\partial)u \neq 0$.

Step 4 The exponential-polynomial part is $\mathcal{R}G(\partial)u$

Since $\mathcal{R}$ is a domain,

$$\operatorname{Ann}\big(G(\partial)u\big) = (I : G) = \{Q : \ QG \in GJ\} = J.$$

$V(J)$ is finite, so $G(\partial)u \in \mathrm{EP}$ and $\mathcal{R}G(\partial)u \subseteq \mathrm{EP}$. Conversely, let $w = Q(\partial)u \in \mathrm{EP}$. For each prime factor P_i of G, with multiplicity f_i, A1(F) gives $h \in \operatorname{Ann}(w)$ with $P_i \nmid h$. Then $hQ \in I \subseteq (G)$, so $P_i^{f_i} \mid hQ$ and $P_i^{f_i} \mid Q$. Hence $G \mid Q$ and $w \in \mathcal{R}G(\partial)u$. The map $Q \mapsto (QG)(\partial)u$ is onto with kernel J, so the space is $\cong \mathcal{R}/J$.

Step 5 (iii) and (iv)

$J = \mathcal{R} \iff V(J) = \emptyset$ (A1(iii)) $\iff \sigma_p(u) = \emptyset$ (Steps 2–3). Also $J = \mathcal{R} \iff I = (G)$ is principal, and $W \cap \mathrm{EP} \cong \mathcal{R}/J = 0 \iff J = \mathcal{R}$. Finally $\operatorname{Ann}(u) = (P)$ iff I is principal and $G \sim P$.

Why it works. the finite part J of the annihilator is exactly the module of exponentials inside $\mathcal{R}u$; its points are the eigenvalues.

■

Uses: A1, A2.

An empty spectrum alone is not enough

Take $u = e^{x^2}$ and $P = \tau^2$. For $Q = Q_0(\xi) + \tau Q_1$ one has $Q(\partial)u = Q_0(\partial_x)e^{x^2}$, which vanishes only if $Q_0 = 0$. So $\operatorname{Ann}(u) = (\tau)$ is principal and $\sigma_p(u) = \emptyset$, but $\operatorname{Ann}(u) \neq (\tau^2)$. The spectrum sees J_u, not G_u; condition (iv) needs both.

1.3 Finite data

Dim. A6 **C3-T2**

Bézout threshold and jet threshold [proved]

(C3-T2) Let $u = \sum_{k=1}^{M} c_k\, e^{\langle z_k, \cdot \rangle}$ with distinct $z_k \in V(P)$ and $c_k \neq 0$, P squarefree, $n \geq \deg P$. If

$$\big|\{z_k\} \cap V(P_i)\big| > n \deg P_i \qquad \text{for every } i,$$

then $\mathrm{Ann}(u) \cap \mathcal{R}_{\leq n} = P \cdot \mathcal{R}_{\leq n - \deg P}$. The threshold cannot be lowered uniformly.

(C3-T3) Let $0 \neq u \in \mathrm{EP}(\Omega)$, $d = \dim \mathcal{R}u$, $p_0 \in \Omega$.

(i) $\mathrm{Ann}(u) \cap \mathcal{R}_{\leq n} = \{Q \in \mathcal{R}_{\leq n} : \partial^\beta (Q(\partial)u)(p_0) = 0 \ \forall |\beta| \leq d-1\}$.

(ii) $\mathrm{Ann}(u)$ is generated in degree $\leq d$, so the jet of order $2d-1$ determines it.

(iii) The order $2d-1$ is optimal in the worst case.

Stated in the paper as no. 5.2, p. 25

Idea. For modes: an annihilator vanishes at every exponent, and Bézout bounds how many exponents a curve that does not contain $V(P_i)$ can pass through. For jets: inside the d-dimensional space $\mathcal{R}u$, the subspaces of elements vanishing to order k at p_0 shrink strictly, so after $d-1$ steps they are zero.

Proof.

Step 1 Annihilators of a sum of modes

By A2(a)(b), $Q(\partial)u = \sum_k c_k Q(z_k)\, e^{\langle z_k, \cdot \rangle}$, which is zero iff $Q(z_k) = 0$ for all k.

Step 2 Bézout forces every P_i to divide Q

Let $Q \in \mathrm{Ann}(u)$ with $\deg Q \leq n$. If $Q = 0$ there is nothing to prove; a nonzero constant cannot kill $u \neq 0$. Suppose $P_i \nmid Q$. Then Q and P_i are coprime, and Bézout's theorem (the homogenisations have no common component, since the line at infinity divides neither) gives

$$\big|V(Q) \cap V(P_i)\big| \leq \deg Q \cdot \deg P_i \leq n \deg P_i.$$

But by Step 1, $\{z_k\} \cap V(P_i) \subseteq V(Q) \cap V(P_i)$ has more points. So $P_i \mid Q$ for every i, and since the P_i are distinct primes, $P \mid Q$.

Step 3 The threshold is sharp

Take $Z_n = \{(j^2, \pm j) : j = 1, \dots, n\}$, $2n$ points on $\tau = \xi^2$, and

$$Q_n = \prod_{j=1}^{n} (\tau - j^2).$$

It has degree n, vanishes on Z_n, and is not a multiple of $\tau - \xi^2$ because it does not involve ξ.

Step 4 (i) Vanishing subspaces shrink strictly

Let $W = \mathcal{R}u$ and

$$K_k = \{w \in W : \partial^\beta w(p_0) = 0 \text{ for all } |\beta| \le k\}.$$

If $K_k = K_{k+1}$, then K_k is stable under derivatives (if $w \in K_{k+1}$ then $\partial w \in K_k$), so every $w \in K_k$ has all derivatives zero at p_0; being analytic on a connected set, $w = 0$. So the dimensions drop strictly until they reach 0. Since $\dim K_0 = d - 1$, we get $K_{d-1} = 0$. For $Q \in \mathcal{R}_{\le n}$, $Q(\partial)u \in W$ vanishes iff it lies in K_{d-1}. Any upper bound $d' \ge d$ works as well.

Step 5 (ii) Generators of low degree

$\mathcal{R}/\operatorname{Ann}(u) \cong W$ has dimension d. In a graded monomial order, the standard monomials form a set of d elements closed under division. A standard monomial $\tau^a \xi^b$ has $(a+1)(b+1) \ge a+b+1$ divisors, all standard, so $a+b+1 \le d$. The minimal generators of the initial ideal therefore have degree at most d, and so do the elements of the reduced Gröbner basis, which generate $\operatorname{Ann}(u)$. Apply (i) with $n = d$: the jet of order $2d-1$ suffices.

Step 6 (iii) Optimality

Let $u = \sum_{j=1}^{d} e^{jx}$, so $d = \dim \mathcal{R}u$, and $n = d$. Then $\operatorname{Ann}(u) \cap \mathbb{C}[\xi]_{\le d} = \mathbb{C} \cdot Q_0$ with $Q_0 = \prod_j (\xi - j)$. Put $m_k = \partial_x^k u(p_0)$. With only $|\beta| \le d-2$ the test of (i) is the Hankel system

$$\sum_{i=0}^{d} q_i \, m_{k+i} = 0, \qquad k = 0, \dots, d-2,$$

which has $d-1$ equations in $d+1$ unknowns, so a kernel N of dimension at least 2. Take $Q_1 \in N \setminus \mathbb{C}Q_0$ and $Q' = Q_0 + sQ_1$ with s small: it has degree d and d simple roots μ_j. Choose w_j with $\sum_j w_j \mu_j^k = m_k$ for $k < d$ (Vandermonde) and set

$$u' = \sum_j w_j \, e^{\mu_j (x - x_0)}.$$

The moment sequences of u and u' satisfy the same recursion given by Q' and agree for $k \le d-1$, hence for $k \le 2d-2$. So u and u' have the same jet of order $2d-2$, but $Q' \in \operatorname{Ann}(u') \setminus \operatorname{Ann}(u)$.

Why it works. finitely many exponentials are finitely many points; Bézout counts how many points a curve must miss, and the strict drop of vanishing subspaces counts how many derivatives are needed.

■

Uses: A1, A2.

Dim. A7 **C3-P1**

No finite certificate

[proved] (jets) / [from cited source] (functionals)

Let Ω be connected, $p_0 \in \Omega$, P nonconstant, u smooth with $P(\partial)u = 0$, $K \geq 0$, and $E_P = \ker P \cap \mathrm{EP}(\Omega)$.

(i) $\bigcap_{v \in E_P} \mathrm{Ann}(v) = (P)$.

(ii) There is $v \in E_P$ with the same jet of order K at p_0 as u; in particular $\mathrm{Ann}(v) \supsetneq (P)$.

On convex Ω the same holds for any finitely many continuous linear functionals (Malgrange); on a non-convex region it can fail.

Stated in the paper as no. 5.3, p. 26

Idea. By duality, the possible jets of solutions are cut out by the multiples of P. Exponential-polynomial solutions already cut out the same space, because for every Q that is not a multiple of P one builds a solution $s^{e-1}e^{\langle z,\cdot\rangle}$ that Q does not kill.

Proof.

Pair an operator Q of degree at most K with a jet by

$$\langle Q, J_K w\rangle = \big(Q(\partial)w\big)(p_0).$$

Step 1 **Vanishing on all of E_P means annihilating all of E_P**

If $(Q(\partial)v)(p_0) = 0$ for every $v \in E_P$, then, since E_P is stable under derivatives, all derivatives of the exponential-polynomial $Q(\partial)v$ vanish at p_0. It is entire with zero Taylor series, so it is 0. Hence $Q \in \bigcap_{v \in E_P} \mathrm{Ann}(v)$.

Step 2 **(i) A non-multiple of P misses some exponential solution**

Let $Q \neq 0$ with $P \nmid Q$. Some i has $f = v_{P_i}(Q) < e_i$; write $Q = P_i^f Q'$ with $P_i \nmid Q'$. The sets $V(P_i) \cap V(\nabla P_i)$ and $V(P_i) \cap V(Q')$ are finite (A1(V)), while $V(P_i)$ is infinite. Pick $z \in V(P_i)$ with

$$\nabla P_i(z) \neq 0, \qquad Q'(z) \neq 0,$$

and a coordinate $s \in \{t, x\}$ with $\partial_s P_i(z) \neq 0$. On polynomials in s, $P_i(z + \partial)$ lowers the degree by exactly one and $Q'(z + \partial)$ preserves it. Let $v = s^{e_i-1}e^{\langle z,\cdot\rangle}$. Then $P(z + \partial)s^{e_i-1} = 0$, so $v \in E_P$, while $Q(z + \partial)s^{e_i-1}$ has degree $e_i - 1 - f \geq 0$ and is nonzero. So $Q \notin \mathrm{Ann}(v)$.

Step 3 **(ii) Every jet of a solution is a jet of an exponential solution**

By Steps 1–2, every Q of degree $\leq K$ orthogonal to the jets of E_P is a multiple of P, hence orthogonal to the jet of u. In finite dimension this means

$$J_K(u) \in J_K(E_P),$$

so a suitable $v \in E_P$ exists. Finally $\mathrm{Ann}(v) \supsetneq (P)$: if $v = 0$ it is $\mathcal{R}$; otherwise $V(\mathrm{Ann}\, v)$ is finite while $V(P)$ is infinite.

Step 4 General functionals and convexity

For general continuous functionals the analogue on convex Ω is Malgrange's approximation theorem (Ann. Inst. Fourier 6, Chap. I §2 Thm 2) [from cited source]. Without convexity it fails: on a U-shaped region take $P = \tau$, $u = \phi(x)\mathbf{1}_{t>2}$, and

$$\ell(w) = w(5/2, 5/2) - w(1/2, 5/2).$$

Then $\ell(u) = 1$ while $\ell(v) = 0$ for every $v \in \ker \partial_t$ with $\mathrm{Ann}(v) \neq (\tau)$.

Why it works. finite data see a finite-dimensional shadow of $\ker P$, and exponential solutions already fill every such shadow.

■

Uses: A1, A2.

Dim. A8 **P4-T4**

Decidability with a bound [proved]

In the model M2$^+$ (exact jets at p_0 in a computable field K, plus a zero-test oracle):

(i) for every finite-dimensional $\mathcal{L} \subseteq \mathcal{R}$ with a basis over K, $\mathrm{Ann}(u) \cap \mathcal{L}$ is computable;

(ii) "$\mathrm{Ann}(u) \neq (P)$" is semi-decidable;

(iii) if an integer $B \geq \dim(\mathcal{R}u \cap \mathrm{EP})$ is given, then with $N = B + \deg P - 1$,

$$\mathrm{Ann}(u) = (P) \iff \mathrm{Ann}(u) \cap \mathcal{R}_{\leq N} = (P) \cap \mathcal{R}_{\leq N},$$

which is decidable. The test "$Q(\partial)u \in \mathrm{EP}$?" is then decidable, and G_u is computable.

Stated in the paper as no. 5.4, p. 27

Idea. Linear algebra on jets gives a decreasing chain of candidate spaces that stabilises at the true annihilators; the zero oracle confirms when it has stabilised. If identification fails, a witness of small degree exists, and the bound B says how small.

Proof.

Step 1 (i) A decreasing chain of linear spaces

Let

$$A_k = \{Q \in \mathcal{L} : \partial^\beta (Q(\partial)u)(p_0) = 0 \text{ for all } |\beta| \leq k\}.$$

It is the kernel of a matrix with entries $\partial^{\beta+\alpha}u(p_0) \in K$, so it has a computable basis. The chain $A_0 \supseteq A_1 \supseteq \cdots \supseteq \mathrm{Ann}(u) \cap \mathcal{L}$ stabilises, and its limit is $\mathrm{Ann}(u) \cap \mathcal{L}$, because $Q(\partial)u$ is analytic with zero Taylor series at p_0. Algorithm: for $k = 0, 1, \ldots$ compute a basis of A_k and ask the oracle about each element; stop when all answers are "zero". It stops only when $A_k \subseteq \mathrm{Ann}(u)$, so it is correct, and it stops at the stabilisation index.

Step 2 (ii) Semi-decision

For $n = \deg P, \deg P + 1, \ldots$ compute $\mathrm{Ann}(u) \cap \mathcal{R}_{\le n}$ by (i) and compare its dimension with that of $P \cdot \mathcal{R}_{\le n-\deg P}$. If $\mathrm{Ann}(u) \neq (P)$, a witness Q exists and the loop stops at $n = \max(\deg Q, \deg P)$.

Step 3 (iii) A witness of degree at most N

Suppose $\mathrm{Ann}(u) \neq (P)$. By A3, some $w_i = (P/P_i)(\partial)u$ is an exponential-polynomial in $\mathcal{R}u$, so

$$d_i = \dim \mathcal{R}w_i \le B.$$

By A6(ii), $\mathrm{Ann}(w_i)$ is generated in degree $\le d_i$, and not all generators are divisible by P_i (A1(F)). Pick such a generator h, so $\deg h \le B$. Then

$$Q = h \cdot \frac{P}{P_i} \in \mathrm{Ann}(u) \setminus (P), \qquad \deg Q \le B + \deg P - 1 = N.$$

The comparison in degree N is computable by (i).

Step 4 The EP test from the bound

Let $w = Q(\partial)u$; its oracles follow from those of u. If $w \in \mathrm{EP}$, then $\dim \mathcal{R}w \le B$, so the minimal polynomials of ∂_t and ∂_x on $\mathcal{R}w$ have degree $\le B$: there are nonzero $a(\tau), b(\xi)$ of degree $\le B$ in $\mathrm{Ann}(w)$. Conversely, such a, b give a finite zero set and $w \in \mathrm{EP}$ (A2). Both searches are computable by (i). With this test, A4 computes G_u.

Why it works. a failure of identification is always witnessed in bounded degree, and the bound B makes that degree explicit.

■

Uses: A1, A2, A3, A4, A6.

Dim. A9 **P4-T5**

Black-box impossibility [proved]

Let $P \in K[\tau, \xi]$ be nonconstant (K the algebraic numbers), Ω connected, $p_0 \in \Omega$ with coordinates in $K \cap \mathbb{R}$, and u real-analytic with $P(\partial)u = 0$, jets in K, and $\mathrm{Ann}(u) = (P)$ (such u exists for every P, (P4-L7)). For every order k_0 and every finite set $\mathcal{Q} \subset K[\tau, \xi]$ there is an exponential-polynomial $v \in \ker P$ with

- the same jet of order k_0 at p_0 as u;
- for every $Q \in \mathcal{Q}$: $Q(\partial)v = 0 \iff Q(\partial)u = 0$;
- $\mathrm{Ann}(v) \supsetneq (P)$.

Hence no algorithm in the model M2$^+$ decides "$\mathrm{Ann}(\cdot) = (P)$" correctly on all inputs.

Stated in the paper as no. 5.3, p. 26

Idea. Run a correct algorithm on u: it halts after finitely many questions. An adversary builds v among the exponential-polynomial solutions so that it gives the same answer to every one of those questions. First match the jet, then, for each Q with $Q(\partial)u \neq 0$, add a solution with zero jet that Q does not kill, with generic coefficients.

Proof.

Step 1 Witness points with algebraic coordinates

Each $V(P_i)$ has infinitely many points with coordinates in K: roots of $P_i(\cdot, \xi_0)$ for $\xi_0 \in K$, or $K \times \{\xi_0\}$ if P_i does not involve τ. For such z with $\nabla P_i(z) \neq 0$, pick a coordinate s_z with $\partial_s P_i(z) \neq 0$ and put

$$g_{i,z,j} = s_z^j \, e^{\langle z, \cdot - p_0 \rangle}, \qquad j < e_i.$$

As in A7, Step 2, $P(z+\partial)s_z^j = 0$, so $g_{i,z,j} \in \ker P$.

Step 2 Matching the jet

Let V_{k_0} be the span of the jets of order k_0 of all $g_{i,z,j}$. With the pairing of A7,

$$V_{k_0}^{\perp} = (P) \cap \mathcal{R}_{\leq k_0};$$

the inclusion "$\subseteq$" is A7, Step 2, with points chosen in K^2. Since $\langle PA, J_{k_0} u \rangle = 0$, $J_{k_0}(u) \in V_{k_0}$. A linear system with coefficients in K that is solvable over $\mathbb{C}$ is solvable over K, so some combination v_1 of the g's has the jet of u.

Step 3 A zero-jet solution for each test

For $Q \in \mathcal{Q} \setminus (P)$ (so $Q(\partial)u \neq 0$), pick i with $f = v_{P_i}(Q) < e_i$, $Q = P_i^f Q'$, and more than $\dim \mathcal{R}_{\leq k_0}$ points $z \in V(P_i) \cap K^2$ with $\nabla P_i(z) \neq 0$ and $Q'(z) \neq 0$. The map

$$c \;\mapsto\; J_{k_0}\Big(\sum_z c_z \, s_z^{e_i - 1} e^{\langle z, \cdot - p_0 \rangle} \Big)$$

goes from a larger space to a smaller one, so it has a nonzero kernel vector c. Let e_Q be the corresponding sum: its jet is 0, while $Q(\partial)e_Q \neq 0$ by A2(b).

Step 4 A generic combination answers like u

Let $v_\lambda = v_1 + \sum_Q \lambda_Q \, e_Q$. It has the jet of u for every λ. For each Q with $Q(\partial)u \neq 0$, the set $\{\lambda : \; Q(\partial)v_\lambda = 0\}$ is a proper affine subspace. Finitely many proper subspaces do not cover the space, so some λ avoids all of them. For $Q \in (P)$ both answers are "zero". So $v = v_\lambda$ answers every question as u does, and $v \in \mathrm{EP} \cap \ker P$ gives $\mathrm{Ann}(v) \supsetneq (P)$.

Step 5 Conclusion

The run of the algorithm on u depends only on the answers it receives. They coincide for v, so it outputs "Ann $= (P)$" on v as well, which is wrong.

Why it works. a finite conversation with a black box can always be imitated by a finite sum of exponentials, and such sums never identify.

■

Uses: A2, A7.

CHAPTER 2

Decomposition and regularity

In this chapter

Starting from	$P = c\prod_{j=1}^{s} Q_j$ with $Q_j = P_j^{m_j}$ pairwise coprime; $\Omega \subset \mathbb{R}^2$ open; $\mathcal{A}$ is $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$; the toolkit of Chapter 1.
Why	to prove that a solution splits into one piece per factor, with an ambiguity of exactly known size, that holes in Ω can break this, and exactly when the pieces keep the Sobolev regularity of u.
Role	Dim. A10–A18. They prove Chapters 6 and 7 of the paper.

Notation for the whole chapter

For a module M over $\mathcal{R}$, $\mathrm{Sol}_{\mathcal{A}}(M) = \mathrm{Hom}_{\mathcal{R}}(M, \mathcal{A})$ is its solution space; for example $\mathrm{Sol}(\mathcal{R}/(f)) = \ker_{\mathcal{A}} f$. For $z = (p, q) \in \mathbb{C}^2$ we write $e_z = e^{pt+qx}$ and $D_t = \partial_t - p$, $D_x = \partial_x - q$. Note that $D_t(e_z \tilde{w}) = e_z\, \partial_t \tilde{w}$.

2.1 Terms and factors

Dim. A10 — **R2-T3**

Terms do not give components

[proved]

Let $L = L_1 + L_2 \in \mathcal{R}$.

(a) In every $\mathcal{R}$-module $\mathcal{A}$:

$$\ker L \cap \big(\ker L_1 + \ker L_2\big) \subseteq \mathrm{Sol}\big(\mathcal{R}/(L,\ L_1L_2)\big) = \mathrm{Sol}\big(\mathcal{R}/(L_1 + L_2,\ L_1^2)\big).$$

(b) If $\mathcal{A} = C^\infty$ or $\mathcal{D}'$ of a convex Ω and L_1, L_2 are coprime, equality holds.

(c) $V(L_1 + L_2,\ L_1^2) = V(L_1) \cap V(L_2)$. For coprime L_1, L_2 the space is finite-dimensional, of dimension $2\sum_p I_p(L_1, L_2)$, and consists of exponential-polynomials with exponents at the common zeros.

Stated in the paper as no. 6.1.1, p. 30

Idea. A solution that splits by terms satisfies one extra equation, $L_1L_2u = 0$, because the operators commute. Modulo L that extra equation is L_1^2, and the common zeros of L and L_1^2 are where both terms vanish.

Proof.

Step 1 (a) The extra equation

Let $u = u_1 + u_2$ with $L_1 u_1 = 0$ and $L_2 u_2 = 0$. Since the operators commute,

$$L_1 L_2\, u = L_2(L_1 u_1) + L_1(L_2 u_2) = 0.$$

Together with $Lu = 0$ this gives $u \in \mathrm{Sol}(\mathcal{R}/(L, L_1 L_2))$. Modulo L, $L_2 \equiv -L_1$, so $L_1 L_2 \equiv -L_1^2$ and the two ideals coincide.

Step 2 (b) Equality from the decomposition theorem

For coprime L_1, L_2 the finite module $\mathcal{R}/(L_1, L_2)$ has no obstruction (A12), so the sequence of A13 is exact: every element of $\mathrm{Sol}(\mathcal{R}/(L, L_1 L_2))$ is killed by $L_1 L_2$, hence by A13 splits as a sum of solutions of L_1 and of L_2.

Step 3 (c) The variety

On $V(L_1)$ we have $L_1 + L_2 = 0 \iff L_2 = 0$. Off $V(L_1)$, $L_1^2 \neq 0$. So

$$V(L_1 + L_2,\ L_1^2) = V(L_1) \cap V(L_2).$$

For coprime terms this set is finite (A1(V)). The dimension follows from the additivity of intersection numbers, $I_p(F, GH) = I_p(F, G) + I_p(F, H)$, and $I_p(L_1 + L_2, L_1) = I_p(L_2, L_1)$.

Why it works. commuting terms turn a split by terms into an extra equation, which cuts the solution space down to the finitely many points where both terms vanish.

■

Uses: A1, A12, A13.

Example 2.1 The heat equation as a sum of two terms

Take $L_1 = \partial_x^2$ and $L_2 = -\partial_t$. The ideal is $(\xi^2 - \tau,\ \xi^4) = (\tau^2,\ \xi^2 - \tau)$, of dimension $4 = 2 \cdot I_0(\xi^2, \tau)$. The solutions that split by terms are

$$1, \qquad x, \qquad x^2 + 2t, \qquad x^3 + 6tx.$$

The generic heat solution $e^{s^2 t + sx}$ with $s \neq 0$ has $L_1 L_2 u = -s^4 u \neq 0$, so it does not split.

2.2 The factor decomposition

Dim. A11 **C12-L3**

The finite quotient of two coprime factors [proved]

Let $f, g, h \in \mathcal{R} \setminus \{0\}$.

(a) If f, g are coprime, (f, g) contains a nonzero $r(\xi)$ and a nonzero $s(\tau)$; hence $\dim \mathcal{R}/(f, g) < \infty$.

(b) If f is coprime to g and to h, then

$$\dim \mathcal{R}/(f, gh) = \dim \mathcal{R}/(f, g) + \dim \mathcal{R}/(f, h).$$

(c) $(f, g) = \mathcal{R} \iff V(f) \cap V(g) = \emptyset$.

Stated in the paper as no. 6.2, p. 32

Idea. (a) is the Bézout identity of A1, Step 1. (b) is an exact sequence of finite-dimensional spaces. (c) uses that commuting operators on a nonzero finite-dimensional space have a common eigenvector.

Proof.

Step 1 (a) A finite spanning set

A1, Step 1 gives $r(\xi) \in (f, g)$ and symmetrically $s(\tau) \in (f, g)$, both nonzero. Then $\mathcal{R}/(f, g)$ is spanned by the monomials

$$\tau^a \xi^b, \qquad a < \deg s, \quad b < \deg r.$$

Step 2 (b) An exact sequence

The sequence

$$0 \to \mathcal{R}/(f, h) \xrightarrow{\cdot g} \mathcal{R}/(f, gh) \to \mathcal{R}/(f, g) \to 0$$

is well defined because $g\,(f, h) \subseteq (f, gh)$, and its middle is exact because the kernel of the projection is $(f, g)/(f, gh)$, the image of multiplication by g. Injectivity: if $ga = \alpha f + \beta gh$, then $g(a - \beta h) = \alpha f$; since $\gcd(f, g) = 1$, $f \mid a - \beta h$, so $a \in (f, h)$. Dimensions add.

Step 3 (c) A common eigenvector gives a common zero

($\Leftarrow$) If f, g shared a nonconstant factor π, then $\emptyset \neq V(\pi) \subseteq V(f) \cap V(g)$ (A1(F)). So disjoint zero sets force coprimality, and (a) applies. If $(f, g) \neq \mathcal{R}$, then $N = \mathcal{R}/(f, g)$ is nonzero and finite-dimensional. The commuting operators τ, ξ on N have a common eigenvector v, with eigenvalues (p, q). Then $fv = gv = 0$ gives $f(p, q) = g(p, q) = 0$, a common zero.

($\Rightarrow$) If (p, q) is a common zero, $(f, g) \subseteq \mathfrak{m}_{(p,q)} \neq \mathcal{R}$.

Why it works. coprime curves meet in finitely many points, and the quotient ring counts them with multiplicity.

■

Uses: A1.

Dim. A12 **C12-L1'**

Twisted Poincaré lemma; no obstruction for finite modules [proved]

Let Ω be open, connected, nonempty, with $H^1_{dR}(\Omega) = 0$.

(i) **(C12-L1′)** If $\varphi, \psi \in \mathcal{A}$ satisfy $D_x\varphi = D_t\psi$, there is $w \in \mathcal{A}$ with

$$D_t w = \varphi, \qquad D_x w = \psi.$$

Moreover $\{w \in \mathcal{A} : D_t w = D_x w = 0\} = \mathbb{C}\, e_z$.

(ii) **(C12-L2′)** For every $\mathcal{R}$-module M with $n = \dim M < \infty$:

$$\operatorname{Ext}^1_{\mathcal{R}}(M, \mathcal{A}) = 0, \qquad \dim \operatorname{Sol}_{\mathcal{A}}(M) = n,$$

and every solution is the restriction of a unique exponential-polynomial solution on $\mathbb{R}^2$.

Stated in the paper as no. 6.3, p. 34

Idea. Multiplying by e_z^{-1} turns the twisted derivatives into ordinary ones: (i) becomes "a closed 1-form is exact". For distributions, solve locally on squares and glue; the gluing error is a smooth closed form, exact by hypothesis. (ii) follows from (i) for the simplest modules and by induction along a composition series.

Proof.

Step 1 Removing the twist

Put $\tilde\varphi = e_z^{-1}\varphi$ and $\tilde\psi = e_z^{-1}\psi$. The condition becomes

$$\partial_x \tilde\varphi = \partial_t \tilde\psi,$$

and we look for $\tilde w$ with $\partial_t \tilde w = \tilde\varphi$, $\partial_x \tilde w = \tilde\psi$; then $w = e_z \tilde w$.

Step 2 Smooth case

The 1-form $\eta = \tilde\varphi\, \mathrm{d}t + \tilde\psi\, \mathrm{d}x$ is closed, hence exact because $H^1_{dR}(\Omega) = 0$.

Step 3 Distributions: local primitives glued by a partition of unity

Cover Ω by open squares U_j with closure in Ω, and take a locally finite partition of unity $\chi_j \in C^\infty_c(U_j)$. On each square a primitive w_j exists (integration in one variable, as in A2, Step 6). On an overlap $w_j - w_k$ has zero gradient, so it is a constant c_{jk}. Set $w = \sum_j \chi_j w_j$. On U_k,

$$w - w_k = \sum_j \chi_j\, c_{jk} =: h_k \quad \text{is smooth.}$$

So $a = \tilde\varphi - \partial_t w$ and $b = \tilde\psi - \partial_x w$ equal $-\partial_t h_k$, $-\partial_x h_k$ locally. They form a smooth closed 1-form, which is $d\theta$ for a smooth θ by hypothesis. Then $w + \theta$ is the required primitive.

Step 4 The kernel

If $\partial_t \tilde{w} = \partial_x \tilde{w} = 0$, then $\tilde{w}$ is locally constant, hence constant on the connected Ω.

Step 5 (ii) The simplest module

For $M = \mathcal{R}/\mathfrak{m}_z$ use the Koszul resolution

$$0 \to \mathcal{R} \xrightarrow{\beta} \mathcal{R}^2 \xrightarrow{\alpha} \mathcal{R} \to \mathcal{R}/\mathfrak{m}_z \to 0.$$

Applying $\mathrm{Hom}(\cdot, \mathcal{A})$, the solutions are $\ker(D_t, D_x) = \mathbb{C}e_z$ (Step 4) and $\mathrm{Ext}^1 = \ker \beta^* / \operatorname{im} \alpha^*$, which is exactly the compatibility problem of (i): it vanishes.

Step 6 (ii) Induction on the dimension

A finite-dimensional M has a composition series with factors $\mathcal{R}/\mathfrak{m}_z$. The long exact sequence of Ext and induction on n give $\mathrm{Ext}^1(M, \mathcal{A}) = 0$ and $\dim \mathrm{Sol}(M) = n$. On $\mathbb{R}^2$ the solutions are exponential-polynomials; restriction to Ω is injective (analytic functions vanishing on an open set vanish), and the dimensions agree.

Why it works. on a region without holes, closed forms are exact, so finite systems of first-order equations are always solvable.

■

Uses: A2.

Dim. A13 **C12-T1′**

Factor decomposition [proved]

Let Ω be open, connected, nonempty, with $H^1_{dR}(\Omega) = 0$, $\mathcal{A} \in \{\mathcal{D}'(\Omega), C^\infty(\Omega)\}$, $P = c \prod_{j=1}^{s} Q_j$ with pairwise coprime Q_j.

(i) $\ker_{\mathcal{A}} P = \ker_{\mathcal{A}} Q_1 + \cdots + \ker_{\mathcal{A}} Q_s$.

(ii) The kernel $\mathcal{N}$ of the sum map $\Sigma : \bigoplus_j \ker Q_j \to \ker P$ has dimension

$$\delta(P) = \sum_{i<j} \dim \mathcal{R}/(Q_i, Q_j) < \infty,$$

and its elements are tuples of exponential-polynomials. For $s = 2$, $\mathcal{N} = \{(w, -w) : w \in \ker Q_1 \cap \ker Q_2\}$.

(iii) The decomposition is unique $\iff$ $\delta(P) = 0$ $\iff$ $V(Q_i) \cap V(Q_j) = \emptyset$ for all $i \neq j$.

(iv) If u is smooth, every decomposition of u in $\mathcal{D}'$ consists of smooth functions.

Stated in the paper as no. 6.3, p. 33

Idea. For two coprime factors there is a short exact sequence of modules. Applying $\mathrm{Hom}(\cdot, \mathcal{A})$ gives a sequence of solution spaces that ends with Ext^1 of the finite module $\mathcal{R}/(f, g)$. That group vanishes on a region without holes (A12), so the sum map is onto, and its kernel is the solution space of the finite quotient.

Proof.

Step 1 Two factors: the module sequence

Let $f = Q_1$, $g = Q_2$. In a unique factorisation domain $(f) \cap (g) = (fg)$, and the sequence

$$0 \to \mathcal{R}/(fg) \xrightarrow{\iota} \mathcal{R}/(f) \oplus \mathcal{R}/(g) \xrightarrow{\pi} \mathcal{R}/(f,g) \to 0, \qquad \iota(1) = (1,1), \quad \pi(a,b) = a - b,$$

is exact.

Step 2 Two factors: the solution sequence

Since $\ker_{\mathcal{A}} f = \mathrm{Hom}(\mathcal{R}/(f), \mathcal{A})$, applying $\mathrm{Hom}(\cdot, \mathcal{A})$ gives the exact sequence

$$0 \to \mathrm{Sol}\big(\mathcal{R}/(f,g)\big) \xrightarrow{\pi^*} \ker f \oplus \ker g \xrightarrow{\iota^*} \ker fg \to \mathrm{Ext}^1_{\mathcal{R}}\big(\mathcal{R}/(f,g), \mathcal{A}\big),$$

with $\iota^*(u_1, u_2) = u_1 + u_2$ and $\pi^*(w) = (w, -w)$.

Step 3 Two factors: the obstruction vanishes

By A11(a), $\mathcal{R}/(f,g)$ is finite-dimensional. By A12(ii), $\mathrm{Ext}^1 = 0$ and $\dim \mathrm{Sol} = \dim \mathcal{R}/(f,g)$. So ι^* is onto, which is (i), and its kernel has dimension $\dim \mathcal{R}/(f,g)$, which is (ii) for $s = 2$.

Step 4 Any number of factors: induction

$P' = Q_2 \cdots Q_s$ is coprime to Q_1, so $\ker P = \ker Q_1 + \ker P'$, and by induction $\ker P' = \sum_{j \geq 2} \ker Q_j$. Write $\Sigma = \Sigma_2 \circ (\mathrm{id} \oplus \Sigma')$ with Σ' onto. For B onto, $\dim \ker(AB) = \dim \ker B + \dim \ker A$, and A11(b) gives $\dim \mathcal{R}/(Q_1, P') = \sum_{j \geq 2} \dim \mathcal{R}/(Q_1, Q_j)$. Hence

$$\delta(P) = \delta(P') + \sum_{j \geq 2} \dim \mathcal{R}/(Q_1, Q_j) = \sum_{i<j} \dim \mathcal{R}/(Q_i, Q_j).$$

Step 5 The ambiguity consists of exponential-polynomials

If $(w_j) \in \mathcal{N}$, then

$$w_j = -\sum_{i \neq j} w_i \ \in \ \ker Q_j \cap \ker \prod_{i \neq j} Q_i = \mathrm{Sol}\Big(\mathcal{R}/\big(Q_j, \textstyle\prod_{i \neq j} Q_i\big)\Big),$$

a finite-dimensional solution space, made of exponential-polynomials by A12(ii).

Step 6 (iii) and (iv)

$\delta(P) = 0 \iff$ every $\mathcal{R}/(Q_i, Q_j) = 0 \iff (Q_i, Q_j) = \mathcal{R} \iff V(Q_i) \cap V(Q_j) = \emptyset$ (A11(c)). For (iv), take a smooth decomposition from (i) in C^∞; any other decomposition in $\mathcal{D}'$ differs from it by an element of $\mathcal{N}$, which is smooth.

Why it works. the Chinese remainder sequence is exact as modules; its only possible failure on functions is an Ext^1 of a finite module, and that vanishes when Ω has no holes.

■

Uses: A11, A12.

Where the exact number $\delta(P)$ comes from

In terms of local intersection numbers,

$$\delta(P) = \sum_{i<j} m_i m_j \sum_z I_z(P_i, P_j).$$

For $s \geq 3$ the pairwise formula is exact, with no inclusion–exclusion correction; it was checked on three concurrent lines, three tangent parabolas, a triple point with multiplicity, and four factors.

Dim. A14 **C12-P10**

Holes break the decomposition [proved]

Let Ω contain a closed curve γ and a point $\alpha = (t_0, x_0) \notin \Omega$ with winding number $n = \mathrm{Ind}_\gamma(\alpha) \neq 0$. For

$$P = \tau^2 + \xi^2 = (\tau - i\xi)(\tau + i\xi) = Q_1 Q_2,$$

the function $u = \log\left((t-t_0)^2 + (x-x_0)^2\right)$ lies in $\ker_{C^\infty(\Omega)} P$ but not in $\ker_{\mathcal{D}'} Q_1 + \ker_{\mathcal{D}'} Q_2$.

Stated in the paper as no. 6.4, p. 34

Idea. A splitting $u = h + k$ would make dh equal to the form dw/w with $w = \overline{z - \alpha}$. An exact form integrates to zero around γ, but dw/w integrates to $-2\pi i\, n \neq 0$.

Proof.

Step 1 **The candidate component**

u is harmonic off α. Put $w = (t - t_0) - i(x - x_0)$. Then

$$F = Q_2 u = \frac{2}{w}$$

is smooth on Ω.

Step 2 **A splitting would make a form exact**

Suppose $u = h + k$ with $Q_1 h = 0$ and $Q_2 k = 0$. Then $Q_2 h = F$ and $Q_1 h = 0$, so

$$\partial_t h = \frac{F}{2}, \qquad \partial_x h = \frac{F}{2i}.$$

So the smooth form $\eta = \frac{F}{2}\,\mathrm{d}t + \frac{F}{2i}\,\mathrm{d}x = \dfrac{dw}{w}$ equals dh.

Step 3 The integral around γ

Mollify: $\eta_\epsilon = \eta * \rho_\epsilon = d(h * \rho_\epsilon)$ is exact near γ, so $\oint_\gamma \eta_\epsilon = 0$. Since η is smooth, $\eta_\epsilon \to \eta$ uniformly on γ, hence $\oint_\gamma dw/w = 0$. But

$$\oint_\gamma \frac{dw}{w} = \overline{\oint_\gamma \frac{dz}{z-\alpha}} = -2\pi i\, n \neq 0,$$

a contradiction.

Why it works. the logarithm has monodromy around the hole, and no single-valued component can carry it.

■

Uses: none.

For a fixed P the condition is subtler

On the punctured disc, $P = \tau\xi$ decomposes in C^∞ but not in $\mathcal{D}'$ (take $u = H(x)H(t)$, with H the Heaviside function). The hypothesis $H^1_{dR} = 0$ is what makes the theorem hold for *every* P; the exact condition for a fixed P is the cohomological criterion (OPF-T4) (Part II).

2.3 Regularity

Dim. A15 **C12-C1**

A derivative can be lost [false] (counterexample proved)

Let $\Omega = (0,1)^2$, $f = \tau - \xi$, $g = \tau - \xi - 1$. For each $m \in \mathbb{N}$ there is

$$u \in H^m(\Omega) \cap C^\infty(\Omega), \qquad f(\partial)g(\partial)u = 0,$$

whose unique decomposition $u = u_1 + u_2$ ($u_1 \in \ker f$, $u_2 \in \ker g$) has $u_1, u_2 \notin H^m(\Omega)$, while $u_2 = f(\partial)u \in H^{m-1}(\Omega)$. The loss is exactly one derivative.

Stated in the paper as no. 7.1, p. 37

Idea. Near the corner $(0,0)$ the characteristic chords of direction $(1,1)$ are short. On a short chord the shapes 1 and e^t of the two components are nearly equal, so a function that is small on the chord can have large components. A suitable profile along the chords makes u smooth while its components blow up by one derivative.

Proof.

Step 1 Characteristic coordinates

Use (t,s) with $s = x + t$ (determinant 1, so Sobolev norms are equivalent). Then

$$f = \partial_t\big|_s, \qquad g = \partial_t\big|_s - 1.$$

Near the corner the region is $\{0 < t < s < 1\}$: the chord at height s has length s.

Step 2 The function

Let $G_0(s) = 1/s$ and $G_m(s) = s^{m-1}\log s$ for $m \geq 1$, and

$$u = G(s)\,(e^t - 1).$$

Then $fu = G(s)e^t$ and $g(Ge^t) = 0$, so $fgu = 0$. Since $f - g = 1$, $(f, g) = \mathcal{R}$ and the decomposition is unique:

$$u_1 = -g(\partial)u = -G(s), \qquad u_2 = f(\partial)u = e^t\,G(s).$$

Step 3 u is in H^m

The derivatives are $\partial_t^k \partial_s^j u = G^{(j)}(s)\,\partial_t^k(e^t - 1)$. Integrating over $t \in (0, s)$ and using $|e^t - 1| \leq et$:

$$\int_0^s |\partial_t^k \partial_s^j u|^2 \,\mathrm{d}t \leq \begin{cases} e^2 s\,|G^{(j)}(s)|^2, & k \geq 1, \\ \frac{e^2}{3} s^3\,|G^{(j)}(s)|^2, & k = 0. \end{cases}$$

By induction $G^{(j)} = s^{m-1-j}(\alpha_j \log s + \beta_j)$ for $j \leq m-1$ and $G^{(m)} = (m-1)!/s$. So the integrals $\int_0^1 |G^{(j)}|^2 s\,\mathrm{d}s$ $(j \leq m-1)$ and $\int_0^1 |G^{(j)}|^2 s^3\,\mathrm{d}s$ $(j \leq m)$ are all finite.

Step 4 u_2 is not in H^m

$\partial_s^m u_2 = e^t G^{(m)}$ and

$$\int \left|e^t G^{(m)}\right|^2 \;\geq\; c \int_0^1 s^{-2}\, s\,\mathrm{d}s = \infty.$$

Hence also $u_1 = u - u_2 \notin H^m$, while $u_2 = f(\partial)u \in H^{m-1}$.

Why it works. the component shapes become collinear on short chords, and the profile G is chosen to sit exactly at the edge of integrability.

■

Uses: none.

Dim. A16 **C12-T3**

Chord criterion

[proved]

Let Ω be bounded, open and convex, $c \in \mathbb{R}$, $d_1 \neq d_2$, $f = \partial_t + c\partial_x - d_1$, $g = \partial_t + c\partial_x - d_2$. In coordinates (t, s) with $s = x - ct$, let $\ell(s)$ be the length of the chord at height s, and $\mathcal{H}_i = \ker_{L^2} f_i$. Equivalent:

(a) $\inf_s \ell(s) > 0$;

(b) every $u \in L^2$ with $fgu = 0$ has components in L^2 with $\|u_1\|^2 + \|u_2\|^2 \leq C\|u\|^2$;

(c) $\mathcal{H}_1 + \mathcal{H}_2$ is closed;

(d) the Friedrichs cosine $c_F(\mathcal{H}_1, \mathcal{H}_2) < 1$.

Stated in the paper as no. 7.2, p. 38

Idea. On each chord u is $F(s)e^{d_1 t} + G(s)e^{d_2 t}$. The squared L^2 norm on the chord is a Gram quadratic form in (F, G), uniformly positive exactly when the chords do not shrink. The other equivalences are Hilbert-space facts.

Proof.

Step 1 The components on each chord

Since $f - g = \kappa := d_2 - d_1 \neq 0$, the decomposition is unique:

$$u_2 = \kappa^{-1} f(\partial) u, \qquad u_1 = -\kappa^{-1} g(\partial) u,$$

and $u_1 = e^{d_1 t} F(s)$, $u_2 = e^{d_2 t} G(s)$ with $F, G \in L^2_{loc}$.

Step 2 (a)⇒(b): a uniform Gram bound

Let $\ell_0 = \inf \ell > 0$. The Gram matrix

$$\Gamma(\alpha, \beta) = \Big(\int_\alpha^\beta e^{d_a t}\, \overline{e^{d_b t}}\, \mathrm{d}t \Big)_{a,b=1,2}$$

is positive definite and continuous; on the compact set $\{-M \le \alpha,\ \alpha + \ell_0 \le \beta \le M\}$ its smallest eigenvalue is at least some $\gamma > 0$. By Fubini,

$$\int_{\alpha(s)}^{\beta(s)} |u|^2 \, \mathrm{d}t \ \ge\ \gamma \left(|F(s)|^2 + |G(s)|^2 \right),$$

while $\|e^{d_i t} F\|^2 \le B \int |F|^2$. Integrating in s gives (b).

Step 3 (b)⇒(c)⇒(d)

$\ker_{L^2} fg$ is closed, and by (b) it equals $\mathcal{H}_1 + \mathcal{H}_2$. If the sum is closed, the sum map $\mathcal{H}_1 \oplus \mathcal{H}_2 \to \mathcal{H}_1 + \mathcal{H}_2$ is a continuous bijection of Banach spaces (the intersection is $\{0\}$), so by the open mapping theorem $\|x\|^2 + \|y\|^2 \le C\|x + y\|^2$. If $c_F = 1$, unit vectors x_k, y_k with $\mathrm{Re}\langle x_k, y_k \rangle \to -1$ would give $\|x_k + y_k\| \to 0$, a contradiction.

Step 4 (d)⇒(a), by contraposition

Suppose $\inf \ell = 0$. Since ℓ is concave, $\ell \to 0$ at an end s_-. Choose disjoint intervals $E_k \to s_-$ with $\sup_{E_k} \ell^2 \le 4^{-k}$, constants $c_k = (\int_{E_k} \ell)^{-1/2}$, and $G = \sum_k c_k \mathbf{1}_{E_k}$. With

$$h_s(t) = e^{d_2 t} - e^{\lambda \alpha(s)} e^{d_1 t}, \qquad |h_s| \le C_1 \ell(s) \text{ on the chord},$$

the function $u = G(s) h_s(t) \in \ker fg$ has

$$\|u\|^2 \le C_1^2 \sum_k c_k^2 \int_{E_k} \ell^3 \le C_1^2 \sum_k 4^{-k} < \infty, \qquad \|e^{d_2 t} G\|^2 \ge e^{-2|d_2| M} \sum_k c_k^2 \int_{E_k} \ell = \infty.$$

So u is a limit of its truncations, which lie in $\mathcal{H}_1 + \mathcal{H}_2$, but u itself does not: the sum is not closed, and $c_F = 1$.

Why it works. the conditioning of the split on a chord is a 2×2 Gram matrix whose smallest eigenvalue is controlled by the chord length.

■

Uses: A13.

Dim. A17 **C12-P7**

Regularity criterion: closed and dense [proved]

Let $P = c\prod_{j=1}^{N} Q_j$ with pairwise coprime Q_j, $\mathcal{H}_j = \ker_{H^m(\Omega)} Q_j$, $W = \bigoplus_j \mathcal{H}_j$, $S(\omega) = \sum_j \omega_j$, and

$$\gamma_N = \inf \sigma\big(S^*S\big|_{(\ker S)^\perp}\big).$$

(a) Every $u \in \ker_{H^m} P$ has components in $H^m \iff \sum_j \mathcal{H}_j$ is closed and dense in $\ker_{H^m} P$.

(b) $\sum_j \mathcal{H}_j$ is closed $\iff \gamma_N > 0$; then the minimal-energy split satisfies, optimally,

$$\sum_j \|u_j\|_{H^m}^2 \le \gamma_N^{-1} \|u\|_{H^m}^2.$$

(c) For $N = 2$: $\gamma_2 = 1 - c_F$, the Friedrichs cosine after removing $\mathcal{I} = \mathcal{H}_1 \cap \mathcal{H}_2$.

(d) For $N \ge 3$ the pairwise cosines do not decide.

Stated in the paper as no. 7.3, p. 39

Idea. The components exist in H^m iff the sum map is onto $\ker_{H^m} P$. A bounded map has closed range iff it is bounded below on the orthogonal complement of its kernel, and γ_N is the square of that lower bound. For (d), glue infinitely many 3×3 examples whose conditioning tends to 0 while all pairwise angles stay large.

Proof.

Step 1 (a) Closed and dense means equal

$\ker_{H^m} P$ is closed in H^m (because $P(\partial) : H^m \to \mathcal{D}'$ is continuous) and contains the sum. A closed dense subspace is the whole space; the converse is clear.

Step 2 (b) Closed range iff bounded below

S^*S is self-adjoint, positive, and leaves $(\ker S)^\perp$ invariant. On it

$$\|S\omega\|^2 = \langle S^*S\omega, \omega\rangle \ge \gamma_N \|\omega\|^2.$$

If $\gamma_N > 0$, S is bounded below on $(\ker S)^\perp$, so its range is closed. If the range is closed, $S : (\ker S)^\perp \to \operatorname{im} S$ is a continuous bijection between Banach spaces, so its inverse is bounded, $\|S\omega\| \ge c\|\omega\|$, and $\gamma_N \ge c^2 > 0$. The minimal-energy preimage $S^\dagger u$ satisfies $\|S^\dagger u\|^2 \le \gamma_N^{-1}\|u\|^2$, with near-equality along spectral vectors near γ_N.

Step 3 (c) Two spaces

$(\ker S)^\perp = (A \oplus B) \oplus \{(z,z) : z \in \mathcal{I}\}$ with $A = \mathcal{H}_1 \ominus \mathcal{I}$, $B = \mathcal{H}_2 \ominus \mathcal{I}$. On the second summand $S^*S = 2$. On $A \oplus B$,

$$S^*S = I + K, \qquad K(a,b) = (\Pi_A b,\ \Pi_B a),$$

with $\|K\| = c_F$ and spectrum symmetric about 0. So the bottom of the spectrum is $1 - c_F$.

Step 4 (d) Three lines in $\mathbb{C}^3$

For $\varepsilon > 0$ let

$$v_1 = \frac{(1, 0, \varepsilon)}{\sqrt{1+\varepsilon^2}}, \qquad v_2 = \Big(-\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}, 0\Big), \qquad v_3 = \Big(-\tfrac{1}{2}, -\tfrac{\sqrt{3}}{2}, 0\Big),$$

and $\ell_i = \mathbb{C}v_i$. Here S^*S is the Gram matrix of (v_1, v_2, v_3), with

$$\det = \frac{3\varepsilon^2}{4(1+\varepsilon^2)} > 0, \qquad \gamma_3(\varepsilon) \sim \frac{\varepsilon^2}{3} \to 0,$$

while the pairwise cosines are at most $\frac{1}{2}$.

Step 5 (d) Gluing the blocks

Take the Hilbert direct sum over blocks with $\varepsilon_k \to 0$, and $\mathcal{L}_i = \bigoplus_k \ell_i^{(\varepsilon_k)}$. Pairwise cosines are suprema over blocks, so they stay $\le \frac{1}{2}$ and all pairwise sums are closed. But $S^*S = \bigoplus_k G_{\varepsilon_k}$ has bottom of spectrum 0 and trivial kernel, so the triple sum is not closed.

Why it works. regularity of the components is the bounded inverse theorem for the sum map, and γ_N is its smallest singular value squared.

■

Uses: A13.

Dim. A18 **OPF5B-T1**

Density on bounded convex regions

[proved mod. Hörmander 10.2.1, Stein VI.3]

Let $\Omega \subset \mathbb{R}^2$ be bounded, open, convex, $P = c\prod_{j=1}^{N} Q_j \neq 0$ as in A13, $m \ge 0$, and $\mathcal{H}_j = \ker_{H^m(\Omega)} Q_j$. Then $\sum_j \mathcal{H}_j$ is dense in $\ker_{H^m(\Omega)} P$. Consequently the components of every $u \in \ker_{H^m} P$ lie in H^m iff $\gamma_N > 0$.

Stated in the paper as no. 7.3, p. 40

Idea. Approximate u by solutions of P defined on a neighbourhood of $\bar{\Omega}$, then apply the decomposition A13 on a slightly larger convex set, where the components are smooth up to $\partial\Omega$. The approximants are built by shifting u slightly inward near each boundary point, gluing with a partition of unity, and correcting the gluing error with a fundamental solution. The correction gains exactly as much as the error loses, for every P.

Tools for this proof

Write $p(\zeta) = P(i\zeta)$, $p^{(\alpha)} = \partial_\zeta^\alpha p$, and

$$\tilde{P}(\zeta) = \Big(\sum_\alpha |p^{(\alpha)}(\zeta)|^2\Big)^{1/2}, \qquad k_m(\zeta) = (1+|\zeta|^2)^{m/2}.$$

$B_{2,k}$ is the space of f with $k\hat{f} \in L^2$; $B_{2,k_m} = H^m$. A weight k is *temperate* if $k(\zeta+\eta) \le C(1+|\eta|)^N k(\zeta)$.

- **[F1]** (Hörmander, ALPDO II, Thm. 10.2.1) [from cited source]: P has a fundamental solution E with $\varphi E \in B_{\infty,\tilde{P}}$ for every $\varphi \in C_c^\infty$.
- **[F2]** (Stein, Ch. VI §3, Thm. 5) [from cited source]: a bounded convex set has a bounded extension operator $H^m(\Omega) \to H^m(\mathbb{R}^2)$.
- **(L1)** $\tilde{P}$ and $1/\tilde{P}$ are temperate; $|p^{(\alpha)}| \le \tilde{P}$, so $P^{(\alpha)}(\partial)$ maps B_{2,k_m} to $B_{2,k}$ with $k = k_m/\tilde{P}$; multiplication by $\varphi \in C_c^\infty$ is bounded on $B_{2,k}$ for temperate k; and $\|(\varphi E) * g\|_{2,k\tilde{P}} \le \sup|\tilde{P}\,\widehat{\varphi E}| \cdot \|g\|_{2,k}$.

Proof.

Let $u \in \ker_{H^m(\Omega)} P$ and $\varepsilon \in (0,1]$.

Step 1 Fixed choices

Fix $o \in \Omega$ and $\rho > 0$ with $B(o,\rho) \subset \Omega$. Cover $\partial\Omega$ by balls $B(p_i,\rho/8)$, $i = 1,\dots,n$. Take $\Omega_0 \Subset \Omega$ containing the rest of $\bar{\Omega}$, and a partition of unity $\theta_0 \in C_c^\infty(\Omega_0)$, $\theta_i \in C_c^\infty(B(p_i,\rho/4))$, with $\sum_i \theta_i = 1$ on a neighbourhood V of $\bar{\Omega}$. Put

$$\tau_0 = 0, \qquad \tau_i = \varepsilon\,(o - p_i).$$

Step 2 Geometry: inward shifts stay inside

For $y_0 \in \bar{\Omega} \cap \bar{B}(p_i,\rho/2)$,

$$B\big(y_0 + \tau_i,\ \varepsilon\rho/2\big) \subset \Omega.$$

Indeed $y_0 + \tau_i + q = (1-\varepsilon)y_0 + \varepsilon(z + q/\varepsilon)$ with $z = o + y_0 - p_i \in \bar{B}(o,\rho/2)$, and $z + q/\varepsilon \in B(o,\rho) \subset \Omega$ for $|q| < \varepsilon\rho/2$; convexity does the rest. Consequently, on a neighbourhood V_ε of $\bar{\Omega}$, every point of $\operatorname{supp}\theta_i$ shifted by τ_i lies in Ω.

Step 3 Shifted copies solve P where they are used

Let $\tilde{u} = \operatorname{Ext} u \in H^m(\mathbb{R}^2)$ [F2], $\tilde{u}_i = \tilde{u}(\cdot + \tau_i)$, $w_i = \tilde{u}_i - \tilde{u}$. Translations commute with $P(\partial)$, so $P(\partial)\tilde{u}_i = 0$ wherever the shifted point lies in Ω, in particular on $V_\varepsilon \cap \operatorname{supp}\theta_i$.

Step 4 The shifts are small

By continuity of translations in H^m (in Fourier variables, dominated convergence with $|\tau_i| \le \varepsilon \operatorname{diam}\Omega$),

$$\|w_i\|_{H^m(\mathbb{R}^2)} \to 0 \qquad (\varepsilon \to 0).$$

Step 5 The gluing error

Let

$$g = \sum_{i=0}^{n} [P(\partial), \theta_i]\, w_i = \sum_i \sum_{\alpha \ne 0} \frac{\partial^\alpha \theta_i}{\alpha!}\, (\partial_z^\alpha P)(\partial)\, w_i.$$

It is supported in a fixed compact set, and by (L1)

$$\|g\|_{2,k} \le C \sum_i \|w_i\|_{H^m}, \qquad k = \frac{k_m}{\tilde{P}},$$

with C independent of ε.

Step 6 The correction gains what the error loses

Let $z = E * g$, so $P(\partial)z = g$. With $\varphi = 1$ near the relevant bounded set, $z = (\varphi E) * g$ near $\bar{\Omega}$, and by (L1) and [F1]

$$\|z\|_{H^m(\Omega)} \le C' \sup |\tilde{P}\,\widehat{\varphi E}| \cdot \|g\|_{2,k} \le C'' \sum_i \|w_i\|_{H^m}.$$

Step 7 A solution on a neighbourhood of $\bar{\Omega}$, close to u

Put $v = \sum_i \theta_i \tilde{u}_i - z$. On V_ε,

$$P(\partial)\Big(\sum_i \theta_i \tilde{u}_i\Big) = \sum_i [P, \theta_i]\tilde{u}_i = \sum_i [P, \theta_i] w_i + \Big[P, \sum_i \theta_i\Big]\tilde{u} = g,$$

because $\sum_i \theta_i \equiv 1$ there. So $P(\partial)v = 0$ on $V_\varepsilon \supset \bar{\Omega}$. On Ω, $v - u = \sum_i \theta_i w_i - z$, hence

$$\|v - u\|_{H^m(\Omega)} \le C''' \sum_i \|w_i\|_{H^m} \to 0.$$

Step 8 Decompose on a slightly larger convex set

For fixed ε, a metric neighbourhood W' of $\bar{\Omega}$ inside V_ε is convex. Mollify, $v_\eta = v * \rho_\eta$: it is smooth on W', solves P there, and $v_\eta \to v$ in $H^m(\Omega)$. By A13 on W', $v_\eta = \sum_j v_{\eta,j}$ with smooth $v_{\eta,j} \in \ker Q_j$. Since $\bar{\Omega} \subset W'$ is compact, the $v_{\eta,j}$ have bounded derivatives on Ω, so $v_{\eta,j}|_\Omega \in \mathcal{H}_j$. Choosing first ε, then η, makes $\|v_\eta - u\|_{H^m(\Omega)}$ as small as we like.

Why it works. the commutator error contains only derivatives of P, which are dominated by $\tilde{P}$, and Hörmander's fundamental solution gains exactly $\tilde{P}$; no assumption on the factors is needed.

■

Uses: A13; [F1] Hörmander, [F2] Stein.

CHAPTER 3

The interaction term

In this chapter

Starting from	an evolution factor $Q = \tau - a(\xi)$, $a = \sum_{n=1}^{N} a_n$ with $a_n \in \mathbb{C}[\xi]$ the terms; a bounded rectangle $\Omega = I \times J$; $u \in \ker Q$, base time t_0, $u_0 = u(t_0, \cdot)$, $s = t - t_0$.
Why	the interaction term compares the full solution with the sum of the single-term evolutions. We prove its exact form and cost in a class of exponential type, that it is not continuous in Sobolev norms and cannot be regularised, that the Dyson series converges, and that non-commuting terms with an infinite adjoint orbit admit no finite product formula.
Role	Dim. A19–A24. They prove Chapter 8 of the paper.

The class $\mathfrak{B}$ and two lemmas used throughout

For $\rho > 0$,

$$\|f\|_{B_\rho} = \sup_{j \geq 0} \rho^{-j} \|f^{(j)}\|_{L^2(J)}, \qquad B_\rho(J) = \{f : \|f\|_{B_\rho} < \infty\},$$

and $\mathfrak{B}_\rho(\Omega)$ likewise with $\sup_{i,j} \rho^{-i-j} \|\partial_t^i \partial_x^j u\|_{L^2}$; $\mathfrak{B} = \bigcup_\rho \mathfrak{B}_\rho$. For $A(\xi) = \sum_l \alpha_l \xi^l$ write $|A|(\rho) = \sum_l |\alpha_l| \rho^l$.

(P1-L1) B_ρ and $\mathfrak{B}_\rho$ are Banach spaces with $\|\partial_x\| \leq \rho$; $\mathfrak{B}_\rho \hookrightarrow H^m$ for every m; every exponential-polynomial lies in $\mathfrak{B}$; for $u \in \mathfrak{B}_\rho(\Omega)$ the trace $t \mapsto u(t, \cdot)$ is continuous into $B_{\rho'}(J)$ for every $\rho' > \rho$ [proved].

(P1-L2) On B_ρ, $\|A(\partial_x)\| \leq |A|(\rho)$, and $e^{sA(\partial_x)} = \sum_k \frac{s^k}{k!} A(\partial_x)^k$ converges in operator norm. These exponentials commute, satisfy $e^{s(A+A')(\partial_x)} = e^{sA(\partial_x)} e^{sA'(\partial_x)}$, and

$$e^{sA(\partial_x)} \big(x^l e^{\xi x}\big) = \partial_\xi^l \big(e^{sA(\xi)} e^{\xi x}\big) \qquad \text{[proved]}.$$

Both are proved in Part II (Dim. B for P1-L1, P1-L2).

3.1 Exact form and cost

Dim. A19 **P1-T2**

Closed form of the interaction [proved]

Let $u \in \mathfrak{B}_\rho(\Omega)$ with $Q(\partial)u = 0$, $\rho' > \rho$, $u_n(t) = e^{s\,a_n(\partial_x)}u_0$.

(i) u_n is the unique solution in $\mathfrak{B}$ of $(\partial_t - a_n(\partial_x))v = 0$ with $v(t_0) = u_0$, and $u(t) = e^{s\,a(\partial_x)}u_0$.

(ii) In $B_{\rho'}(J)$,

$$E_{\mathrm{int}}^{\mathrm{add}}(t) = \sum_{|S|\geq 2} \prod_{n\in S} \left(e^{s\,a_n(\partial_x)} - \mathrm{Id}\right) u_0, \qquad E_{\mathrm{int}}^{\mathrm{mult}} \equiv 0.$$

(iii) If $u_0 = \int e^{\zeta x}\,\mathrm{d}\mu_0(\zeta)$ with μ_0 compactly supported, then with $z_n = e^{s\,a_n(\zeta)} - 1$,

$$E_{\mathrm{int}}^{\mathrm{add}}(t) = \int e_{\geq 2}(z)\, e^{\zeta x}\,\mathrm{d}\mu_0(\zeta), \qquad e_{\geq 2}(z) = \prod_n (1+z_n) - 1 - \sum_n z_n.$$

Stated in the paper as no. 8.2, p. 43

Idea. Everything happens in a commutative Banach algebra of bounded operators. The single-term flows commute and their product is the full flow. Subtracting the "first-order" part $\mathrm{Id} + \sum_n (e^{s a_n} - \mathrm{Id})$ from the expanded product leaves exactly the subsets with at least two elements.

Proof.

Step 1 (i) Flows are unique in $\mathfrak{B}$

By P1-L1, u is a continuous curve in $B_{\rho'}$. A solution in $\mathfrak{B}$ of $\partial_t v = A(\partial_x)v$ with datum $v(t_0)$ is $e^{(t-t_0)A}v(t_0)$; uniqueness follows from the group property on B_{ρ_2} for large ρ_2. Apply this with $A = a$ and with $A = a_n$.

Step 2 (ii) The full flow is the product of the single flows

By P1-L2,

$$e^{s\,a(\partial_x)} = \prod_{n=1}^{N} e^{s\,a_n(\partial_x)},$$

so $E_{\mathrm{int}}^{\mathrm{mult}} = 0$.

Step 3 (ii) Expanding the product

Write $e^{s\,a_n} = \mathrm{Id} + \Delta_n$ with $\Delta_n = e^{s\,a_n} - \mathrm{Id}$. The Δ_n commute, so

$$\prod_{n=1}^{N} (\mathrm{Id} + \Delta_n) = \sum_{S\subseteq\{1,\dots,N\}} \prod_{n\in S} \Delta_n.$$

The empty set gives u_0; the singletons give $u_n(t) - u_0$. Subtracting them from $u(t)$ leaves the sum over $|S| \geq 2$.

Step 4 (iii) Mode by mode

$\zeta \mapsto e^{\zeta x} \in B_{\rho'}$ is continuous on the support of μ_0, so the integral is a Bochner integral and bounded operators pass under it. On one mode,

$$\prod_{n\in S}\left(e^{s\,a_n(\partial_x)} - \mathrm{Id}\right)e^{\zeta x} = \prod_{n\in S} z_n\; e^{\zeta x}.$$

Summing over $|S| \geq 2$ gives $e_{\geq 2}(z)$.

Why it works. commuting bounded operators obey the algebra of numbers, and the interaction is the part of a product of N factors that involves at least two of them.

■

Uses: P1-L1, P1-L2.

Example 3.1 A stationary state is pure interaction

Take $a_1 = -c\xi$, $a_2 = D\xi^2$ and the stationary mode $u = e^{cx/D}$, so $a(c/D) = 0$. With $\theta = c^2 s/D$:

$$z_1 = e^{-\theta} - 1, \qquad z_2 = e^{\theta} - 1, \qquad E_{\mathrm{int}}^{\mathrm{add}} = z_1 z_2\, u = -4\sinh^2\!\left(\frac{\theta}{2}\right) e^{cx/D}.$$

Dim. A20 **P1-T3**

Cost: $O(N)$ versus $\Theta(2^N)$ [proved]

Model: $u_0 = \sum_{k=1}^{M} p_k(x) e^{\zeta_k x}$ with $\deg p_k \leq L$; terms a_n of degree d_n; unit-cost arithmetic and exp.

(a) One value of $E_{\mathrm{int}}^{\mathrm{add}}(t)$ costs

$$O\Big(M\big[N(L+1)^2 + (L+1)\textstyle\sum_n (d_n+1)\big]\Big).$$

(b) All order aggregates $E^{(k)} = \sum_{|S|=k}\prod_{n\in S}\Delta_n u_0$, $2 \leq k \leq N$, cost $O\big(M[N^2(L+1)^2 + (L+1)\sum_n(d_n+1)]\big)$.

(c) The full table of cross terms has $M(2^N - N - 1)$ entries and costs $\Theta\big(M\,2^N(L+1)^2\big)$.

Stated in the paper as no. 8.3, p. 45

Idea. By A19(iii), each mode contributes $e_{\geq 2}(z)$ times the mode, and $e_{\geq 2}$ is a product minus a sum: it can be evaluated without ever expanding the product. Polynomial prefactors are handled by truncated power series in ζ.

Proof.

Step 1 The quantity to compute for one summand

By A19(iii) and P1-L2, the part of E_{int} coming from $x^l e^{\zeta_k x}$ is

$$\partial_\zeta^l \left[e_{\geq 2}\big(z(\zeta)\big)\, e^{\zeta x} \right]_{\zeta=\zeta_k}.$$

So we need the Taylor coefficients of $e_{\geq 2}(z(\zeta))$ at ζ_k up to order L. Put $\eta = \zeta - \zeta_k$ and work modulo η^{L+1}.

Step 2 (a) Four operations per summand

(1) $a_n(\zeta_k + \eta) \bmod \eta^{L+1}$ by repeated synthetic division: $O((L+1)(d_n+1))$.
(2) $\exp\big(s\, a_n\big) \bmod \eta^{L+1}$ by the recurrence $f' = g'f$: $O((L+1)^2)$.
(3) $\prod_n (1+z_n)$: $N-1$ truncated products, $O(N(L+1)^2)$.
(4) Subtract $1 + \sum_n z_n$ and read off the coefficients.

Summing over n and over the M summands gives (a).

Step 3 (b) All orders by dynamic programming

The aggregate of order k is the elementary symmetric function $e_k(z)$, the coefficient of λ^k in $\prod_n (1+\lambda z_n)$. Multiplying in one factor at a time, step n updates at most n coefficients, each a truncated product: $O(N^2(L+1)^2)$ per summand.

Step 4 (c) The full table

Each cross term c_S is a separate output, so the output alone has $M(2^N - N - 1)$ entries of size $L+1$. They are obtained by $c_S = c_{S\setminus\{\max S\}} \cdot z_{\max S}$, one truncated product per subset, which matches the lower bound up to the factor $L+1$.

Why it works. a product can be evaluated in linear time; only listing its expansion is exponential.

■

Uses: A19, P1-L2.

3.2 Instability

Dim. A21 **P1-P1(e)**

The interaction is not continuous in H^m [false] (continuity is false; proof of the counterexample)

Let $Q = \tau + c\xi - D\xi^2$ with $c, D > 0$, terms $a_1 = -c\xi$ and $a_2 = D\xi^2$, $I = (0, T)$, and any base time $t_0 \in \bar{I}$. For every $m \in \mathbb{N}$,

$$\sup \left\{ \frac{\|E_{\mathrm{int}}^{\mathrm{add}}\|_{L^2(\Omega)}}{\|u\|_{H^m(\Omega)}} : u \in \mathrm{EP},\ Qu = 0,\ u \neq 0 \right\} = \infty.$$

Hence $u \mapsto E_{\mathrm{int}}^{\mathrm{add}}$ has no continuous extension from $\mathfrak{B} \cap \ker Q$ to $\ker_{H^m} Q$.

Stated in the paper as no. 8.5, p. 46

Idea. On a bounded interval, real exponentials $e^{\zeta x}$ with large ζ are legitimate data. The diffusion flow alone amplifies them by $e^{D\zeta^2 s}$, while the true solution grows only like $e^{(D\zeta^2 - c\zeta)s}$. The interaction term must cancel the difference, and the ratio grows like $e^{c\zeta T}$, faster than any power of ζ.

Proof.

Step 1 The test solutions

For real $\zeta > c/D$ let

$$u_\zeta = e^{\alpha t + \zeta x}, \qquad \alpha = D\zeta^2 - c\zeta > 0.$$

Then $Qu_\zeta = 0$ and $\|u_\zeta\|_{H^m(\Omega)} \le C_m (1+\zeta)^m\, e^{\alpha T}\, \|e^{\zeta x}\|_{L^2(J)}$.

Step 2 The interaction of one mode

By A19(iii),

$$E_{\mathrm{int}}^{\mathrm{add}} = e^{\alpha t_0}\, z_1 z_2\, e^{\zeta x}, \qquad z_1 = e^{-c\zeta s} - 1, \quad z_2 = e^{D\zeta^2 s} - 1.$$

Step 3 Case $t_0 < T$

Put $T' = T - t_0 > 0$. For $s \in [T'/2, T']$ and ζ large, $|z_1| \ge \frac{1}{2}$ and $|z_2| \ge \frac{1}{2} e^{D\zeta^2 s}$. Integrating over $t \in [T - \zeta^{-2}, T]$:

$$\|E_{\mathrm{int}}^{\mathrm{add}}\|^2 \ge c\, \zeta^{-2}\, e^{2\alpha t_0 + 2D\zeta^2 T'}\, \|e^{\zeta x}\|^2.$$

Dividing by $\|u_\zeta\|_{H^m}^2$, and using $2D\zeta^2 T' - 2\alpha(T - t_0) = 2c\zeta T'$, the ratio of squares is at least

$$c_m\, \zeta^{-4m-2}\, e^{2c\zeta(T - t_0)} \longrightarrow \infty.$$

Step 4 Case $t_0 = T$

Now $s \in [-T, 0]$, $z_1 = e^{c\zeta|s|} - 1$, and $|z_2| = 1 - e^{-D\zeta^2|s|} \ge \frac{1}{2}$ for large ζ. For $t \in [0, T/2]$, $|z_1| \ge \frac{1}{2} e^{c\zeta T/2}$, and the ratio is at least $c_m'(1+\zeta)^{-4m} e^{c\zeta T} \to \infty$.

Step 5 No continuous extension

$u \mapsto E_{\text{int}}^{\text{add}}$ is linear and unbounded on a dense subspace, so it has no continuous extension.

Why it works. a bounded interval admits real exponential modes, and on them one term alone grows much faster than the full equation.

■

Uses: A19.

Where this does not happen

On $\mathbb{R}$ with L^2 data the modes are $e^{i\kappa x}$, every $|z_n| \le 2$ for dissipative or dispersive terms, and the interaction is bounded: $\|E_{\text{int}}\| \le 2\|u_0\|$ for advection–diffusion (P2-P5). The failure is a feature of *bounded* domains.

Dim. A22 **P2-T2**

No regularisation removes the interaction [proved]

(P2-T2) Let $\Omega = I \times J$, $a = \sum_{n=1}^{N} a_n$ with distinct a_n and $a - a_n \neq 0$ for all n, $D_n = \ker_{H^m}(\partial_t - a_n(\partial_x))$, $M = \overline{\sum_n D_n}$, and $u \in \ker_{H^m}(\partial_t - a(\partial_x))$. Put

$$g = \prod_{n=1}^{N}(a - a_n) \in \mathbb{C}[\xi], \qquad K = \deg g, \qquad Q = \prod_{n}(\tau - a_n), \qquad k = \deg Q.$$

(a) $Q(\partial)u = g(\partial_x)u$, and $\|E^{\perp}(u)\|_{H^m} \ge \|g(\partial_x)u\|_{H^{m-k}}/C_Q$, where $E^{\perp}(u)$ is the component of u orthogonal to M.

(b) $E^{\perp}(u) = 0 \iff g(\partial_x)u = 0$; this space has dimension K and consists of exponential-polynomials.

(c) Every split by terms, with any regulariser, leaves a residual of norm at least $\|E^{\perp}(u)\|$.

(P2-T4) If maps R_α with Lipschitz constants L_α converge pointwise to Φ on a set V, then $\mathrm{Lip}(\Phi) \le \liminf L_\alpha$. Every exact selection of components has Lipschitz constant at least $\gamma_N^{-1/2}$.

Stated in the paper as no. 8.5, p. 46

Idea. Every per-term piece is killed by $Q = \prod_n(\tau - a_n)$, so Q kills the closed span M. On a solution of the full equation, Q acts as the polynomial $g(\partial_x)$. So $\|Q(\partial)u\|$ measures the distance of u from M, and only the finitely many solutions killed by g have distance zero. P2-T4 is the fact that a pointwise limit of uniformly Lipschitz maps is Lipschitz with the same constant.

Proof.

Step 1 (a) Q acts as g on solutions

Divide $Q \in \mathbb{C}[\xi][\tau]$ by the monic $\tau - a(\xi)$:

$$Q = (\tau - a)H + r(\xi), \qquad r(\xi) = Q(a(\xi), \xi) = \prod_n (a - a_n)(\xi) = g(\xi).$$

Since constant-coefficient operators commute in $\mathcal{D}'$, $Q(\partial)u - g(\partial_x)u = H(\partial)(\partial_t - a(\partial_x))u = 0$.

Step 2 (a) The distance from M

Each $v \in D_n$ is killed by its own factor $\tau - a_n$ of Q, hence by Q; by continuity Q kills M. Write $u = P_M u + E^{\perp}(u)$. The map $Q(\partial) : H^m \to H^{m-k}$ is bounded with norm C_Q ((P2-L1)), so

$$\|g(\partial_x)u\|_{H^{m-k}} = \|Q(\partial)E^{\perp}(u)\|_{H^{m-k}} \le C_Q \, \|E^{\perp}(u)\|_{H^m}.$$

Step 3 (b) The exceptional space

If $E^{\perp}(u) = 0$, Step 2 gives $g(\partial_x)u = 0$, so u is killed by the ideal $\mathfrak{J} = (\tau - a,\ g)$. The map $\mathcal{R} \to \mathbb{C}[\xi]$, $\tau \mapsto a(\xi)$, is onto with kernel $(\tau - a)$, so

$$\mathcal{R}/\mathfrak{J} \cong \mathbb{C}[\xi]/(g), \qquad \dim = K.$$

By A12, the solutions form a K-dimensional space of exponential-polynomials. Conversely, such u split by factors of Q (A13) with components in $\mathfrak{B} \subset H^m$ ((P1-P2(c))), so $u \in M$.

Step 4 (c) Any regulariser

For any representation $S\omega$, $u - S\omega \in u + M$, so $\|u - S\omega\| \ge \operatorname{dist}(u, M) = \|E^{\perp}(u)\|$.

Step 5 (P2-T4) Lipschitz constants pass to the limit

For $v, w \in V$: $\|R_\alpha v - R_\alpha w\| \le L_\alpha \|v - w\|$, and the left side tends to $\|\Phi v - \Phi w\|$. Taking the supremum over $v \ne w$ gives $\operatorname{Lip}(\Phi) \le \liminf_\alpha L_\alpha$. For an exact selection σ ($S\sigma(u) = u$), $\sigma(u) - \sigma(0)$ is a preimage of u, so its norm is at least that of the minimal preimage, $\|S^{\dagger}u\|$; hence $\operatorname{Lip}(\sigma) \ge \|S^{\dagger}\| = \gamma_N^{-1/2}$ (A17).

Why it works. the per-term pieces all satisfy one extra equation, and on solutions of the full equation that extra equation is a nonzero polynomial in ∂_x.

■

Uses: A12, A13, A17, P2-L1, P1-P2.

Example 3.2 Advection–diffusion

For $a_1 = -c\xi$ and $a_2 = D\xi^2$: $g = (D\xi^2)(-c\xi) = -cD\xi^3$, $K = 3$, and $Q(\partial)u = -cD\, u_{xxx}$. The exceptional space is

$$\{E^{\perp} = 0\} = \operatorname{span}\{1,\ x - ct,\ (x - ct)^2 + 2Dt\},$$

with, for example,

$$(x-ct)^2+2Dt=\underbrace{\Big[(x-ct)^2-\tfrac{2D}{c}(x-ct)\Big]}_{\in D_1}+\underbrace{\tfrac{2D}{c}\,x}_{\in D_2}.$$

3.3 Perturbation and non-commuting terms

Dim. A23 **P3-T1**

Dyson–Phillips converges for every time [proved]

(a) On $X=L^2(\mathbb{R})$ let $A_0=a(\partial_x)$, $a(\xi)=\sum_{l\le q}\alpha_l\xi^l$, $\alpha_q\neq 0$, and $V=\sum_{k\le r}b_k(x)\partial_x^k$ with bounded b_k and $r<q$.

(a1) The condition (Par): $\operatorname{Re}a(i\kappa)\le\omega-\delta|\kappa|^q$ for some $\delta>0$, holds iff q is even and $\operatorname{Re}(\alpha_q i^q)<0$.

(a2) Under (Par), the Dyson–Phillips series of $e^{t(A_0+\varepsilon V)}$ converges in operator norm, uniformly on every $[0,T]$, is entire in ε, and its sum is the semigroup generated by $A_0+\varepsilon V$.

(b) The same holds on $L^2(0,L)$ for $A_0=D\partial_x^2$ with Dirichlet conditions and $V=b_1\partial_x+b_0$.

Stated in the paper as no. 8.6, p. 48

Idea. The principal term smooths: $\|\partial^k S_0(s)\|$ blows up only like $s^{-k/q}$. Since the perturbation has order $r<q$, the singularity $s^{-r/q}$ is integrable. Iterating gives Beta integrals, and the series is bounded by a Mittag-Leffler function, which is entire.

Proof.

Step 1 The abstract estimate (P3-L2)

Suppose $\|S_0(t)\|\le Me^{\omega t}$ and, for $s\in(0,T]$,

$$\|VS_0(s)\|\le K\,s^{-\beta}e^{\omega s},\qquad \beta\in[0,1),\quad \gamma=1-\beta.$$

Define $S^{(0)}=S_0$ and $S^{(n)}(t)x=\int_0^t S^{(n-1)}(t-s)\,VS_0(s)x\,\mathrm{d}s$. By induction,

$$\|S^{(n)}(t)\|\le Me^{\omega t}\,\frac{\big(K\Gamma(\gamma)\big)^n\,t^{n\gamma}}{\Gamma(1+n\gamma)}.$$

The induction step is the Beta integral

$$\int_0^t(t-s)^{(n-1)\gamma}s^{\gamma-1}\,\mathrm{d}s=t^{n\gamma}\,\frac{\Gamma\big((n-1)\gamma+1\big)\Gamma(\gamma)}{\Gamma(n\gamma+1)}.$$

Step 2 Convergence and analyticity

The bound sums to $Me^{\omega t}E_\gamma\big(K\Gamma(\gamma)t^\gamma\big)$, where $E_\gamma(w)=\sum_n w^n/\Gamma(1+n\gamma)$ is the Mittag-Leffler function, entire by Stirling. Replacing V by εV gives a power series in ε with infinite radius.

Step 3 The sum is the right semigroup

By Miyadera–Voigt (Engel–Nagel, Thm. III.3.14, Cor. III.3.15–16) [from cited source], $A_0+\varepsilon V$ generates a semigroup S satisfying the variation-of-constants formula. The series satisfies the same formula; the difference Δ satisfies $\Delta(t)=\int_0^t \Delta(t-s)\,\varepsilon V S_0(s)\,\mathrm{d}s$, and iterating with the Beta integral gives $\|\Delta\|\le C\,(|\varepsilon|K\Gamma(\gamma))^n t^{n\gamma}/\Gamma(1+n\gamma)\to 0$.

Step 4 (a1) The symbol criterion

Let $p(\kappa)=\operatorname{Re}a(i\kappa)=\operatorname{Re}(\alpha_q i^q)\kappa^q+p_{<q}(\kappa)$. If q is even and $\operatorname{Re}(\alpha_q i^q)=-2\delta<0$, then $p(\kappa)\le -2\delta\kappa^q+C(1+|\kappa|^{q-1})\le\omega-\delta|\kappa|^q$. Conversely, (Par) forces $p\to-\infty$ like $-\delta|\kappa|^q$ on both sides, so $\deg p=q$, the leading coefficient is negative, and q cannot be odd.

Step 5 (a2) The hypothesis of Step 1 holds with $\beta=r/q$

By Plancherel, for $s>0$,

$$\|\partial^k S_0(s)\|=\sup_\kappa|\kappa|^k e^{s\,p(\kappa)}\le e^{\omega s}\sup_{y\ge0}y^k e^{-s\delta y^q}=e^{\omega s}\Big(\frac{k}{q\delta e\,s}\Big)^{k/q}.$$

With $\|b_k\cdot\|\le\|b_k\|_\infty$ and $s^{-k/q}\le T^{(r-k)/q}s^{-r/q}$ on $(0,T]$, this gives the bound of Step 1 with $\beta=r/q<1$.

Step 6 (b) Dirichlet

In the sine basis $e_n=\sqrt{2/L}\sin(n\pi x/L)$,

$$\|\partial S_0(s)f\|^2=\sum_n|f_n|^2\mu_n^2e^{-2D\mu_n^2 s}\le\frac{\|f\|^2}{2eDs},$$

so Step 1 applies with $\beta=\frac12$.

Why it works. a smoothing principal term pays for a lower-order perturbation, and the payment is an integrable singularity in time.

■

Uses: Engel–Nagel (Miyadera–Voigt).

The hypotheses cannot be dropped (P3-C1)

With a perturbation of the same order ($A_0=\partial^2$, $V=-2\partial^2$) the terms have norm $(2n)^ne^{-n}/n!\sim 2^n/\sqrt{2\pi n}$ and the series diverges. With a dispersive principal term ($A_0=i\partial^2$, $V=i\partial$) the sum does not even generate a semigroup on L^2.

Dim. A24 **P1-T5**

No finite product of flows for an infinite adjoint orbit [proved]

Let $\mathcal{A}$ be a unital associative $\mathbb{C}$-algebra and $X, Y \in \mathcal{A}$ such that

$$X, \quad Y_0 = Y, \quad Y_1 = \operatorname{ad}_X Y, \quad Y_2 = \operatorname{ad}_X^2 Y, \ \dots$$

are linearly independent. Then there is no finite word $(Z_1, \dots, Z_d) \in \{X, Y\}^d$, no common polydisc $U = \{|t| < r_0\} \times \{|\varepsilon| < \varepsilon_0\}$, and no functions g_i holomorphic on U with $g_i(0, \varepsilon) = 0$ and $g_i(t, 0) \equiv 0$ whenever $Z_i = Y$, such that

$$e^{t(X+\varepsilon Y)} = \prod_{i=1}^{d} e^{g_i(t,\varepsilon) Z_i} \qquad \text{in } \mathcal{A}[[t]] \text{ for all } |\varepsilon| < \varepsilon_0.$$

Stated in the paper as no. 8.7, p. 48

Idea. Differentiate the identity in ε at $\varepsilon = 0$. The left side produces the moments $t^{k+1}/(k+1)$ of the independent elements Y_k. The right side, being a finite word, produces a sum of at most p exponential-type terms, whose Hankel matrices have rank at most p. But the Hankel matrices of the moments are Hilbert matrices, of full rank.

Proof.

Step 1 At zero coupling

For $\varepsilon = 0$ the Y-factors are the identity, so $e^{tX} = e^{G(t)X}$ with $G = \sum_{Z_i = X} g_i(t, 0)$. If $G \neq t$, let m be the order of $G - t$; the coefficient of t^m in $e^{(G-t)X} = 1$ gives $(G - t)_m X = 0$, impossible since $X \neq 0$. So $G = t$.

Step 2 Derivative in ε: the left side

Coefficients of t^m on both sides live in the finite-dimensional span of words of length $\leq m$, and depend holomorphically on ε, so we may differentiate. By the formal Duhamel formula,

$$\partial_\varepsilon e^{t(X+\varepsilon Y)}\Big|_{\varepsilon=0} = e^{tX} \int_0^t e^{-\sigma X} Y e^{\sigma X} \, \mathrm{d}\sigma = e^{tX} \sum_{k \geq 0} (-1)^k \frac{t^{k+1}}{(k+1)!} Y_k.$$

Step 3 Derivative in ε: the right side

At $\varepsilon = 0$ all factors are exponentials of X. The X-factors contribute $r(t) X e^{tX}$ with $r = \sum_{Z_i = X} \partial_\varepsilon g_i(t, 0)$. The Y-factors contribute $q_i(t)\, e^{G_i X} Y e^{(t - G_i)X}$, with $q_i = \partial_\varepsilon g_i(t, 0)$ and G_i the sum of the earlier X-times. Multiply on the left by e^{-tX} and use

$$e^{-\tau X}\, Y\, e^{\tau X} = \sum_k (-1)^k \frac{\tau^k}{k!} Y_k, \qquad \tau_i = t - G_i.$$

Step 4 Comparing coefficients

The result is

$$rX + \sum_k (-1)^k Y_k \sum_{i:Z_i=Y} q_i \frac{\tau_i^k}{k!} = \sum_k (-1)^k \frac{t^{k+1}}{(k+1)!} Y_k.$$

By independence, $r = 0$ and, for every $k \geq 0$,

$$\sum_{i:Z_i=Y} q_i(t)\, \tau_i(t)^k = \frac{t^{k+1}}{k+1}.$$

Step 5 A rank contradiction

Fix $t \in (0, r_0)$ and let $p = \#\{i : Z_i = Y\}$. The sequence $m_k = t^{k+1}/(k+1) = \sum_{i=1}^{p} q_i \tau_i^k$ has Hankel matrices

$$H_n = [m_{i+j}]_{i,j<n} = W^{\mathsf{T}} \operatorname{diag}(q)\, W, \qquad W_{ij} = \tau_i^j,$$

of rank at most p. On the other hand

$$H_n = t \operatorname{diag}(t^i) \operatorname{Hilb}_n \operatorname{diag}(t^j), \qquad \operatorname{Hilb}_n = \Big[\frac{1}{i+j+1}\Big],$$

and the Hilbert matrix is positive definite (the Gram matrix of monomials in $L^2(0,1)$), so H_n has rank n. For $n = p+1$ this is a contradiction.

Why it works. a finite product of flows can only produce finitely many "exponential moments", while the true first-order interaction needs infinitely many independent ones.

■

Uses: none.

> **Example 3.3 Where the hypothesis holds, and where it does not**
>
> In the Weyl algebra, $X = x^a\partial$ and $Y = \partial$ with $a \geq 3$ give $Y_k = c_k\, x^{k(a-1)}\partial$ with $c_k \neq 0$: distinct degrees, so the hypothesis holds. For ∂^2 and x^k with $k \geq 3$ it fails (each is locally nilpotent on the other), and the question is open (P1-Q3).

CHAPTER 4

Nonlinear equations

In this chapter

Starting from	scalar evolutions $u_t = \sum_n T_n(u)$ with T_n differential polynomials (Burgers: $T_1 = -uu_x$, $T_2 = \nu u_{xx}$); smooth periodic data u_0; for shocks, the Hopf–Lax and Cole–Hopf representations.
Why	to prove the nonlinear results of the paper: the t^2 jet of the interaction, its closed form before the shock, its sizes $\Theta(\nu)$ and $\nu\lvert\log\nu\rvert$ after the shock, the exact sequence of dictionary identification, and the two identification thresholds (Burgers N-shocks, KdV N-solitons).
Role	Dim. A25–A31. They prove Chapter 9 of the paper. Three proofs (A27, A28, A31) are long in the research files; here they are given as a complete chain of steps with every key formula, and marked *condensed*.

Notation for the whole chapter

For a differential polynomial P in the jet variables $y_0 = u$, $y_1 = u_x$, …, the *evolutionary derivation* is

$$D_P = \sum_k \partial_x^k(P)\,\frac{\partial}{\partial y_k},$$

so that $D_P(Q) = Q'[P]$, the derivative of Q in the direction P. For Burgers with datum u_0 on the circle $\mathbb{T}$ we use the inviscid flow $v = S_B(t)u_0$ (of $-uu_x$ alone), the heat flow $h = S_A(t)u_0$ (of νu_{xx} alone), and the Lagrangian map

$$X(t,a) = a + t\,u_0(a), \qquad J(t,a) = 1 + t\,u_0'(a), \qquad t^* = -\frac{1}{\min u_0'}.$$

[F1] (Lax–Oleinik = Kružkov entropy solution) and **[F2]** (Hopf's formula for viscous Burgers) are classical [from cited source]; [F2] is proved in the research files, [F1] is Kružkov 1970, Thm. 1 §3.

4.1 The interaction term

Dim. A25 **S4T-T1**

Jet of the nonlinear interaction [proved]

Let $\hat{S}_n(s) = \sum_k \frac{s^k}{k!} D_{T_n}^k$ be the formal flow of T_n, $\hat{u}(s)$ the Taylor series of u at t_0, and

$$\hat{E}^{\mathrm{add}}(s) = \hat{u}(s) - u_0 - \sum_n \big(\hat{S}_n(s)u_0 - u_0\big).$$

(i) The coefficient of $s^k/k!$ in $\hat{E}^{\mathrm{add}}$ is $\sum_{\mathbf{w}} D_{\mathbf{w}} u_0$, over the words $\mathbf{w} = (n_1, \dots, n_k)$ with at least two distinct letters, $D_{\mathbf{w}} = D_{T_{n_1}} \cdots D_{T_{n_k}}$. In particular

$$[s^2]\,\hat{E}^{\mathrm{add}} = \frac{1}{2} \sum_{m \neq n} T_n'[u_0]\big(T_m(u_0)\big).$$

(ii) For Burgers ($A = \nu u_{xx}$, $B = -uu_x$):

$$[s^2]\,\hat{E}^{\mathrm{add}} = -\nu\,\big(2u_0'u_0'' + u_0u_0'''\big),$$

and for $u_0 = \varepsilon \sin x$ this is $\frac{3}{2}\nu\varepsilon^2 \sin 2x$.

Stated in the paper as no. 9.1, p. 50

Idea. The full flow is the exponential of the *sum* of the derivations. A power of a sum expands over all words in the letters, while each single-term flow is the exponential of one letter. They agree on the constant words, so the difference is exactly the mixed words.

Proof.

Step 1 The Taylor series of the true solution

Since $u_t = F(u) = \sum_n T_n(u)$, by induction

$$\partial_t^k u(t_0) = D_F^k\, u_0, \qquad D_F = \sum_n D_{T_n},$$

using that $P \mapsto D_P$ is linear.

Step 2 Expanding the power of a sum

In the associative algebra of operators on differential polynomials,

$$\Big(\sum_n D_{T_n}\Big)^k = \sum_{\mathbf{w} \in \{1,\dots,N\}^k} D_{\mathbf{w}}.$$

The constant words $(n, \dots, n)$ contribute $D_{T_n}^k u_0$, which are exactly the Taylor coefficients of $\hat{S}_n(s)u_0$. What remains is the sum over the mixed words: this is (i).

Step 3 Order two

For $k = 2$ the mixed words are the ordered pairs (m, n) with $m \neq n$, and

$$D_{T_m} D_{T_n} u_0 = D_{T_m}(T_n) = T_n'[u_0]\big(T_m(u_0)\big).$$

Dividing by 2! gives the formula.

Step 4 Burgers

With $A'[w] = \nu w_{xx}$ and $B'[w] = -(u_0 w)_x$:

$$A'[B] + B'[A] = -\nu\,(u_0 u_0')'' - \nu\,(u_0 u_0'')' = -\nu\,\big(4u_0' u_0'' + 2u_0 u_0'''\big).$$

Half of it is $-\nu(2u_0' u_0'' + u_0 u_0''')$. For $u_0 = \varepsilon \sin x$,

$$2u_0' u_0'' + u_0 u_0''' = -3\varepsilon^2 \sin x \cos x = -\tfrac{3}{2}\varepsilon^2 \sin 2x.$$

Why it works. the single-term flows reproduce exactly the words with one letter, so the interaction is the sum of the mixed words, which start at order two.

■

Uses: none.

Why every coefficient is $O(\nu\varepsilon^2)$

A word with a letters A and b letters B, applied to $u_0 = \varepsilon w_0$, has degree $1 + b$ in ε and a factor ν^a. A mixed word has $a, b \geq 1$, so every coefficient is $O(\nu\varepsilon^2)$: there is no term of order ε.

Dim. A26 **S4T-T2**

Closed form before the shock

[proved]

Let u_0 be smooth and real, $0 < T < t^*$, $X = X(t,a)$, $J = J(t,a)$. For $t \in [0,T]$,

$$E_{\text{int}}^{\text{add}}(t,X) = \nu t\Big[\frac{u_0''(a)}{J^2} - u_0''(X)\Big] + R^{\text{add}},$$

with $\|R^{\text{add}}(t)\|_\infty \leq \nu^2 C(t)$. In the small-data regime $u_0 = \varepsilon w_0$, $\varepsilon T \max(-w_0') \leq \theta < 1$, the remainder is $\leq C(\theta, w_0, T)\,\nu^2\varepsilon^2 t^3$.

Stated in the paper as no. 9.2.1, p. 52

Idea. Expand the viscous solution around the inviscid one, $u = v + \nu\delta + O(\nu^2)$. The first-order coefficient δ solves a linear transport equation along the characteristics, and in Lagrangian coordinates it integrates exactly. Every remainder is controlled by a maximum principle along characteristics, with constants that do not depend on ν before the shock.

Proof.

Two tools are proved in Part II: the comparison principle (S4T-L0) (for $g_t + bg_x + cg - \nu g_{xx}$ with $|c| \leq K$, $\|g(t)\|_\infty \leq e^{Kt}(\|g(0)\|_\infty + \int_0^t \|Lg\|_\infty)$) and the pre-shock derivative bounds (S4T-L1) ($\|\partial_x^j w\|_\infty \leq K_j$ on $[0,T]$, independent of ν, for both the viscous and the inviscid solution).

Step 1 The difference $e = u - v$ is $O(\nu)$

Since $uu_x - vv_x = ue_x + v_x e$,

$$e_t + u\,e_x + v_x\,e = \nu\,u_{xx}, \qquad e(0) = 0.$$

The comparison principle with $c = v_x$ gives $\|e(t)\|_\infty \le \nu\,\varepsilon_0(t)$, $\varepsilon_0 = K_2 t e^{K_1 t}$. Differentiating once and twice gives in the same way $\|e_x\| \le \nu\varepsilon_1$ and $\|e_{xx}\| \le \nu\varepsilon_2$.

Step 2 The first-order coefficient in Lagrangian form

Let δ solve the linear transport equation

$$\delta_t + (v\,\delta)_x = v_{xx}, \qquad \delta(0) = 0.$$

Along $X(t,a)$ we have $J_t = v_x(t,X)\,J$ and $v_{xx}(t,X) = u_0''(a)/J^3$, so

$$\frac{d}{dt}(\delta J) = J\,v_{xx} = \frac{u_0''(a)}{J^2}, \qquad \delta(t,X) = \frac{t\,u_0''(a)}{J^2}.$$

Step 3 The second-order remainder

$w = e - \nu\delta$ solves $w_t + (vw)_x = \nu e_{xx} - e e_x$, $w(0) = 0$. The comparison principle gives $\|w(t)\|_\infty \le \nu^2\rho(t)$ with $\rho = te^{K_1 t}(\varepsilon_2 + \varepsilon_0\varepsilon_1)$.

Step 4 The heat flow

By Taylor's formula with integral remainder for $h = S_A(t)u_0$,

$$\big\|h(t) - u_0 - \nu t\,u_0''\big\|_\infty \le \frac{\nu^2 t^2}{2}\,\|u_0''''\|_\infty.$$

Step 5 Assembling

$E_{\text{int}}^{\text{add}} = (u - u_0) - (v - u_0) - (h - u_0) = e - (h - u_0)$, so

$$E_{\text{int}}^{\text{add}} = \nu\big(\delta - t\,u_0''\big) + w - \big(h - u_0 - \nu t u_0''\big),$$

and at (t,X) the first term is $\nu t\big[u_0''(a)/J^2 - u_0''(X)\big]$.

Step 6 The small-data refinement

The crude remainder contains $\frac{t^2}{2}\|u_0''''\|$, which is linear in ε. The remainder $\hat{w} = w - (h - u_0 - \nu t u_0'')$ solves

$$\hat{w}_t + (v\hat{w})_x - \nu\hat{w}_{xx} = \nu^2\partial_x^2\big(\delta - tu_0''\big) - (v\,r_h)_x - e\,e_x, \qquad \hat{w}(0) = 0,$$

and each source term is $O(\nu^2 t^2)$ and quadratic in ε: the first by a mixed Taylor expansion in (t,λ) for the datum λu_0 (it vanishes to second order in t and in λ), the other two directly. The comparison principle gives the bound $C\nu^2\varepsilon^2 t^3$.

Why it works. before the shock the characteristics do not cross, so the viscous correction is a linear transport along them that integrates in closed form.

■

Uses: S4T-L0, S4T-L1 (Part II), [F2].

4.2 After the shock *(condensed proofs)*

The Hopf–Lax picture

Put $U_0(y) = \int_0^y u_0$ and, for fixed t,

$$\Phi_x(y) = U_0(y) + \frac{(x-y)^2}{2t}, \qquad m(x) = \min_y \Phi_x(y).$$

[F1]: the inviscid solution is $v(t,x) = (x-a)/t$ with a a minimiser of Φ_x. **[F2]**: the viscous solution is

$$u(t,x) = \frac{x - \langle y \rangle_x}{t}, \qquad \langle f \rangle_x = \frac{\int f(y)\, e^{-\Phi_x(y)/2\nu}\, \mathrm{d}y}{\int e^{-\Phi_x(y)/2\nu}\, \mathrm{d}y}.$$

So $u - v = (a - \langle y \rangle_x)/t$: the whole difference is the distance between the minimiser and the Gibbs mean at temperature 2ν. Hypothesis (H_t): every minimiser has $J > 0$, and only finitely many $x_1, \dots, x_K$ (the shocks) have two minimisers $a_i^- < a_i^+$. Put

$$\rho_i = \tfrac{1}{2} \log \frac{J(a_i^-)}{J(a_i^+)}, \qquad \kappa(y) = \frac{t\, u_0''(y)}{J(y)^2}.$$

Dim. A27 OPA-T1

The interaction after the shock is $\Theta(\nu)$

[proved mod. [F1] Kružkov]

Under (H_t), for fixed $t > 0$ and $\nu \to 0$,

$$\| E_{\mathrm{int}}^{\mathrm{add}}(t) \|_{L^1(\mathbb{T})} = \nu \Lambda_{\mathrm{add}} + o(\nu),$$

$$\Lambda_{\mathrm{add}} = \int_0^{2\pi} t \left| \frac{u_0''(a(x))}{J(a(x))^2} - u_0''(x) \right| \mathrm{d}x + 4 \sum_{i=1}^{K} \log\left(2 \cosh \frac{\rho_i}{2}\right).$$

If $t > t^*$ then $K \geq 1$ and $\Lambda_{\mathrm{add}} \geq 4 \log 2 > 0$: the order is exactly ν. The estimate is not uniform in t ($\Lambda \to \infty$ as $t \downarrow t^*$).

Stated in the paper as no. 9.2.2, p. 52

Idea. Away from the shocks the Gibbs measure concentrates at a single minimiser, and Laplace's method gives $u - v = \nu\kappa + O(\nu^{3/2})$. Near a shock it concentrates on two minimisers, with weights given by a logistic function of $(x - x_i)/\nu$. The difference between that smooth step and the sharp jump integrates to $4\nu \log(2\cosh(\rho_i/2))$.

Proof.

Step 1 Uniform structure (compactness)

Under (H_t) there are $\delta, j_0, \eta, r_0 > 0$ such that: $J \geq j_0$ near every minimiser; away from the shock intervals $(x_i - r_0, x_i + r_0)$ the minimiser $a(x)$ is unique and $\Phi_x - m(x) \geq \eta$ at distance $\geq \delta$ from it; near x_i there are two C^3 branches $a_i^{\pm}(x)$, and

$$\Delta_i(x) = \Phi_x(a_i^+(x)) - \Phi_x(a_i^-(x)), \qquad \Delta_i(x_i) = 0, \qquad \Delta_i'(x) = -\frac{a_i^+ - a_i^-}{t} \leq -\frac{4\delta}{t}.$$

Step 2 Laplace with explicit remainders

If $\psi(0) = \psi'(0) = 0$ and $\lambda \leq \psi'' \leq \Lambda$ near 0, then with $\langle\cdot\rangle$ the mean for $e^{-\psi/\varepsilon}$,

$$\langle z \rangle = -\varepsilon \frac{\psi'''(0)}{2\psi''(0)^2} + O(\varepsilon^{3/2}), \qquad \langle |z|^k \rangle \leq m_k \varepsilon^{k/2}.$$

(Integrate by parts twice against $e^{-\psi/\varepsilon}$.) The tails where $\Phi_x - m \geq \eta$ contribute $O(e^{-\eta/4\nu})$.

Step 3 Smooth region

With $\varepsilon = 2\nu$, $\psi(z) = \Phi_x(a+z) - m$, $\psi'' = J/t$, $\psi''' = u_0'''$:

$$u - v = \frac{a - \langle y \rangle_x}{t} = \nu\,\kappa(a(x)) + O(\nu^{3/2})$$

uniformly away from the shocks.

Step 4 Shock layer: logistic weights

Near x_i, the Gibbs mass splits between the two wells with weight $\sigma = L(\ell(x))$ on the right one, $L(z) = 1/(1+e^{-z})$, and

$$\ell(x) = -\frac{\Delta_i(x)}{2\nu} + \rho_i(x) + O(\nu^{1/2}),$$

the term ρ_i coming from the ratio of the two Gaussian widths $\sqrt{J(a^{\mp})}$. With $H = \mathbf{1}_{x>x_i}$,

$$u - v = \frac{a^+ - a^-}{t}(H - \sigma) + \nu\kappa(a(x)) + R, \qquad |R| \leq C\nu|H - \sigma| + C\nu^{3/2}.$$

Since $|H - \sigma| \leq Ce^{-2\delta|x-x_i|/t\nu}$, the two pieces overlap only on a set of size $O(\nu|\log\nu|)$ and can be integrated separately.

Step 5 The layer integral

Put $x = x_i + \nu w$. Then $\ell \to D_i w/2 + \rho_i$ with $D_i = (a_i^+ - a_i^-)/t$, and by dominated convergence

$$\int \frac{a^+ - a^-}{t}|H - \sigma|\,\mathrm{d}x = 2\nu\Big[\int_{-\infty}^{0} L(z+\rho_i)\,\mathrm{d}z + \int_0^{\infty}\big(1 - L(z+\rho_i)\big)\,\mathrm{d}z\Big] + o(\nu).$$

Using $\int_{-\infty}^{c} L = \log(1+e^c)$ and $\int_c^{\infty}(1-L) = \log(1+e^{-c})$, this is

$$2\nu\log\big((1+e^{\rho_i})(1+e^{-\rho_i})\big) = 4\nu\log\Big(2\cosh\frac{\rho_i}{2}\Big).$$

Step 6 From $u - v$ to E_{int}

Summing Steps 3–5 gives $\|u - v\|_{L^1} = \nu\Lambda_{uv} + o(\nu)$. Since $E_{\text{int}}^{\text{add}} = (u - v) - (h - u_0)$ and $h - u_0 = \nu t u_0'' + O(\nu^2)$ uniformly (A26, Step 4), the same computation with $\kappa(a(x)) - t u_0''(x)$ in place of $\kappa(a(x))$ gives Λ_{add}.

Why it works. at small viscosity the solution is a Gibbs average over the minimisers of the Hopf–Lax functional; a shock is a tie between two minimisers, smoothed into a logistic step of width ν.

■

Uses: [F1], [F2]; research file `work/OP-ANA`, §2 (Lemmas 1–3).

Dim. A28 **OPF5A-T2**

Uniform rate $\nu|\log\nu|$ across the shock time [proved mod. [F1] Kružkov]

Assume the shock is born at a single generic cusp (condition (G*)): u_0' has a unique global minimum at a^*, nondegenerate, so that $\kappa_3 = u_0'''(a^*) > 0$. Then there are τ_0, C, ν_0 such that, uniformly for $t \in (0, t^* + \tau_0]$ and $\nu \le \nu_0$,

$$\left| \|S(t)u_0 - S_B(t)u_0\|_{L^1} - 2\nu \log \frac{1}{\max(|t - t^*|, \sqrt{\nu})} \right| \le C\nu.$$

The same holds for $\|E_{\text{int}}^{\text{add}}\|_{L^1}$. In particular $\sup_t \|E_{\text{int}}^{\text{add}}\|_{L^1} = \nu|\log\nu| + O(\nu)$, with coefficient exactly 1. At $t = t^*$ the next term is $\Gamma\nu$, with $\Gamma = \Gamma_{\text{out}} + \log\frac{\kappa_3}{4} + P$ and the universal constant $P = 1.188654\ldots$ (OPF5A-T1).

Stated in the paper as no. 9.2.2, p. 52

Idea. Near the cusp the Hopf–Lax functional is quartic instead of quadratic. The Laplace expansion of A27 still holds outside a shrinking window, and there it integrates to a logarithm of J, which vanishes like α^2 at the cusp. Inside the window a rescaling turns the problem into a universal tilted double well, whose contribution is $O(\nu)$.

Proof.

Step 1 Exact expansion of the Gibbs mean

Integration by parts on $\mathbb{R}$ gives $\langle\psi'\rangle = 0$ and $\langle z\psi'\rangle = 2\nu$. With $\psi'(z) = pz + \frac{q}{2}z^2 + \rho_3(z)$, $p = J_t(a)/t$, $q = u_0''(a)$, $|\rho_3| \le C|z|^3$:

$$\langle z\rangle = -\frac{\nu q}{p^2} + E_1, \qquad E_1 = \frac{q^2}{4p^2}\langle z^3\rangle + \frac{q}{2p^2}\langle z\rho_3\rangle - \frac{\langle\rho_3\rangle}{p}.$$

Step 2 Geometry of the cusp

Write $a = a^* + \alpha$ and $\eta = 1 - t/t^*$. For $t \le t^*$,

$$\eta + c\alpha^2 \le J_t \le \eta + C\alpha^2, \qquad c(z^4 + A^2z^2) \le \psi(z) \le C(z^4 + A^2z^2),$$

with $A^2 = \eta + \alpha^2$, so the Gibbs width is $w = \min(\nu^{1/4}, \sqrt{\nu}/A)$ and $\langle|z|^k\rangle \le C_k w^k$.

Step 3 Outer region $A \geq \nu^{1/4}$

There $p \geq cA^2$, $|q| \leq CA$, $|E_1| \leq C\nu^{3/2}/A^5$, and with $\mathrm{d}x = J_t\,\mathrm{d}a$,

$$\int_{\mathcal{O}} |E_1|\,\mathrm{d}x \leq C\nu^{3/2} \int_{A\geq\nu^{1/4}} A^{-3}\,\mathrm{d}\alpha \leq C\nu.$$

The main part is $\nu \int t|u_0''|/J_t\,\mathrm{d}a$. Near a^*, $tu_0'' = \partial_a J_t$, so the integral is a logarithm of J_t between the inner edge ($J_t \asymp \max(\sqrt{\nu}, \eta)$) and $J_t \asymp 1$:

$$\nu \int_{\mathcal{O}} |\kappa_t|\,\mathrm{d}x = 2\nu \log \frac{1}{\max(\sqrt{\nu}, \eta)} + O(\nu).$$

Step 4 Inner region $A < \nu^{1/4}$

Its measure in x is $\leq C\nu^{3/4}$ and there $|u - v| \leq \langle |z| \rangle / t \leq C\nu^{1/4}$: contribution $O(\nu)$. Together with Step 3 this proves the formula for $t \leq t^*$ (the result (OPF5A-T1)).

Step 5 After the cusp: the normal form

For $t = t^* + \tau$ the outer region $|\alpha| \geq K'\sqrt{\tau}$ works as before and gives $2\nu \log(1/\tau) + O(\nu)$, which is also the lower bound. In the inner region rescale

$$a = a^* + \sqrt{\tau}A, \qquad x = X_t(a^*) - t\tau^{3/2}\xi, \qquad \Psi_\tau(\xi, A) = \tau^{-2}\big[\Phi_x(a^* + \sqrt{\tau}A) - \Phi_x(a^*)\big].$$

Then $\Psi_\tau \to \Psi_0 = \xi A + \frac{\kappa_3}{24}A^4 - \frac{A^2}{2t^{*2}}$ in C^4 on compacts: a tilted double well whose global minimisers are nondegenerate. The effective viscosity is $\hat{\nu} = \nu/\tau^2$, and the A27 estimate applied uniformly to this compact family ((OPF5A-L3′)) gives

$$\int_{\text{inner}} |u - v|\,\mathrm{d}x = \tau^2 \int |A_m - \langle A \rangle_\xi|\,\mathrm{d}\xi \leq \tau^2 C\hat{\nu} = C\nu.$$

Step 6 From $u - v$ to E_{int}

As in A27, Step 6, with $\|h - u_0\|_{L^1} \leq 2\pi\nu t\|u_0''\|_\infty + O(\nu^2)$.

Why it works. at the cusp the Lagrangian Jacobian vanishes quadratically, so integrating $|\partial_a J|/J$ produces exactly one logarithm of $1/\nu$, with coefficient one.

■

Uses: A27, [F1], [F2]; research file `work/OP-F5A` (L2, L3′, Lemma U).

4.3 Identification in a dictionary

Dim. A29 **S4ID-T1**

Dictionary criterion [proved]

Let u be smooth on an open Ω and solve $F(u) = 0$ with $F = y_t - N(y, y_x, \dots)$. Let $[F]$ be the differential ideal generated by F, V a linear space of differential polynomials, $I(u) = \{G : G(u) = 0\}$, ψ the normal-form map that eliminates y_t and its derivatives using F (so $\ker \psi = [F]$ and $G - \psi(G) \in [F]$), and $\mathcal{J}_u$ the relations satisfied by u among the remaining jet variables.

(a) $I(u) \cap V = (\psi|_V)^{-1}(\mathcal{J}_u)$.

(b) The sequence

$$0 \to [F] \cap V \to I(u) \cap V \xrightarrow{\psi} \mathcal{J}_u \cap \psi(V) \to 0$$

is exact; for finite-dimensional V, $\dim(I(u) \cap V) = \dim([F] \cap V) + \dim(\mathcal{J}_u \cap \psi(V))$.

(c) $I(u) \cap V = [F] \cap V \iff \mathcal{J}_u \cap \psi(V) = 0$.

Stated in the paper as no. 9.3, p. 54

Idea. On a solution, every differential polynomial equals its normal form, because their difference lies in $[F]$. So a relation of u is a polynomial whose normal form is a relation among the free jet variables. The part that the equation explains ($[F]$) is the kernel; the part that only u explains is the image.

Proof.

Step 1 (a) Reduction to normal forms

For $G \in V$, $G - \psi(G) \in [F]$ vanishes on the solution u, so

$$G(u) = \psi(G)(u).$$

Hence $G \in I(u)$ iff $\psi(G)$ vanishes on u, that is, $\psi(G) \in \mathcal{J}_u$.

Step 2 (b) Exactness

The kernel of ψ on $I(u) \cap V$ is $\ker \psi \cap V = [F] \cap V$. By (a) the image lies in $\mathcal{J}_u \cap \psi(V)$. Conversely, every element of $\mathcal{J}_u \cap \psi(V)$ is $\psi(G)$ with $G \in V$, and $G \in I(u)$ by (a). The dimension formula follows from exactness.

Step 3 (c)

Read off from (b): the middle term equals the left one iff the right one is zero.

Why it works. the normal form splits every relation of u into a part forced by the equation and a part forced only by the particular solution.

■

Uses: none.

The literal statement is false

In a complete dictionary $\mathcal{D}_{d,k}$, $[F] \cap V$ already contains $\partial_x F$ and other consequences, so "$I(u) \cap V = \mathbb{C}F$" fails for every u, except in the special case $(d,k) = (2,n)$ (S4ID-P1). The correct statement is (c).

Dim. A30 **S4ID-T2**

Burgers with N shocks

[proved]

Let $F_B = y_t + y_0 y_1 - \nu y_2$, distinct nonzero $k_1, \dots, k_N$, $E_i = \exp(k_i x + \nu k_i^2 t + c_i)$, $\varphi = 1 + \sum_i E_i$ (nonvanishing on Ω), non-resonance (NR): $\sum_i m_i(\nu k_i^2, k_i) \neq 0$ for $m \in \mathbb{Z}^N \setminus \{0\}$, and $u = -2\nu\varphi_x/\varphi$. Let e_m be defined by $z\prod_i(z-k_i) = \sum_{m=1}^{N+1} e_m z^m$, Y_m the complete Bell polynomials, and

$$R_N = \sum_{m=1}^{N+1} e_m\, Y_m\Big(-\frac{y_0}{2\nu}, \dots, -\frac{y_{m-1}}{2\nu}\Big).$$

(a) $\mathcal{J}_u \cap \mathbb{C}[y_0, \dots, y_{N-1}] = 0$.

(b) $\mathcal{J}_u \cap \mathbb{C}[y_0, \dots, y_N] = R_N \cdot \mathbb{C}[y_0, \dots, y_N]$, with $\deg R_N = N+1$.

(c) In the evolutionary dictionary of order $K = N$ and degree d, u identifies Burgers iff $d \leq N$; with the conjecture S4ID-K1, now proved as (OP-T7), the same holds for every $K \geq N$.

Stated in the paper as no. 9.3, p. 54

Idea. By Cole–Hopf, the jet of $w = u/(-2\nu)$ is the sequence of cumulants of a measure with N atoms at the k_i (plus the atom 0). N atoms leave N free parameters, so the first N jet variables are independent, and the moments of $N+1$ atoms satisfy exactly one linear relation, which in cumulants is R_N.

Proof.

Step 1 Reduction to formal variables

By (NR) the monomials E^m have distinct exponents, so they are linearly independent (A2). Hence $X_i \mapsto E_i$ embeds $\mathbb{C}(X)$ into the meromorphic functions on Ω, intertwining $D(X_i) = k_i X_i$ with ∂_x. So $p \in \mathcal{J}_u$ iff p vanishes on the formal jet of $u = -2\nu\, D\varphi/\varphi$, $\varphi = 1 + \sum_i X_i$.

Step 2 Moments

Put $p_i = X_i/\varphi$ and $r_j = D^j\varphi/\varphi = \sum_i k_i^j p_i$. The p_i generate $\mathbb{C}(X)$, so they are algebraically independent. From $Dr_j = r_{j+1} - w\, r_j$ with $w = D\varphi/\varphi$, induction gives

$$r_j = Y_j(w, Dw, \dots), \qquad Y_j = w_{j-1} + (\text{polynomial in } w_0, \dots, w_{j-2}).$$

Step 3 (a) Independence

By the triangular form, $\mathbb{C}[w,\dots,D^{N-1}w] = \mathbb{C}[r_1,\dots,r_N]$. And $(r_1,\dots,r_N)^{\mathsf{T}} = V\,(p_1,\dots,p_N)^{\mathsf{T}}$ with the Vandermonde matrix $V = (k_i^j)$, invertible. So $r_1,\dots,r_N$, hence $w,\dots,D^{N-1}w$, are algebraically independent.

Step 4 (b) The relation

The moments satisfy

$$\sum_{m=1}^{N+1} e_m\, r_m = \sum_i p_i \sum_m e_m k_i^m = \sum_i p_i\, k_i \prod_l (k_i - k_l) = 0.$$

So $R_N \in \mathcal{J}_u$. Only Y_{N+1} contains w_N, with coefficient 1, and $e_{N+1} = 1$: $R_N = -y_N/(2\nu) + \beta(y_0,\dots,y_{N-1})$, of degree $N+1$. For $p \in \mathcal{J}_u \cap \mathbb{C}[y_0,\dots,y_N]$, divide by R_N as a polynomial of degree one in y_N: the remainder lies in $\mathbb{C}[y_0,\dots,y_{N-1}] \cap \mathcal{J}_u = 0$ by (a).

Step 5 (c) The criterion

By A29(c), identification in $V = \mathbb{C}y_t \oplus A_{\le d,\le N}$ holds iff $\mathcal{J}_u \cap A_{\le d,\le N} = 0$. By (b) this is the set of multiples of R_N of degree $\le d$, which is zero iff $d < N+1$.

Why it works. Cole–Hopf turns the N-shock solution into a mixture of N exponentials, and N exponentials satisfy exactly one moment relation of order N.

■

Uses: A2, A29; (OP-T7) for $K > N$ (Part II).

Dim. A31 **OPF5C-T2**

KdV N-solitons: degree $(N+2)2^{N-1}$ [proved mod. Novikov 1974]

For the N-soliton of KdV with generic distinct wave numbers ($k \in U_0$), the relations among the jet variables $y_0,\dots,y_N$ form a principal ideal (Π_N). Normalising Π_N monic in y_N:

$$\deg_{y_N} \Pi_N = 2^N, \qquad \deg \Pi_N = (N+2)\,2^{N-1}.$$

($N = 1,2,3$: degrees 3, 8, 20.) Consequently the evolutionary dictionary of order N and degree d identifies KdV iff $d < (N+2)2^{N-1}$.

Stated in the paper as no. 9.3, p. 54

Idea. The jets of N-solitons lie on a variety cut out by Novikov's stationary equations, written in Mumford coordinates. A weighted Bézout argument shows that this variety is a free module of rank 2^N over the first N jet variables: y_N takes 2^N values over each point, one for each choice of signs $k_i \mapsto \pm k_i$. The minimal polynomial of y_N is therefore of degree 2^N in y_N, and its total degree is read from the weights.

Proof.

Step 1 Principal ideal

As in A30, Step 1, reduce to formal variables. The first N jet variables are independent (Vandermonde in $2k_i^2 k_i^j$), and $y_0, \dots, y_N$ are $N+1$ elements of a field of transcendence degree N. So the ideal of relations in $\mathbb{C}[y_0, \dots, y_N]$ is a nonzero prime of height one, hence principal, (Π_N) (S4ID-T3(b)).

Step 2 Novikov's variety (Nov′)

In Mumford coordinates (U, V), with $f_k(\lambda) = \lambda \prod_i (\lambda + \kappa_i)^2$, the jets of the N-soliton satisfy

$$U \mid V^2 + f_k$$

(Novikov 1974) [from cited source]. The change of variables from jets $(y_0, \dots, y_{2N-1})$ to Mumford coordinates is a weighted triangular automorphism with weights $\operatorname{wt} y_j = j + 2$ (OPF5C-L1).

Step 3 A free module of rank 2^N

The N equations of the variety are weighted homogeneous of weights $4N + 2 - 2m$. Their only common zero with $k = 0$ and $y_0 = \cdots = y_{N-1} = 0$ is the origin (OPF5C-L3). By the weighted Bézout theorem (Hironaka's criterion) the coordinate ring is free over $\mathbb{C}[k, y_0, \dots, y_{N-1}]$ of rank

$$\frac{\prod_{m=0}^{N-1} (4N + 2 - 2m)}{\prod_{j=N}^{2N-1} (j+2)} = \frac{\prod_{i=N+2}^{2N+1} 2i}{\prod_{i=N+2}^{2N+1} i} = 2^N.$$

Step 4 The 2^N branches are distinct

Replacing k_i by $\pm k_i$ does not change f_k, so the 2^N sign choices give 2^N local sections of the variety near the zero potential. Their values of y_N have distinct linear parts (the multisets $\{\pm k_i\}$ differ), so the characteristic polynomial χ_k of multiplication by y_N is squarefree (OPF5C-L5). Hence $\Pi_N = \chi_k$, of degree 2^N in y_N.

Step 5 Total degree from the weights

Multiplication by y_N has weight $N + 2$, so the coefficient of $z^{2^N - i}$ in the universal χ has weight $i(N+2)$, and every monomial has weight at most $W = (N+2)2^N$. Since $\operatorname{wt} y_j \geq 2$, the degree is at most $W/2$. The monomial $y_0^{W/2}$ appears with a nonzero constant coefficient, because no point of the rational fibre has jet $(c, 0, \dots, 0)$ with $c \neq 0$ (OPF5C-L4). So $\deg \Pi_N = W/2 = (N+2)2^{N-1}$.

Step 6 The criterion

As in A30, Step 5, using A29(c) and Step 1.

Why it works. each soliton can be read with either sign of its wave number, which gives 2^N branches over the jet of order $N - 1$, and the weighted grading turns that count into the total degree.

■

Uses: A29, A30 (Step 1), Novikov (Nov′), Blum-Smith Cor. 2.4.19 (Hironaka criterion); research file `work/OP-F5C`, §1.

Part II

The other results

CHAPTER 5

Phases 0–1: the ODE prototype, identification, decidability

In this chapter	
Starting from	the toolkit of Part I, Chapter 1 (A1–A9). For the companion matrix: $P \in \mathbb{C}[\xi][\tau]$ monic in τ of degree k, $C(\xi)$ its companion matrix (ones on the superdiagonal, last row $(-p_0, \dots, -p_{k-1})$), $v(\tau) = (1, \tau, \dots, \tau^{k-1})^{\mathsf{T}}$.
Why	to prove the remaining results of Chapters 3–5 of the paper: the one-variable prototype, the projectors of the companion matrix, the refinements of identification (evolutions, waves, symmetries, local data) and the decidability results beyond A8–A9.
Role	Dim. B1–B21. Two elementary facts are used throughout: (F0) $C(\xi)v(\tau) = \tau v(\tau) - P(\tau, \xi)e_k$, so $v(\tau_0)$ is an eigenvector when $P(\tau_0, \xi_0) = 0$; (F1′) e_k is a cyclic vector of C, so the minimal polynomial of C is P.

5.1 The one-variable prototype and the companion matrix

Dim. B1 — **R4-T1**

ODE identification [proved]

Let $u \in \ker p(D)$, $D = d/dx$, $\deg p = m$. The minimal annihilator of u has degree $\dim \operatorname{span}\{u, u', u'', \dots\}$, the number of basis functions $x^j e^{\lambda x}$ actually present in u. It equals p iff the Wronskian of $u, \dots, u^{(m-1)}$ does not vanish.

Stated in the paper as no. 3.4.1, p. 14

Proof.

Step 1 The derivative space

In one variable every ideal of $\mathbb{C}[\xi]$ is principal, so $\operatorname{Ann}(u) = (q)$ with q the minimal annihilator, and $\mathbb{C}[\xi]u \cong \mathbb{C}[\xi]/(q)$ has dimension $\deg q$.

Step 2 Which basis functions are present

Write $u = \sum_j q_j(x)e^{\lambda_j x}$. By A2(c) in one variable, $(D - \lambda_j)^{\deg q_j + 1}$ kills the j-th summand and no lower power does, so $q = \prod_j (\xi - \lambda_j)^{\deg q_j + 1}$, of degree equal to the number of functions $x^l e^{\lambda_j x}$ present.

Step 3 The Wronskian

$q = p$ iff $\deg q = m$ iff $u, \dots, u^{(m-1)}$ are linearly independent iff their Wronskian is not identically zero.

Why it works. in one variable the derivative space of u is the quotient by its minimal annihilator.

■

Uses: A2.

Dim. B2 — **II-L1**

First-order reduction as a module equivalence [proof written, not audited]

(a) $\mathcal{R}^{1\times k}/\mathcal{R}^{1\times k}(\tau I - C(\xi)) \cong \mathcal{R}/(P)$, with $\bar{e}_1 \mapsto \bar{1}$. (b) The t-jet map $J : u \mapsto (u, \partial_t u, \dots, \partial_t^{k-1} u)$ is a bijection from $\ker P$ onto the solutions of $(\partial_t - C(\partial_x))U = 0$, with inverse $U \mapsto U_1$, and $J(f(\partial)w) = f(\partial_x, C(\partial_x))J(w)$.

Proof.

Step 1 (a) The presentation

Map $e_i \mapsto \tau^{i-1}$. The rows $\tau e_i - e_{i+1}$ go to 0, and the last row $\tau e_k + \sum_j p_j e_{j+1}$ goes to $P \equiv 0$, so the map passes to the quotient and is onto. In the quotient $e_{i+1} \equiv \tau e_i$, so the module is cyclic, generated by $\bar{e}_1$, and its annihilator is exactly (P).

Step 2 (b) The jet map

If $Pu = 0$, $U = J(u)$ satisfies $\partial_t U_i = U_{i+1}$ and $\partial_t U_k = -\sum_j p_j(\partial_x)U_{j+1}$: this is $(\partial_t - C(\partial_x))U = 0$. Conversely the first $k-1$ equations give $U_i = \partial_t^{i-1} U_1$ and the last gives $P(\partial)U_1 = 0$. Intertwining holds for $f = \tau$ by the equation itself, and for $f = \xi$ trivially.

Why it works. the companion system is a presentation of the same cyclic module $\mathcal{R}/(P)$.

■

Uses: none.

Dim. B3 — **II-T1**

When the companion form exists [proof written, not audited]

(i) $\mathcal{R}/(P)$ is a finitely generated $\mathbb{C}[\xi]$-module iff the leading coefficient of P in τ is a nonzero constant; then it is free with basis $1, \tau, \dots, \tau^{k-1}$ and τ acts by $C(\xi)^{\mathsf{T}}$. (ii) For P, Q monic in τ: $C_P \sim C_Q \iff P = Q \iff \mathcal{R}/(P) \cong \mathcal{R}/(Q)$. (iii) If the principal symbol does not vanish at $(1, 0)$, (i) holds with $k = \deg P$; the converse is false (heat).

Proof.

Step 1 (i)

For monic P, division by P in $\mathbb{C}[\xi][\tau]$ gives the basis. Conversely, if the module is generated by $g_1, \dots, g_n$ over $\mathbb{C}[\xi]$, then $\tau g_i = \sum a_{ij} g_j$, and the determinant trick gives a monic $h(\tau) = \det(\tau I - (a_{ij}))$ killing the module, so $h = Pq$ and $1 = \mathrm{lc}_\tau(P)\,\mathrm{lc}_\tau(q)$.

Step 2 (ii)

$\det(\tau I - C_P) = P$, similarity preserves it, and an isomorphism of modules preserves the annihilator (P).

Step 3 (iii)

The coefficient of τ^m is $\sigma_m(1,0)$. Counterexample to the converse: $P = \tau - \xi^2$ has $\sigma_2(1,0) = 0$ but a companion form of size one.

Why it works. a companion form is a choice of a non-characteristic time direction.

■

Uses: B2.

Dim. B4 II-P2

The principal symbol sees only infinity [proof written, not audited]

(a) $\sigma_m(P+Q) = \sigma_m(P)$ if $\deg Q < m$: the principal symbol is not a complete invariant. (b) The projective closure of $V(P)$ meets the line at infinity exactly at the zeros of σ_m. (c) σ_m transforms as a binary form under linear changes of variables.

Proof.

Step 1 (a)

The top-degree part does not change. Example: $\partial_t - \partial_x^2$ and $\partial_t - \partial_x^2 + c$ have the same $\sigma_2 = -\xi^2$ but different kernels.

Step 2 (b), (c)

$P^h(\tau, \xi, \zeta) = \sum_d \zeta^{m-d} \sigma_d$, so at $\zeta = 0$ only σ_m remains; the top-degree part commutes with linear substitution.

Why it works. the principal symbol is the restriction of the equation to the line at infinity.

■

Uses: none.

Dim. B5 II-T2

Projectors of the companion matrix, and their poles [proved]

Let $P = Q_1 \cdots Q_s$ with Q_j monic in τ and pairwise coprime over $K = \mathbb{C}(\xi)$; $B_j = P/Q_j$, $r_j = \mathrm{Res}_\tau(Q_j, B_j) \in \mathbb{C}[\xi]$.

(a) $r_j \neq 0$, and $U_j Q_j + V_j B_j = r_j$ with $U_j, V_j \in \mathbb{C}[\xi][\tau]$.

(b) $\Pi_j = (V_j B_j)(\xi, C(\xi))/r_j(\xi)$ are orthogonal idempotents commuting with C, summing to I; Π_j projects onto $\ker Q_j(C)$ along $\ker B_j(C)$. At a point ξ_0 with $r_j(\xi_0) \neq 0$ the same holds with kernel $\ker B_j(\xi_0, C_0)$.

(c) The poles of Π_j are exactly the ξ-coordinates of the points of $V(Q_j) \cap V(B_j)$.

(d) On solutions: if $u = \sum_i u_i$ with $u_i \in \ker Q_i$, then $(V_j B_j)(\partial)u = r_j(\partial_x)u_j$.

Stated in the paper as no. 3.4.2, p. 15

Idea. This is the Chinese remainder theorem over the field $\mathbb{C}(\xi)$, evaluated at the companion matrix; the resultant is the common denominator, and it vanishes exactly where two factor curves meet.

Proof.

Step 1 (a) Resultant identity

Q_j and B_j are monic and coprime over K, so their resultant is nonzero, and Cramer's rule on the Sylvester matrix gives $U_j Q_j + V_j B_j = r_j$ with coefficients in $\mathbb{C}[\xi]$.

Step 2 (b) Idempotents

$e_j = V_j B_j / r_j \equiv 1 \pmod{Q_j}$ and $\equiv 0 \pmod{Q_i}$ for $i \neq j$. So $e_j^2 \equiv e_j$, $e_i e_j \equiv 0$, $\sum e_j \equiv 1$ modulo P. Since $P(C) = 0$ (F1′, Cayley–Hamilton), $f \mapsto f(C)$ is a ring homomorphism from $K[\tau]/(P)$, and the $\Pi_j = e_j(C)$ inherit these relations. If $Q_j(C)w = 0$ then $\Pi_j w = w$; and $Q_j(C)\Pi_j = (V_j P)(C)/r_j = 0$. So the image is $\ker Q_j(C)$; $B_j(C)(I - \Pi_j) = (U_j P)(C)/r_j = 0$ gives the kernel.

Step 3 (c) Poles

If no point of $V(Q_j) \cap V(B_j)$ has abscissa ξ_0, the specialised polynomials are coprime and $r_j(\xi_0) \neq 0$. Conversely, if Π_j were holomorphic at a common point (τ_0, ξ_0), its value Π_0 would be an idempotent commuting with the cyclic C_0, hence a polynomial in C_0, hence a sum of spectral projectors; but τ_0 is a root of both Q_j and B_j at ξ_0, and C_0 has a single Jordan block there, so Π_0 cannot both keep and kill it.

Step 4 (d) On solutions

For $i \neq j$, $(V_j B_j)(\partial)u_i = 0$ because $Q_i \mid B_j$. And $(V_j B_j)(\partial)u_j = (r_j - U_j Q_j)(\partial)u_j = r_j(\partial_x)u_j$.

Why it works. the resultant is the universal denominator of the Chinese remainder idempotents.

■

Uses: B2.

Example 5.1 Wave

For $(\tau-\xi)(\tau+\xi)$: $r=2\xi$ and

$$\Pi_1=\begin{pmatrix}\frac{1}{2} & \frac{1}{2\xi}\\ \frac{\xi}{2} & \frac{1}{2}\end{pmatrix},$$

with a pole at $\xi=0$, the abscissa of the crossing point of the two lines.

Dim. B6 **II-T3**

Polynomial projectors iff disjoint varieties [proved]

For $P=P_1P_2$ and any Ω, equivalent: (i) $V(P_1)\cap V(P_2)=\emptyset$; (ii) $(P_1,P_2)=\mathcal{R}$; (iii) $\mathcal{R}/(P)\cong\mathcal{R}/(P_1)\times\mathcal{R}/(P_2)$; (iv) some $E_1\in\mathcal{R}$ is the identity on $\ker P_1$ and zero on $\ker P_2$; (v) $\ker P_1\cap\ker P_2=\{0\}$. Under (ii), $\ker P=\ker P_1\oplus\ker P_2$ on every Ω. **(II-C2)** If (i) holds for nonconstant P_1,P_2, their principal symbols share a linear factor.

Stated in the paper as no. 6.5, p. 36

Proof.

Step 1 (i) $\Longleftrightarrow$ (ii) $\Longleftrightarrow$ (iii)

Weak Nullstellensatz; Chinese remainder theorem for comaximal ideals.

Step 2 (ii)⇒(iv),(v), and the splitting

From $aP_1+bP_2=1$ put $E_1=bP_2$: $E_1u_1=(1-aP_1)u_1=u_1$ and $E_1u_2=0$. Every $u\in\ker P$ splits as $(bP_2)(\partial)u+(aP_1)(\partial)u$, with the two pieces in $\ker P_1$ and $\ker P_2$.

Step 3 (iv),(v)⇒(i)

A common zero (τ_0,ξ_0) gives $w=e^{\tau_0 t+\xi_0 x}\in\ker P_1\cap\ker P_2$, $w\neq 0$, contradicting (v) and, via $w=E_1w=0$, (iv).

Step 4 (II-C2)

Two projective plane curves always meet (Bézout). Under (i) they meet only at infinity, that is (B4(b)) at a common zero of the principal symbols.

Why it works. a polynomial projector is a Bézout identity, which exists exactly when the curves never meet.

■

Uses: B4.

Dim. B7 **II-T4**

Only factors give single-valued projectors [proved]

For P monic in τ and a set S of roots $\tau_a(\xi)$ (with full multiplicities), the spectral projector Π_S of $C(\xi)$ is rational in ξ iff $Q_S = \prod_{a\in S}(\tau-\tau_a)^{\mu_a}$ lies in $\mathbb{C}[\xi][\tau]$, that is, iff S is a union of irreducible factors of P. There are exactly 2^r rational projectors, r the number of distinct irreducible factors.

Proof.

Step 1 Rational projector ⇒ rational factor

$\Pi_S = g(C)$ with g of degree $< k$; since $I,\dots,C^{k-1}$ are independent with coefficients in K, Cramer gives $g \in K[\tau]$. Then $Q_S = \gcd(P, g-1)$ (with multiplicities) is computed by Euclid in $K[\tau]$, so $Q_S \in K[\tau]$, and by Gauss in $\mathbb{C}[\xi][\tau]$.

Step 2 Rational factor ⇒ rational projector

B5(b) with $s = 2$.

Why it works. the branches of an irreducible factor are permuted by monodromy, so no single-valued projector separates them.

■

Uses: B5.

Dim. B8 **II-P5**

Normality of the companion matrix [proved]

For a companion matrix C of $p(s) = s^k + \sum_{j<k} p_j s^j$, equivalent: (1) C normal; (2) C has an orthonormal eigenbasis; (3) the roots are simple and $\lambda_a\bar\lambda_b$ is a nontrivial k-th root of unity for all $a \neq b$; (4) $k = 1$; or $k = 2$ with $|p_0| = 1$ and $\bar p_0 p_1 = -\bar p_1$; or $k \geq 3$ with $p = s^k - c$, $|c| = 1$. **The earlier claim "normal iff $p = s^k - c$" is false for** $k = 2$ [false]: $s^2 + s - 1$ has the symmetric companion matrix $\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$.

Proof.

Step 1 Normal forces simple roots

A normal matrix is diagonalisable, and a cyclic one (F1′) is diagonalisable only with simple roots.

Step 2 Orthogonality of eigenvectors

With simple roots the eigenvectors are $v(\lambda_a)$, and

$$\langle v(\lambda_a), v(\lambda_b)\rangle = \sum_{j<k} z^j, \qquad z = \lambda_a\bar\lambda_b,$$

which vanishes iff $z \neq 1$ and $z^k = 1$. This proves (1) $\iff$ (3).

Step 3 $k \geq 3$

With three indices, $|\lambda_a|^2 = |\lambda_a\lambda_b||\lambda_a\lambda_c|/|\lambda_b\lambda_c| = 1$, so the roots are λ_1 times all k-th roots of unity and $p = s^k - \lambda_1^k$.

Step 4 $k = 2$

Direct computation:

$$CC^* - C^*C = \begin{pmatrix} 1 - |p_0|^2 & -p_1\bar{p}_0 - \bar{p}_1 \\ -p_0\bar{p}_1 - p_1 & |p_0|^2 - 1 \end{pmatrix}.$$

Why it works. the eigenvectors of a companion matrix are Vandermonde vectors, orthogonal only for roots on a rotated circle of roots of unity.

■

Uses: none.

5.2 Refinements of identification

Dim. B9 II-C1

The t-jet is a cyclic vector [proved]

For P monic in τ of degree k: $\mathrm{Ann}(u) = (P)$ iff no nonzero $(b_0, \dots, b_{k-1}) \in \mathbb{C}[\xi]^k$ has $\sum_j b_j(\partial_x)\partial_t^j u = 0$.

Stated in the paper as no. 3.4.3, p. 15

Proof.

Step 1 Division by P

This is A3(d): every Q is $AP + \sum_{j<k} b_j(\xi)\tau^j$ uniquely, $Q \in \mathrm{Ann}(u)$ iff $\sum b_j(\partial_x)\partial_t^j u = 0$, and $Q \in (P)$ iff all $b_j = 0$.

Why it works. modulo P, every operator is a $\mathbb{C}[\xi]$-combination of the first k time derivatives.

■

Uses: A3.

Dim. B10 **C3-T4**

Evolutions: Hermite interpolation [proved]

Let $P = \tau - a(\xi)$. (a) $\mathrm{Ann}(u) = (P) \iff u \notin \mathrm{EP}$. (b) If $u = \sum_k q_k e^{\langle z_k, \cdot \rangle}$ (distinct z_k, $q_k \neq 0$), $\delta_k = \deg_x q_k$, $D = \sum_k (\delta_k + 1)$ and $\deg a \leq n$, then

$$\{b \in \mathbb{C}[\xi]_{\leq n} : \tau - b \in \mathrm{Ann}(u)\} = a + \prod_k (\xi - \xi_k)^{\delta_k + 1} \cdot \mathbb{C}[\xi]_{\leq n - D}.$$

Stated in the paper as no. 4.4.1, p. 22

Proof.

Step 1 (a)

A3(a) with one factor and $P/P_1 = 1$.

Step 2 (b) Reduction to an ODE in x

$\tau - b = P + (a - b)(\xi)$, so $\tau - b \in \mathrm{Ann}(u)$ iff $c(\partial_x)u = 0$ with $c = a - b$, $\deg c \leq n$.

Step 3 (b) Triangular conditions

By A2(a),(b), $c(\partial_x)u = 0$ iff for each k

$$c(\xi_k + \partial_x) q_k = \sum_j \frac{c^{(j)}(\xi_k)}{j!} \partial_x^j q_k = 0.$$

Since $\partial_x^j q_k$ has x-degree exactly $\delta_k - j$, the system is triangular and equivalent to $c^{(j)}(\xi_k) = 0$ for $j \leq \delta_k$, that is, $\prod_k (\xi - \xi_k)^{\delta_k + 1} \mid c$.

Why it works. each mode with a polynomial prefactor of degree δ fixes $\delta + 1$ Hermite data of the symbol.

■

Uses: A2, A3.

Dim. B11 **C3-T5**

Wave, Klein–Gordon, telegraph, beam [proved]

(a) Wave $P = (\tau - c\xi)(\tau + c\xi)$, $c \in \mathbb{R} \setminus \{0\}$, $u = f(x - ct) + g(x + ct)$: $\mathrm{Ann}(u) = (P) \iff f \notin \mathrm{EP}$ and $g \notin \mathrm{EP}$. (b) $\tau^2 - \xi^2 + \mu^2$ ($\mu \neq 0$) and $\tau^2 + \alpha\tau - c^2\xi^2$ ($\alpha, c \neq 0$) are irreducible: identified iff $u \notin \mathrm{EP}$. (c) Beam $\tau^2 + \xi^4$: identified iff $(\partial_t \pm i\partial_x^2)u \notin \mathrm{EP}$ for both signs.

Stated in the paper as no. 4.2, p. 19

Proof.

Step 1 (a) The partial quotients

$$w_1 = (\partial_t + c\partial_x)u = 2c\,g'(x+ct), \qquad w_2 = (\partial_t - c\partial_x)u = -2c\,f'(x-ct).$$

Step 2 (a) Pull-backs

$F(x-ct) \in \mathrm{EP}(\Omega)$ iff $F \in \mathrm{EP}(J_F)$: if an exponential-polynomial E equals $F(x-ct)$ on Ω, then $(\partial_t + c\partial_x)E = 0$ on Ω, hence everywhere (A2(b)), so $E = F_0(x-ct)$ with $F_0 \in \mathrm{EP}$, and $F = F_0$ on J_F. In one variable $f \in \mathrm{EP} \iff f' \in \mathrm{EP}$. Apply A3(a).

Step 3 (b), (c)

A nondegenerate conic is irreducible (bordered determinant $-\mu^2$, resp. $c^2\alpha^2/4$, nonzero). For the beam apply A3(a) to the two factors.

Why it works. each factor of the wave operator carries one profile, and each profile must be non-exponential.

■

Uses: A2, A3.

Dim. B12 **C3-C1**

The spectrum of an expansion is not intrinsic [false] (counterexample proved)

The claim "if $u = \sum_{z\in Z} c_z e^{\langle z,\cdot\rangle}$ on a bounded Ω with all $c_z \neq 0$ and $Z \cap V(P_i)$ infinite for each i, then $\mathrm{Ann}(u) = (P)$" is false.

Stated in the paper as no. 4.4.2, p. 22

Proof.

Step 1 The counterexample

$P = \tau + \xi$, $\Omega \subset \{|x-t| < \pi\}$. On $(-\pi, \pi)$,

$$e^s = \sum_{k\in\mathbb{Z}} c_k e^{iks}, \qquad c_k = \frac{\sinh(\pi)\,(-1)^k}{\pi(1-ik)} \neq 0,$$

in L^2, hence in $\mathcal{D}'(\Omega)$. So $u = e^{x-t}$ has infinitely many exponents $(-ik, ik)$ on $\tau + \xi = 0$, but $\tau + 1$, $\xi - 1 \in \mathrm{Ann}(u)$.

Step 2 Diagnosis

The expansion is not unique: $(-1, 1)$ also lies on the line, giving a nontrivial relation. On a window of length $\geq 2\pi$, Fourier uniqueness restores the criterion (C3-P4).

Why it works. on a bounded domain a function has many exponential expansions, so their exponents are not invariants.

■

Uses: none.

Dim. B13 C3-C2

Exponential-polynomial data never identify [proof written, not audited]

If $u \in \mathrm{EP}(\Omega)$ and $P(\partial)u = 0$ with P nonconstant, then $\mathrm{Ann}(u) \neq (P)$.

Proof.

Step 1 Finite versus infinite

$V(\mathrm{Ann}\, u)$ is finite (A2(c)), while $V(P)$ is infinite (A1(F)).

Why it works. finitely many exponentials sit on finitely many points of a curve.

■

Uses: A1, A2.

Dim. B14 C3-P3

Translation symmetry [proof written, not audited]

Let $\ell = \alpha\tau + \beta\xi \neq 0$, $\ell(\partial)u = 0$ and $P(\partial)u = 0$, P nonconstant. (i) If $P \not\sim \ell$, $\mathrm{Ann}(u) \neq (P)$. (ii) If $\ell \nmid P$, $u \in \mathrm{EP}$. (iii) Every dictionary containing the monomials of P and ℓ has a Gram kernel of dimension ≥ 2.

Stated in the paper as no. 4.4.3, p. 23

Proof.

Step 1 (i)

$\ell \in \mathrm{Ann}(u)$; if $\mathrm{Ann}(u) = (P)$ then $P \mid \ell$, so $P \sim \ell$.

Step 2 (ii)

ℓ is irreducible and does not divide P, so they are coprime; $V(\mathrm{Ann}\, u) \subseteq V(P, \ell)$ is finite and $u \in \mathrm{EP}$ (A2).

Step 3 (iii)

P and ℓ are independent elements of the annihilator.

Why it works. a constant-coefficient symmetry is itself an equation, and two independent equations leave only finitely many exponentials.

■

Uses: A1, A2.

Dim. B15 **C3-P4**

Spectral criterion under uniqueness of the expansion

[proof written, not audited]

Let P be squarefree, $u = \sum_{z\in Z} c_z e^{\langle z,\cdot\rangle}$ in $\mathcal{D}'(\Omega)$, $Z \subset V(P)$ countable, $c_z \neq 0$, and assume uniqueness of such expansions (H-uniq). Then $\mathrm{Ann}(u) = \{Q : Q|_Z = 0\}$, and $\mathrm{Ann}(u) = (P)$ iff $Z \cap V(P_i)$ is infinite for every i.

Proof.

Step 1 **Annihilators**

$Q(\partial)u = \sum c_z Q(z) e^{\langle z,\cdot\rangle}$, zero iff $Q|_Z = 0$ by (H-uniq).

Step 2 **($\Leftarrow$)**

If $Q|_Z = 0$ and $Z \cap V(P_i)$ is infinite, then Q and P_i have infinitely many common zeros, so $P_i \mid Q$ (A1(V)); hence $P \mid Q$.

Step 3 **($\Rightarrow$)**

If $F = Z \cap V(P_i)$ is finite, $Q = (P/P_i)\prod_{z\in F} \ell_z$ with linear $\ell_z(z) = 0$ vanishes on Z and is not a multiple of P.

Why it works. with a unique expansion, the exponents are the points where annihilators must vanish.

■

Uses: A1.

Dim. B16 **C3-P5**

Local data

[proof written, not audited] (a) / [from cited source] (b)

(a) For $P = \tau + \xi$, $F(s) = e^s + \varphi(s)$ with a nonzero bump $\varphi \in C_c^\infty(2,3)$ and $u = F(x-t)$: on $(0,1)\times(-5,5)$, $\mathrm{Ann}(u) = (\tau+\xi)$; on $(0,1)^2$, $u = e^{x-t}$ and Ann is zero-dimensional. (b) For elliptic P, $\mathrm{Ann}(u|_{\Omega'}) = \mathrm{Ann}(u)$ for every connected $\Omega' \subset \Omega$ (Petrowsky analyticity) [from cited source].

Stated in the paper as no. 4.4.3, p. 23

Proof.

Step 1 **(a)**

If $u \in$ EP on the large rectangle, F would be an exponential-polynomial on $(-6,5)$, analytic, equal to e^s on $(-1,1)$ hence everywhere, but $F \neq e^s$ on $(2,3)$. On the small square $x - t \in (-1,1)$ and $u = e^{x-t}$.

Step 2 **(b)**

$Q(\partial)u$ solves the elliptic equation, so it is analytic; if it vanishes on Ω' it vanishes on Ω.

Why it works. a non-analytic bump can hide outside a smaller window; analyticity forbids it.

■

Uses: A2.

5.3 Decidability and variable coefficients

Dim. B17 **P4-C1**

Restriction does not identify in the Weyl algebra [proof written, not audited]

(a) For every nonzero smooth u of one variable, the action $A_1 \to \mathrm{End}(A_1 \cdot u)$ is injective.
(b) For $u = e^x$ and $\mathcal{L} = \mathrm{span}\{1, \partial\}$ the restriction to $\mathcal{L}$ is injective, yet $\partial - 1 \in \mathrm{Ann}(u) \cap \mathcal{L}$.
(c) The correct object is the evaluation $L \mapsto Lu$, whose kernel is $\mathrm{Ann}(u) \cap \mathcal{L}$.

Stated in the paper as no. 5.6, p. 28

Proof.

Step 1 (a)

The kernel of the action is a two-sided ideal not containing 1; A_1 is simple, so it is 0.

Step 2 (b)

$(\partial - 1)e^x = 0$. If $c_0 + c_1\partial$ acts as zero on $\mathbb{C}[x]e^x$, applying it to xe^x gives $c_0 x + c_1(1 + x) = 0$, so $c_0 = c_1 = 0$.

Why it works. in a noncommutative ring, killing the generator does not kill the cyclic module.

■

Uses: none.

Dim. B18 **OP-ALG-T1**

Decidability in the holonomic model [proved mod. (AC)]

In the analytic holonomic model M3 (and for analytic solutions with D-finite Cauchy data, (P4-T6)), under (AC): "$L_* u = 0$?" is decidable for every operator with polynomial coefficients; "$Q(\partial)u \in \mathrm{EP}$?" is decidable; "$\mathrm{Ann}(u) = (P)$?" and "$V(\mathrm{Ann}\, u) = V(P)$?" are decidable and G_u is computable; a bound B and generators of $\mathrm{Ann}(u)$ are computable.

Stated in the paper as no. 5.5, p. 27

Proof.

Step 1 Zero test

Holonomic functions are closed under the operators involved; a holonomic function given with enough initial values is zero iff those values are zero (OP-ALG-P1), which is a finite check in a computable field under (AC).

Step 2 EP test

For $w = Q(\partial)u$ compute eliminating operators in t alone and in x alone; $w \in \mathrm{EP}$ iff their constant-coefficient parts $a^*(\partial_t)$, $b^*(\partial_x)$ kill w (OP-ALG-L4, P2), two zero tests.

Step 3 Identification and G_u

A3 and A4 with at most $\sum_i e_i$ EP tests.

Step 4 Bound and generators

$w_0 = G_u(\partial)u \in \mathrm{EP}$ has $\mathcal{R}u \cap \mathrm{EP} = \mathcal{R}w_0$ (A5); its eliminants give a finite-dimensional stable space containing $\mathcal{R}w_0$, hence B; A6(ii) and A8(i) give generators.

Why it works. holonomic functions carry finite certificates of their own equations, which supply the bound that a black box lacks.

■

Uses: A3, A4, A5, A6, A8; research file `work/OP-ALG`.

Dim. B19 — **OP-ALG-T3**

The minimal sufficient datum [proof written, not audited]

The minimal datum that makes "$\mathrm{Ann}(u) = (P)$" decidable in $\mathrm{M2}^+$ is equivalent to a degree D such that, if identification fails, a witness of degree $\leq D$ exists. The bound B of A8 is equivalent to knowing G_u, and it is strictly stronger than the minimal datum on some regions when $\sum_i e_i \geq 2$.

Proof.

Step 1 D suffices and is computable from any sufficient datum

Given D, compare annihilators in degree $\max(D, \deg P)$ (A8(i)). Conversely, from the answer, run the semi-decision A8(ii), which returns a witness degree when identification fails.

Step 2 $B \equiv G_u$

A8(iii) computes G_u from B; conversely, from G_u, $w_0 = G_u(\partial)u$ is an exponential-polynomial and searching annihilators $a(\tau)$, $b(\xi)$ of w_0 gives $B = \deg a \cdot \deg b$.

Why it works. deciding needs only the degree of a witness, while B describes the whole exponential part.

■

Uses: A5, A8.

Dim. B20 **OP-ALG-T4**

Saturated identification criterion

[proof written, not audited]

Let $\mathfrak{D}$ be a ring of operators $\mathfrak{A}[\partial_t]$ with $\mathfrak{A}$ a domain of operators in x, P monic in ∂_t of degree m, $Pu = 0$. Equivalent: (a) $\mathrm{Ann}_{\mathfrak{D}}(u) = \mathfrak{D}P$; (b) the t-jet is free over $\mathfrak{A}$; (c) $\mathrm{rank}_{\mathfrak{A}}(\mathfrak{D}u) = m$; (d) no $Q \notin \mathfrak{D}P$ has Qu in the $\mathfrak{A}$-torsion.

Stated in the paper as no. 5.6, p. 28

Proof.

Step 1 Division

$Q = AP + Q_0$ with $\deg_{\partial_t} Q_0 < m$, $Qu = Q_0 u$, and $Q \in \mathfrak{D}P$ iff $Q_0 = 0$.

Step 2 The equivalences

(a) $\Longleftrightarrow$ (b): a strictly larger annihilator is a nonzero Q_0 of degree $< m$ killing u. (b) $\Longleftrightarrow$ (c): $\mathfrak{D}u \cong \mathfrak{A}^m/N$ with N the module of relations, torsion-free. (b) $\Longleftrightarrow$ (d): a relation is a Q_0 with $Q_0 u = 0$; conversely $bQ_0 u = 0$ with $b \neq 0$ gives the relation bQ_0.

Why it works. as in B9, modulo P everything is a combination of the first m time derivatives.

■

Uses: B9.

Dim. B21 **OP-ALG-C1**

With variable coefficients the finite test fails

[proof written, not audited]

With $\theta = x\partial_x$, $P_1 = \partial_t - \theta$, $P_2 = \partial_t - \theta + 1$, $P = P_1 P_2$, $u_2 = e^{-t}\exp(x^2 e^{2t})$ and $u^{(\lambda)} = (1 + \lambda\partial_x)u_2$: $\mathrm{Ann}(u^{(\lambda)}) \neq \mathfrak{D}P$ for every λ, yet neither partial quotient $P_1 u^{(\lambda)} = -u_2$ nor $P_2 u^{(\lambda)} = \lambda\partial_x u_2$ is torsion. So the literal transfer of A3(a) is false. **(OP-ALG-P5)** In the full Weyl algebra A_2, a holonomic solution never identifies its equation.

Proof.

Step 1 The identities

$P_1 P_2 = P_2 P_1$, $P_2 - P_1 = 1$, $P_1\partial_x = \partial_x P_2$, so $(1 + \lambda\partial_x)P = P_1 Q_\lambda$ with $Q_\lambda = P_2 + \lambda\partial_x P_1$, and $Q_\lambda u^{(\lambda)} = 0$ with $Q_\lambda \notin \mathfrak{D}P$.

Step 2 (P5)

A holonomic u is killed by a nonzero L in x alone, so u is torsion and B20(d) fails with $Q = 1$.

Why it works. without a gcd, annihilators of noncommutative operators are not controlled by finitely many quotients.

■

Uses: B20.

5.4 Results used without proof

C3-P1[gen] Malgrange density: on convex Ω exponential solutions are dense in $\ker P$; finitely many continuous functionals never certify identification. [from cited source]

Malgrange, Ann. Inst. Fourier 6, Chap. I §2 Thm 2 (primary source checked).

II-T9 With variable coefficients in $\mathbb{C}(x)[\partial]$, identification needs the pair (module, cyclic vector). [from cited source]

Ore 1933; van der Put–Singer.

R6-T3 "The companion matrix is normal iff $p = s^m - c$, $|c| = 1$" for every m. [false]

Counterexample $s^2 + s - 1$ in B8; true for $m \geq 3$.

CHAPTER 6

Phase 2: decomposition and regularity

In this chapter	
Starting from	Part I, Chapter 2 (A10–A18): the finite quotient, the twisted Poincaré lemma, the factor decomposition, the regularity criterion and density.
Why	to prove the remaining decomposition results: the one-dimensional lemmas on rectangles, constructive projectors, the Vandermonde formula, interior regularity, the cases where the H^m split holds, the cohomological criterion for regions with holes, and the closedness results for families of transports. The chapter ends with the complements on variable-coefficient identification proved in the same research package.
Role	Dim. B22–B42.

6.1 Rectangles and convex sets

Dim. B22 — **C12-L1**

One-dimensional lemmas on a rectangle [proved]

On $\Omega = I \times J$: (a) $\{\varphi \in \mathcal{D}'(\Omega) : \partial_x \varphi = 0\} = \{c \otimes 1 : c \in \mathcal{D}'(I)\}$; (b) there is a linear $\mathcal{J}$ with $\partial_x \mathcal{J} = \mathrm{id}$ and $[\partial_t, \mathcal{J}] = 0$; (c) the twisted Poincaré lemma holds on Ω; (d) $\{w : D_t w = D_x w = 0\} = \mathbb{C}\, e_{(p,q)}$.

Proof.

Step 1 (a), (b)

Fix $\theta \in C_c^\infty(J)$ with $\int \theta = 1$. Every test function splits as $\psi = \partial_x K\psi + \Psi \otimes \theta$ with $\Psi(t) = \int \psi(t,x)\,\mathrm{d}x$ and $K\psi$ compactly supported. If $\partial_x \varphi = 0$, then $\langle \varphi, \psi \rangle = \langle c, \Psi \rangle$ with $\langle c, \chi \rangle = \langle \varphi, \chi \otimes \theta \rangle$. Define $\langle \mathcal{J}\varphi, \psi \rangle = -\langle \varphi, K\psi \rangle$.

Step 2 (c), (d)

Remove the twist by $e = e_{(p,q)}$. With $w_0 = \mathcal{J}\tilde{\psi}$, $\partial_x(\partial_t w_0 - \tilde{\varphi}) = 0$, so $\partial_t w_0 - \tilde{\varphi} = c \otimes 1$ by (a); subtract a primitive $C \otimes 1$ of $c \otimes 1$. For (d), $\tilde{w} = c \otimes 1$ with $c' = 0$.

Why it works. on a rectangle every equation in one variable can be integrated with the other variable as a parameter.

■

Uses: none.

Dim. B23 **C12-L2**

Finite-dimensional modules on a rectangle [proved]

For a finite-dimensional $\mathcal{R}$-module M and $\mathcal{A}$ on a rectangle: $\operatorname{Ext}^1_{\mathcal{R}}(M,\mathcal{A}) = 0$, $\dim \operatorname{Sol}_{\mathcal{A}}(M) = \dim M$, and every solution is an exponential-polynomial.

Proof.

Step 1 Simple modules

For $M = \mathcal{R}/\mathfrak{m}_z$ the Koszul resolution reduces both claims to B22(c), (d).

Step 2 Induction

A common eigenvector of τ, ξ on M gives a simple submodule S; the long exact sequence for $0 \to S \to M \to M/S \to 0$ and induction give the two claims.

Step 3 Exponential-polynomials

The minimal polynomials $s(\tau)$, $r(\xi)$ of τ, ξ on M kill every solution value, and A2, Steps 5–7, applies.

Why it works. finite modules are built from points, and each point contributes one exponential.

■

Uses: B22, A2.

Dim. B24 **C12-L5**

Functions constant along a direction on a convex set [proved]

If $D \subset \mathbb{R}^2_{(s,r)}$ is open and convex and $\varphi \in \mathcal{D}'(D)$ has $\partial_r \varphi = 0$, then $\varphi = F(s)$ for a unique $F \in \mathcal{D}'(\mathrm{pr}_s D)$; if $\varphi \in L^2_{loc}$, so is F.

Proof.

Step 1 Local form and gluing

On each rectangle $Q \subset D$, B22(a) gives $\varphi = F_Q \otimes 1$. Two rectangles whose projections overlap are joined by a thin rectangle along a segment inside D (convexity), so the F_Q agree and glue.

Step 2 L^2

$F = \int \varphi(\cdot, r)\theta(r)\,dr$ locally, and Cauchy–Schwarz.

Why it works. convexity makes every fibre of the projection connected.

■

Uses: B22.

Dim. B25 **C12-T1**

Factor decomposition on rectangles and convex sets [proved]

On a rectangle (and on every convex set), statements (i)–(iv) of A13 hold.

Proof.

Step 1 Reduction

Rectangles and convex sets are simply connected, so this is a special case of A13. The original proof replaces A12 by B23.

Why it works. as for A13.

■

Uses: A13, B23.

Dim. B26 **C12-P1**

Constructive projectors [proved]

(a) For $s = 2$, $Q_2(\partial)u \in \ker Q_1$, and the components u_1 are exactly the solutions of $\{Q_1 w = 0,\ Q_2 w = Q_2 u\}$, unique up to $\ker Q_1 \cap \ker Q_2$. (b) If all factor curves are disjoint, $u_j = E_j(\partial)u$ with Chinese remainder idempotents $E_j \in \mathcal{R}$, and $u \in H^m \Rightarrow u_j \in H^{m-\deg E_j}$. (c) If $r = AQ_1 + BQ_2 \in \mathbb{C}[\xi]$ and $s = A'Q_1 + B'Q_2 \in \mathbb{C}[\tau]$, every component satisfies

$$r(\partial_x)u_1 = B(\partial)Q_2(\partial)u, \qquad s(\partial_t)u_1 = B'(\partial)Q_2(\partial)u.$$

Proof.

Step 1 (a)

$Q_1 Q_2 u = 0$; if $u = u_1 + u_2$ then $Q_2 u_1 = Q_2 u$, and two solutions differ by an element of $\ker Q_1 \cap \ker Q_2$.

Step 2 (b)

From $A_i Q_i + B_i \prod_{j \neq i} Q_j = 1$ put $E_i = B_i \prod_{j \neq i} Q_j$; then $Q_i E_i = B_i P$ and $\sum_i E_i \equiv 1 \pmod P$. Each derivative costs one Sobolev order.

Step 3 (c)

$r(\partial_x)u_1 = (AQ_1 + BQ_2)u_1 = BQ_2 u_1 = BQ_2 u$.

Why it works. every element of the ideal (Q_1, Q_2) turns into an equation that a component must satisfy.

■

Uses: A11.

Dim. B27 **C12-T2**

Vandermonde formula [proved]

For $P = \prod_{j=1}^{k}(\tau - a_j(\xi))$ with distinct a_j, $V(\xi) = [a_j^l]$, $\Delta = \det V = \prod_{i<j}(a_j - a_i)$:

$$\Delta(\partial_x)\, u_j = \sum_{l=0}^{k-1} \operatorname{adj}(V)_{jl}(\partial_x)\, \partial_t^l u, \qquad \delta(P) = \deg \Delta.$$

The split is unique iff Δ is a nonzero constant, and then $u_j = \Delta^{-1} \sum_l \operatorname{adj}(V)_{jl}(\partial_x)\partial_t^l u$.

Proof.

Step 1 The jet in terms of the components

$\partial_t u_j = a_j(\partial_x)u_j$, so $\partial_t^l u = \sum_j a_j(\partial_x)^l u_j$, that is $U = V(\partial_x)\mathbf{u}$.

Step 2 Inverting

Multiply by $\operatorname{adj} V$ and use $\operatorname{adj}(V)V = \Delta I$.

Step 3 The ambiguity

$\mathcal{R}/(\tau - a_i, \tau - a_j) \cong \mathbb{C}[\xi]/(a_i - a_j)$, so $\delta(P) = \sum_{i<j} \deg(a_i - a_j) = \deg \Delta$.

Why it works. the time derivatives of u are the components weighted by a Vandermonde matrix in the symbols.

■

Uses: A13.

> **Example 6.1 Wave**
>
> For $(\tau - \xi)(\tau + \xi)$, $\Delta = -2\xi$ and the formula is d'Alembert's $\partial_x u_+ = \frac{1}{2}(\partial_x u + \partial_t u)$.

Dim. B28 **C12-P2**

Reducibility in second-order time; first order [proved] (Lean-checked algebra)

(C12-P2) $P = \tau^2 + b(\xi)\tau + c(\xi)$ is reducible iff $D = b^2 - 4c$ is a square d^2 in $\mathbb{C}[\xi]$; then $P = (\tau - a_+)(\tau - a_-)$, $a_\pm = (-b \pm d)/2$, and $\delta(P) = \deg d$. (C12-P3) An evolution $\tau - a(\xi)$ has exactly one component.

Proof.

Step 1 (P2)

A factor of degree 0 in τ divides the leading coefficient 1, so a nontrivial factorisation has two monic linear factors $\tau - \alpha$, $\tau - \beta$ with $\alpha + \beta = -b$, $\alpha\beta = c$, $(\alpha - \beta)^2 = D$. And $\delta = \deg(a_+ - a_-)$ by B27.

Step 2 (P3)

A primitive polynomial of degree one in τ is irreducible (Gauss).

Why it works. a quadratic in τ factors exactly when its discriminant is a square.

■

Uses: B27.

> **Example 6.2 Klein–Gordon and telegraph**
>
> For $\tau^2 - \xi^2 + m^2$, $D = 4(\xi^2 - m^2)$ is a square iff $m = 0$; for $\tau^2 + a\tau - \xi^2$, $D = a^2 + 4\xi^2$ is a square iff $a = 0$. So both are single components.

Dim. B29 **C12-T4**

Interior regularity of the components [proved]

(C12-L4) If $p(\partial_x)w$ and $q(\partial_t)w$ lie in H^k_{loc}, with $\deg p = \deg q = D \geq 1$, then $w \in H^{k+D}_{loc}$.
(C12-T4) For coprime, non-comaximal Q_1, Q_2, with r, s as in B26(c), $D = \deg r \deg s$ and e the degree of the operators BQ_2, $B'Q_2$: every component of $u \in H^m_{loc} \cap \ker Q_1 Q_2$ lies in H^{m-e+D}_{loc}.

Proof.

Step 1 (L4) A Fourier estimate

$|p(i\kappa)| \geq c|\kappa|^D - C(1 + |\kappa|)^{D-1}$, and likewise for q, so

$$(1 + \omega^2 + \kappa^2)^{D/2} \leq C\big(1 + |p(i\kappa)| + |q(i\omega)|\big).$$

Localise with cutoffs; each commutator costs one order less, so finitely many steps reach H^{k+D}_{loc}.

Step 2 (T4)

Apply B26(c) with $p = r^{\deg s}$, $q = s^{\deg r}$ in the ideal, then (L4).

Why it works. an ODE in x and an ODE in t together form a quasi-elliptic pair, which regularises.

■

Uses: B26.

Interior regularity is never lost; A15 shows the loss happens only at the boundary.

Dim. B30 **C12-P4**

Two cases where the H^m split holds [proved]

(a) $P = \tau\xi$ on a bounded rectangle, every m: $c_F = 0$ after removing the constants. (b) The wave $\tau^2 - \xi^2$ on the square $(0,1)^2$, every m: $\|u_+\|^2 + \|u_-\|^2 \le C\|u\|^2$ after normalisation, and $c_F < 1$.

Proof.

Step 1 (a)

By B25, $u = A(t) \otimes 1 + 1 \otimes B(x)$. Integrating against $\theta(x)$ shows $A \in H^m$, hence $B \in H^m$. If $\int A = 0$, the H^m product of $A \otimes 1$ and $1 \otimes B$ reduces to the $\alpha = 0$ term $\int A \int B = 0$.

Step 2 (b) Characteristic coordinates

With $s = x + t$, $r = x - t$ the square becomes the rhombus $|r| < s < 2 - |r|$, and $u = F(s) + G(r)$ (B24).

Step 3 (b) Five pieces

On the central square $(\frac{1}{2}, \frac{3}{2}) \times (-\frac{1}{2}, \frac{1}{2})$, averaging against test functions controls F and G by u. On each of the four corner triangles one of the two profiles is already controlled (its argument stays in the central range), so the other is u minus it. The five pieces cover the rhombus.

Why it works. when the two families of characteristics are transversal, every point sees one profile on a long chord.

■

Uses: B24, B25.

Dim. B31 **C12-P5**

Variable coefficients: LCLM instead of the product [proved mod. Ore 1933]

For $L_1, L_2 \in \mathbb{C}(x)[\partial]$ on an interval J without singular points: $\ker \mathrm{LCLM}(L_1, L_2) = \ker L_1 + \ker L_2$ and $\ker \mathrm{GCRD}(L_1, L_2) = \ker L_1 \cap \ker L_2$. The product $L_1 L_2$ does not work: $\ker((x\partial - 1)\partial) \neq \ker(x\partial - 1) + \ker \partial$.

Proof.

Step 1 Inclusions

$G = UL_1 + VL_2$, $L_i = W_i G$, $M = AL_1 = BL_2$ hold as identities of operators, and continuity extends them across the poles of the rational coefficients. So $\ker G = \ker L_1 \cap \ker L_2$ and $\ker L_i \subseteq \ker M$.

Step 2 Dimensions

By Ore's identity $\operatorname{ord} M + \operatorname{ord} G = \operatorname{ord} L_1 + \operatorname{ord} L_2$ [from cited source], so $\dim(\ker L_1 + \ker L_2) = \operatorname{ord} M = \dim \ker M$.

Why it works. in a noncommutative ring the least common left multiple plays the role of the product.

■

Uses: Ore 1933.

6.2 Regions with holes

Dim. B32 **OPF-T4**

Cohomological criterion [proof written, not audited] ((i), (ii) need only sheaf theory)

Let Ω be connected, $\mathcal{N}$ the $\delta(P)$-dimensional space of global exponential tuples with sum 0, $\mathcal{K}_j$, $\mathcal{K}_P$ the solution sheaves. (i) $0 \to \underline{\mathcal{N}}_\Omega \to \bigoplus_j \mathcal{K}_j \xrightarrow{\Sigma} \mathcal{K}_P \to 0$ is exact. (ii) $\operatorname{coker} \Sigma_\Omega \cong \ker\left[\iota : H^1(\Omega;\mathbb{C}) \otimes \mathcal{N} \to \bigoplus_j H^1(\Omega;\mathcal{K}_j)\right]$; so the decomposition holds on Ω iff ι is injective, and $\dim \operatorname{coker} \le \delta(P)\, b_1(\Omega)$. (iii) For the Laplacian on a disc with k holes the cokernel has dimension exactly k [from cited source].

Stated in the paper as no. 6.4, p. 35

Proof.

Step 1 (i) Exactness on small discs

On every disc A13 gives surjectivity and kernel $\mathcal{N}$; restriction between discs is the identity on $\mathcal{N}$, so the kernel sheaf is the constant sheaf $\underline{\mathcal{N}}$.

Step 2 (ii) The long exact sequence

$$0 \to \mathcal{N} \to \bigoplus_j \ker Q_j \xrightarrow{\Sigma_\Omega} \ker P \xrightarrow{\partial} H^1(\Omega;\underline{\mathcal{N}}) \xrightarrow{\iota} \bigoplus_j H^1(\Omega;\mathcal{K}_j).$$

So $\operatorname{coker} \Sigma_\Omega \cong \operatorname{im} \partial = \ker \iota$, and $H^1(\Omega;\underline{\mathcal{N}}) \cong H^1(\Omega;\mathbb{C}) \otimes \mathcal{N}$ has dimension $\delta(P)\, b_1(\Omega)$.

Step 3 (iii)

Elliptic factors are surjective on every planar open set [from cited source], so $H^1(\Omega;\mathcal{K}_j) = 0$ and the cokernel is all of $H^1 \otimes \mathcal{N}$; for the Laplacian $\delta = 1$, with basis $\log|z - a_l|^2$.

Why it works. the obstruction to splitting is the monodromy of the shared exponentials around the holes.

■

Uses: A13; sheaf cohomology.

6.3 Closedness of sums of kernels

Dim. B33 **P2-T3**

Two subspaces: exact constants [proved]

For closed $\mathcal{H}_1, \mathcal{H}_2$ with intersection $\mathcal{I}$: $\ker S = \{(z, -z) : z \in \mathcal{I}\}$; on $(\ker S)^\perp$ the spectrum of S^*S lies in $[1 - c_F, 1 + c_F] \cup \{2\}$ and contains $1 \pm c_F$. The minimal-energy split is stable iff $c_F < 1$, with optimal constant $(1 - c_F)^{-1}$. **(P2-C2)** For $N \geq 2$ subspaces the same holds with γ_N; Tikhonov regularisation converges to the minimal-energy split, and its norm blows up when $\gamma_N = 0$.

Proof.

Step 1 **Spectrum**

This is A17, Steps 2–3: $S^*S = I + K$ on $A \oplus B$ with $\|K\| = c_F$ and symmetric spectrum, and $S^*S = 2$ on the diagonal of $\mathcal{I}$.

Step 2 **Tikhonov**

$\omega_\alpha = (S^*S + \alpha)^{-1} S^* u$; in the spectral resolution of S^*S it converges to $S^\dagger u$ when $u \in \operatorname{im} S$, and $\|\omega_\alpha\| \to \infty$ for $u \in \overline{\operatorname{im} S} \setminus \operatorname{im} S$.

Why it works. the sum map is a block operator whose off-diagonal part has norm equal to the cosine of the angle.

■

Uses: A17.

Dim. B34 **P2-C1**

Where regularising the factor split fails [proved]

(a) For two parallel transports on a convex Ω in L^2 the sum is dense in $\ker f_g$; if the chords shrink, some u has no L^2 components, and for every such u the Tikhonov split diverges while its residual tends to 0. (b) On the square, for the function of A15, either every regularisation leaves a residual, or $c_F = 1$ and the regularised components diverge.

Proof.

Step 1 **(a) Density**

In coordinates (t, s), $u = e^{d_1 t} F(s) + e^{d_2 t} G(s)$; truncating F, G to compact intervals of s gives elements of $\mathcal{H}_1 + \mathcal{H}_2$ converging to u.

Step 2 **Divergence**

B33 with $c_F = 1$ (A16).

Why it works. dense but not closed means: approximable, but only by pieces of unbounded size.

■

Uses: A16, B33.

Dim. B35 **OPF-T5**

Density via contracting symmetries

[proof written, not audited]

Let Ω be bounded and convex, and suppose affine maps $\Phi_\lambda(y) = o + A_\lambda(y-o)$, $A_\lambda \to I$, satisfy $P(A_\lambda^{\mathsf{T}} z) = \kappa_\lambda P(z)$ and $\Phi_\lambda(\bar{\Omega}) \subset \Omega$. Then $\sum_j \ker_{H^m} Q_j$ is dense in $\ker_{H^m} P$ for every m. This covers homogeneous P (the wave on every convex set) and quasi-homogeneous P on axis-parallel rectangles (the beam).

Proof.

Step 1 **Shrink, mollify, decompose**

$u \circ \Phi_\lambda \to u$ in H^m and solves P (the symmetry). Mollify on a convex neighbourhood of $\Phi_\lambda(\bar{\Omega})$, pull back: a smooth solution on a convex neighbourhood of $\bar{\Omega}$, which splits by A13 into components smooth up to the boundary.

Why it works. a symmetry of P pulls a solution slightly inward without leaving the solution space.

■

Uses: A13.

This is superseded on convex sets by A18, which needs no symmetry.

Dim. B36 **OPF-P6**

Parallel families: pairwise closed iff totally closed

[proof written, not audited]

For $Q_j = \partial_t + c\partial_x - d_j$ (d_j distinct) on a bounded convex Ω in L^2, equivalent: (a) the chords have $\inf \ell > 0$; (b) $\sum_j \mathcal{H}_j$ closed; (c) every pairwise sum closed; (d) some pairwise sum closed.

Proof.

Step 1 **Structure**

The factors are comaximal, so the split is unique, $u = \sum_j e^{d_j t} F_j(s)$.

Step 2 **(a)⇒(b)**

The $N \times N$ Gram matrix of $\{e^{d_j t}\}$ on chords of length $\geq \ell_0$ is uniformly positive definite, as in A16, Step 2.

Step 3 **(d)⇒(a)**

If $\inf \ell = 0$, A16, Step 4, applied to any pair shows that pair is not closed.

Why it works. for parallel transports everything happens chord by chord, and all pairs see the same chords.

■

Uses: A16.

Dim. B37 **OPF-P7**

Separable families: the sum is always closed

[proof written, not audited]

If every factor depends on t alone or on x alone, $P = A(\tau)B(\xi)$ on a bounded rectangle, then $\sum_j \ker_{H^m} Q_j$ is closed in H^m for every grouping.

Proof.

Step 1 **A bounded projector**

With a basis e_l of $\ker A(\partial_t)$ and biorthogonal $\psi_l \in C_c^\infty(I)$, $\pi_A u = \sum_l e_l(t) \int u(t', x)\psi_l(t')\,\mathrm{d}t'$ is a bounded idempotent on H^m with image $\ker_{H^m} A(\partial_t)$; likewise π_B.

Step 2 **Closed sum**

The t-factors sum to $\operatorname{im}\pi_A$ and the x-factors to $\operatorname{im}\pi_B$; two ranges of commuting bounded projectors have a closed sum, $\operatorname{im}(\pi_A + \pi_B - \pi_A\pi_B)$.

Why it works. separated variables give commuting projectors.

■

Uses: B23.

Dim. B38 **OPR2-T2**

Real transports in L^2

[proof written, not audited] *(condensed)*

For real transports $Q_j = v_j \cdot \nabla - d_j$ on a bounded convex Ω, grouped in classes of parallel directions, assume that of any two classes at least one has chords bounded below. Then $\sum_j \ker_{L^2} Q_j$ is closed iff every class with at least two factors has chords bounded below; in particular pairwise closed implies totally closed.

Stated in the paper as no. 7.3, p. 40

Proof.

Step 1 **Class projectors**

For a class with chords bounded below, B36 gives a closed sum with a chord-by-chord projector whose kernel is bounded.

Step 2 **Compact cross products**

For two transversal classes, the product of their projectors has a kernel integrable over the transversal chords, hence is compact, under the hypothesis on the chords.

Step 3 Closed sum

A finite sum of closed subspaces whose projectors have compact pairwise products and trivial intersections is closed (Fredholm argument). If a class with two factors has shrinking chords, B36 shows the class sum itself is not closed.

Why it works. transversal transports interact compactly, so only parallel ones can destroy closedness.

■

Uses: B36; research file `work/OP-R2`.

Dim. B39 **OPR2-T1**

Heat times transport: density [proof written, not audited]

For $P = (\tau - \xi^2)(\tau + c\xi)$ on a rectangle, $\ker_{H^m}(\partial_t - \partial_x^2) + \ker_{H^m}(\partial_t + c\partial_x)$ is dense in $\ker_{H^m} P$ for every m and c; so the components lie in H^m iff $c_F < 1$.

Proof.

Step 1 From A18

A rectangle is bounded and convex, so A18 applies directly. (The original proof used local inward translations at the corners and the hypoellipticity of the heat operator.)

Why it works. as for A18.

■

Uses: A18.

Dim. B40 **OPF5B-P8**

The tangent model of a wedge is closed [proof written, not audited]

On the infinite wedge $C = \{r\omega(\varphi) : 0 < \varphi < \beta\}$, $\beta < \pi$, with pairwise non-parallel homogeneous transports whose lines all cut C in bounded chords, there is $\gamma_* > 0$ with $\|\sum_j f_j\|^2 \geq \gamma_* \sum_j \|f_j\|^2$ for $f_j \in \mathcal{H}_j$.

Proof.

Step 1 Structure

$f \in \mathcal{H}_j$ iff $f(y) = F(n_j \cdot y)$ with $F \in L^2(s\,\mathrm{d}s)$, since the chords have length proportional to s.

Step 2 Mellin transform

The Mellin transform in r is unitary and maps $\mathcal{H}_j$ to $\{F^\sharp(\sigma)\, a_j(\varphi)^{-1+i\sigma}\}$, with $\|a_j^{-1+i\sigma}\|_{L^2(0,\beta)}$ independent of σ.

Step 3 Fibrewise Gram matrices

The norm of a sum becomes an integral over σ of quadratic forms with the Gram matrix $G(\sigma)$ of the functions $a_j^{-1+i\sigma}$, which is positive definite and has a positive infimum of its smallest eigenvalue (the limits $\sigma \to \pm\infty$ decouple by oscillation).

Why it works. dilation invariance diagonalises the problem, leaving one finite Gram matrix per frequency.

■

Uses: none.

6.4 Complements: variable-coefficient identification

Dim. B41 **OPF5B-T2**

Saturated criterion for non-monic P [proof written, not audited]

Let $P = \sum_{j\le m} p_j \partial_t^j$ with $p_m \in \mathfrak{A} \setminus 0$, $\mathfrak{K}$ the Ore localisation. For $Pu = 0$, equivalent: (a′) $\mathrm{Ann}_{\mathfrak{D}}(u) \subseteq \mathfrak{K}P$; (b) the t-jet is free over $\mathfrak{A}$; (c) $\mathrm{rank}_{\mathfrak{A}}\mathfrak{D}u = m$; (d′) no $Q \notin \mathfrak{K}P$ has Qu torsion. **(OPF5B-P3)** If $p_m = \pi(x)$ is a function and $\mathfrak{D}/\mathfrak{D}P$ has no π-torsion, B20 holds verbatim. **(OPF5B-C4)** The radial heat kernel $t^{-1}e^{-x^2/4t}$ identifies the radial heat operator $x\partial_t - \partial_x x \partial_x$; the Bessel modes $e^{-k^2 t} J_0(kx)$ do not.

Proof.

Step 1 Localise

In $\mathfrak{K}$, $P = p_m \hat{P}$ with $\hat{P}$ monic; divide by $\hat{P}$ and clear denominators with the Ore condition to turn a non-multiple annihilator into a nontrivial relation of degree $< m$, as in B20.

Step 2 Rank

Localisation is exact: $\mathrm{rank}\,\mathfrak{D}u = \dim_K \mathfrak{K}/\mathfrak{K}\,\mathrm{Ann}(u) \le m$, with equality iff $\mathrm{Ann}(u) \subseteq \mathfrak{K}P$.

Step 3 (C4)

The right action of P_r on $\mathfrak{D}/x\mathfrak{D} \cong \mathbb{C}[\eta, T]$ is $f \mapsto Tf' - \eta f - \eta^2 f'$, injective (leading coefficient $-(d+1)c_d$), so there is no x-torsion. The heat kernel is not killed by any nonzero operator in x alone (its derivatives are polynomials of growing degree times the Gaussian); the Bessel modes satisfy Bessel's ODE.

Why it works. a non-monic operator behaves like a monic one after inverting its leading coefficient, provided nothing is lost to torsion.

■

Uses: B20.

Dim. B42 **OPF5B-T6**

Repeated socle: no finite test

[proof written, not audited]

Assume some solution u^* identifies P (monic) and $\mathrm{End}_{\mathfrak{D}}(\mathfrak{D}/\mathfrak{D}P)$ contains infinitely many non-injective endomorphisms E_λ with pairwise trivially intersecting kernels. Then for every finite set of tests $\mathcal{T}_f$ the criterion "$\mathrm{Ann}(u) = \mathfrak{D}P$ iff Tu is not torsion for all $T \in \mathcal{T}_f$" fails for some solution. The hypothesis holds for $P = P_1P_2$, $P_2 = P_1 + c$, whenever a nonzero J with $P_1J \in \mathfrak{D}P_2$ exists (example: $\theta = x\partial_x$, $J = \partial_x$).

Proof.

Step 1 **Solutions from endomorphisms (OPF5B-L5)**

$E(\bar{Q}) = \overline{QR}$ with $PR \in \mathfrak{D}P$; $u_R = Ru^*$ solves P, and u_R identifies iff E_R is injective.

Step 2 **Beating a finite test**

Each test T lies in at most one kernel $\ker E_\lambda$; choose λ outside the finitely many used. Then u_λ passes every test but does not identify.

Step 3 **The example (OPF5B-P7)**

$\mathfrak{D}/\mathfrak{D}P \cong \mathfrak{D}/\mathfrak{D}P_1 \oplus \mathfrak{D}/\mathfrak{D}P_2$ via $\bar{Q} \mapsto (\bar{Q}, \bar{Q})$, and $E_\lambda(x_1, x_2) = (0, x_2 + \lambda J x_1)$ has kernel $\{(x_1, -\lambda J x_1)\}$, pairwise trivially intersecting.

Why it works. a repeated simple factor offers infinitely many ways to fail, more than any finite list of tests can detect.

■

Uses: B20.

6.5 Results used without proof

C12-P6	No orthogonality of components at fixed time in general.	[from cited source]

Shohat, Amer. J. Math. (checked, CIT-AUDIT-2).

C12-Q2	Closedness of the sum for $N \geq 3$ general differential factors.	[open]

Solved for real transports (B38), separable families (B37), wedges in the tangent model (B40).

OPF-Q3	Density for quasi-homogeneous P on non-rectangular convex sets.	closed by A18

A18 covers every bounded convex set.

CHAPTER 7

Phase 3: the interaction term

In this chapter	
Starting from	Part I, Chapter 3 (A19–A24): the class $\mathfrak{B}$, the closed form, the cost, the instability, Dyson–Phillips and the no-go for infinite adjoint orbits.
Why	to prove the supporting lemmas of the class $\mathfrak{B}$, the results on factors and second-order evolutions inside $\mathfrak{B}$, the solvable and non-solvable Lie cases, the interaction created by a boundary, the exact frontier between continuous and discontinuous interaction, the balance decomposition, and the residual-based no-go results.
Role	Dim. B43–B60.

7.1 The class $\mathfrak{B}$

Dim. B43 P1-L1

Properties of $\mathfrak{B}$ [proved]

(a) B_ρ, $\mathfrak{B}_\rho$ are Banach spaces and $\|\partial_x\|_{B_\rho \to B_\rho} \le \rho$. (b) $\mathfrak{B}_\rho \hookrightarrow H^m$ for every m. (c) Every exponential-polynomial lies in $\mathfrak{B}$. (d) For $u \in \mathfrak{B}_\rho$, $t \mapsto u(t,\cdot)$ extends to $\bar{I}$ and lies in $C(\bar{I}; B_{\rho'})$ for $\rho' > \rho$.

Stated in the paper as no. 8.2, p. 43

Proof.

Step 1 (a)

A Cauchy sequence has all derivatives Cauchy in L^2; the limits satisfy $g_{j+1} = g_j'$, so $g_0 \in H^\infty$ with $g_0^{(j)} = g_j$, uniformly in j. Shifting the index gives $\|f'\|_{B_\rho} \le \rho \|f\|_{B_\rho}$.

Step 2 (b), (c)

$\|u\|_{H^m}^2 \le \binom{m+2}{2} \max(1,\rho)^{2m} \|u\|_{\mathfrak{B}_\rho}^2$. For $x^l e^{\zeta x}$ the j-th derivative is $e^{\zeta x}$ times a polynomial with coefficients $O(j^l |\zeta|^j)$, and $j^l (|\zeta|/\rho)^j$ is bounded for $\rho > |\zeta|$.

Step 3 (d)

$H^1(I; L^2(J)) \hookrightarrow C(\bar{I}; L^2(J))$ applied to $\partial_x^j u$, with the geometric decay $(\rho/\rho')^j$.

Why it works. $\mathfrak{B}$ is the space where derivatives grow at most geometrically, so exponentials of polynomials in ∂_x converge.

■

Uses: none.

Dim. B44 **P1-L2**

Flows of terms in $\mathfrak{B}$ [proved]

On B_ρ: $\|A(\partial_x)\| \le |A|(\rho)$; $e^{sA(\partial_x)}$ converges in norm with $\|e^{sA}\| \le e^{|s||A|(\rho)}$; the family is entire in s, a group, independent of ρ; exponentials of different polynomials commute and $e^{s(A+A')} = e^{sA}e^{sA'}$; and $e^{sA(\partial_x)}(x^l e^{\zeta x}) = \partial_\zeta^l\big(e^{sA(\zeta)}e^{\zeta x}\big)$.

Proof.

Step 1 Norm and commutation

$\|A(\partial)\| \le \sum |\alpha_l|\|\partial\|^l$. Polynomials in one operator commute, and in a Banach algebra $e^{X+Y} = e^X e^Y$ for commuting X, Y (Cauchy product).

Step 2 Action on modes

$\zeta \mapsto e^{\zeta x} \in B_\rho$ is holomorphic for $|\zeta| < \rho$ and $A(\partial)^k e^{\zeta x} = A(\zeta)^k e^{\zeta x}$; a bounded operator commutes with ∂_ζ^l.

Why it works. on $\mathfrak{B}$ the derivative is a bounded operator, so its polynomials have exponentials.

■

Uses: B43.

Dim. B45 **P1-L3**

The Cauchy problem in the class [proved]

(P1-L3) A continuous curve v in B_ρ with $(\partial_t - A(\partial_x))v = 0$ is $v(t) = e^{(t-t_0)A}v(t_0)$; conversely these are solutions in $\mathfrak{B}$. (P1-L4) For $r \in \mathbb{C}[\xi]$ the Volterra primitive $w = \int_{x_0}^{x} G(x-y)F(t,y)\,dy$ (G the fundamental solution of $r(\partial)$) satisfies $r(\partial_x)w = F$ and maps $\mathfrak{B}_\rho$ into $\mathfrak{B}_R$. (P1-L5) Formal lemmas in $\mathcal{A}[[t]]$: $e^{gX}Ye^{-gX} = \sum_k g^k \mathrm{ad}_X^k Y/k!$, and for $U = \prod_i e^{g_i X_i}$, $U'U^{-1} = \sum_i g_i' \,\mathrm{Ad}(P_{i-1})X_i$.

Proof.

Step 1 (L3)

Testing the equation against $\varphi \otimes \psi$ shows that the distributional time derivative of v, as a B_ρ-valued distribution, is $A(\partial)v$; a Banach-valued distribution with zero derivative is constant, so $\frac{d}{dt}\big(e^{-(t-t_0)A}v\big) = 0$.

Step 2 (L4)

Variation of constants gives $r(\partial_x)w = F$. For $j \ge \delta = \deg r$, $\partial_x^j w$ is expressed through $\partial_x^{j-\delta}F$ and lower derivatives of w; induction gives $\sup_i \rho^{-i}\|\partial_t^i \partial_x^j w\| \le KR^j$.

Step 3 (L5)

Both sides of each identity solve the same formal ODE with the same initial value.

Why it works. uniqueness in $\mathfrak{B}$ is what excludes Tychonoff and boundary-fed solutions.

■

Uses: B43, B44.

Dim. B46 P1-P1

Single flows are not determined without a class [proved] / [from cited source]
(Tychonoff)

(a) For $a_n = \xi$ the solutions with a given datum on J are $F(x+t)$ for any extension F: infinitely many. (b) For $a_n = \xi^2$ there are nonzero heat solutions with zero datum; those with datum u_0 form an infinite-dimensional affine space. (c) So without a class E_{int} is not a function of u; in $\mathfrak{B}$ it is unique.

Stated in the paper as no. 8.4, p. 46

Proof.

Step 1 (a)

In coordinates $(t, y = x + t)$ the equation is $\partial_t w = 0$ on a parallelogram, so $w = F(y)$, fixed on J and free elsewhere.

Step 2 (b)

Tychonoff's $v(t,x) = \sum_k g^{(k)}(t)\, x^{2k}/(2k)!$ with $g(t) = e^{-t^{-2}}$ for $t > 0$, 0 otherwise, is smooth and vanishes for $t \le 0$ [from cited source]. Its time shifts $v(t-\sigma, x)$ are independent: on the first interval only the earliest survives, and $v(s,0) = g(s) \neq 0$.

Step 3 (c)

B45(L3).

Why it works. on a bounded domain both the transport and the heat flow need information that the datum does not contain.

■

Uses: B45.

7.2 Factors and second-order evolutions in $\mathfrak{B}$

Dim. B47 **P1-P2**

Factors in $\mathfrak{B}$ [proved]

On a bounded rectangle, $u \in \mathfrak{B} \cap \ker P$: (a) if the factor curves are disjoint, the components $E_j(\partial)u$ lie in $\mathfrak{B}$; (b) if one decomposition lies in $\mathfrak{B}$, all do; (c) for $P = \prod_j(\tau - a_j(\xi))$, every decomposition lies in $\mathfrak{B}$. **(P1-P3)** The interaction per term inside each factor is then defined modulo a finite-dimensional space of exponential-polynomials, of dimension at most $\delta(P)$.

Proof.

Step 1 (a), (b)

$\mathfrak{B}$ is closed under derivatives; the ambiguity consists of exponential-polynomials, which lie in $\mathfrak{B}$.

Step 2 (c)

By B27, $\Delta(\partial_x)u_j = F_j \in \mathfrak{B}$. B45(L4) gives $w \in \mathfrak{B}$ with $\Delta(\partial_x)w = F_j$; then $h = u_j - w \in \ker\Delta(\partial_x) = \mathcal{D}'(I) \otimes V_\Delta$, and $(\partial_t - a_j)h \in \mathfrak{B}$ turns the coefficients into solutions of a linear ODE system in t with $\mathfrak{B}$ data, hence of geometric growth.

Step 3 (P3)

Changing decomposition adds $(w_j) \in \mathcal{N}$, which changes each interaction term linearly by an exponential-polynomial.

Why it works. the Vandermonde formula expresses each component through derivatives of u, up to an ODE in x that preserves geometric growth.

■

Uses: B27, B45.

Dim. B48 **P1-P4**

Second-order factors: propagator and Gauss form [proved]

For $Q = \tau^2 + b(\xi)\tau + c(\xi)$ and $u \in \mathfrak{B}_\rho \cap \ker Q$: (a) $U = (u, \partial_t u)$ satisfies $U(t) = e^{(t-t_0)C(\partial_x)}U(t_0)$, with e^{tC} given by the canonical fundamental system φ_1, φ_2 of $\varphi'' + b(\zeta)\varphi' + c(\zeta)\varphi = 0$; (b) for $|t| < T^*(\rho')$, with $T^* \geq \log 2/K(\rho')$, the Gauss factorisation

$$e^{tC} = \begin{pmatrix} 1 & 0 \\ \varphi_1'/\varphi_1 & 1 \end{pmatrix} \begin{pmatrix} \varphi_1 & 0 \\ 0 & e^{-tb}/\varphi_1 \end{pmatrix} \begin{pmatrix} 1 & \varphi_2/\varphi_1 \\ 0 & 1 \end{pmatrix}$$

holds. **(II-T7)** Mode by mode, the Wei–Norman (Gauss) form is global iff $\varphi_1(t;\zeta)$ has no zero.

Proof.

Step 1 (a)

B45(L3) for systems; the entries of $e^{tC(\zeta)}$ solve the scalar ODE in t with the canonical initial values.

Step 2 (b)

Multiply out the three matrices and use the Wronskian $\varphi_1\varphi_2' - \varphi_1'\varphi_2 = e^{-tb}$. An entire function without zeros on $|\zeta| \le \rho'$ is invertible in the closed algebra generated by ∂_x; $\varphi_1(t;\zeta) = 1 + O(t^2)$ has no zero for $|t| \le \log 2/K$.

Why it works. a 2×2 matrix factors into lower, diagonal and upper parts exactly where its top-left entry does not vanish.

■

Uses: B45.

7.3 Non-commuting terms

Dim. B49 **P1-T4**

Solvable Lie algebra: exact global formula [proved] (Lie's theorem [from cited source])

If the terms generate a solvable Lie algebra $\mathfrak{g}$ of dimension d, then for every $A \in \mathfrak{g}$ there are entire functions $g_1, \dots, g_d$, obtained by d quadratures, with $e^{tA} = \prod_{i=1}^{d} e^{g_i(t)X_i}$; hence $E_{\mathrm{int}}^{\mathrm{mult}}$ has a closed form with d scalar functions.

Stated in the paper as no. 8.7, p. 48

Proof.

Step 1 A flag of ideals

By Lie's theorem there is a basis with $\mathfrak{i}_k = \mathrm{span}\{X_k, \dots, X_d\}$ ideals [from cited source]; ad_{X_j} acts on $\mathfrak{i}_k/\mathfrak{i}_{k+1}$ by a scalar λ_{jk}.

Step 2 A triangular Wei–Norman system

By B45(L5), $\mathrm{Ad}(P_{i-1})X_i = \sum_{l \ge i} M_{il} X_l$ with entire M_{il} and $M_{ii} = \exp(\sum_{j<i} \lambda_{ji} g_j) \neq 0$. The system $\sum_{i \le l} g_i' M_{il} = c_l$ is triangular: $g_1 = c_1 t$, and each g_l is a quadrature of an entire function.

Why it works. a solvable algebra can be climbed one ideal at a time, each step a scalar ODE.

■

Uses: B45.

Dim. B50 **P1-P5**

Non-solvable: local only [proved]

For every finite-dimensional $\mathfrak{g}$ the Wei–Norman factorisation exists with analytic germs g_i near $t = 0$; globality can fail: in $\mathfrak{sl}_2$ with $A = E - F$ in Gauss order, $g_{21}(t) = -\tan t$. **(P1-C1)** The earlier claim "infinite Lie algebra ⇒ no finite factorisation of $E_{\text{int}}^{\text{mult}}$" is false as stated [false]: $E_{\text{int}}^{\text{mult}} = e^{-tT_2}e^{-tT_1}e^{t(T_1+T_2)}$ is always a product of three exponentials.

Proof.

Step 1 Local existence

The map $g' \mapsto \sum_i g_i' \mathrm{Ad}(P_{i-1}(g))X_i$ is given by an entire matrix $\Xi(g)$ with $\Xi(0) = I$; the analytic ODE $g' = \Xi(g)^{-1}c$ has a local solution.

Step 2 The example

Direct computation in Mat_2: the Gauss coordinates of $e^{t(E-F)}$, a rotation, involve $\tan t$.

Step 3 (C1)

Definition of $E_{\text{int}}^{\text{mult}}$; the right question is A24.

Why it works. coordinates of the second kind exist near the identity but can run into singularities.

■

Uses: B45.

7.4 Boundaries

Dim. B51 **P1-P6**

A boundary creates an interaction of order one [proved]

On $L^2(0, L)$ with Dirichlet diffusion $A_1 = D\partial^2$, inflow transport $A_2 = -c\partial$ and the advection–diffusion generator A: $S_2(t) = 0$ for $t \geq L/c$, while $S(t)$ is injective. So for $t \geq L/c$, $E_{\text{int}}^{\text{mult}}(t) = S(t)$: the relative splitting error is 1 for every nonzero datum.

Stated in the paper as no. 8.6, p. 48

Proof.

Step 1 The transport empties the interval

$S_2(t)f(x) = f(x - ct)\mathbf{1}_{x>ct}$, which is 0 for $ct \geq L$.

Step 2 The full flow is injective

A is similar, by $e^{-cx/2D}$, to a self-adjoint operator, so $S(t)$ is an analytic semigroup of self-adjoint type, injective.

Why it works. the boundary condition belongs to the sum of the terms and cannot be shared between them.

■

Uses: none.

Dim. B52 **OPA-T2**

Boundary interaction: sharp rate [proved]

For advection–diffusion with Dirichlet conditions on $(0, L)$ and a datum $u_0 \in C^{k+1}$ vanishing at the ends, with contact order k (first nonzero derivative at an endpoint):

$$\|E(t)\| = K_k\, t^{k/2+3/4}\big(1 + O(t^{1/2})\big), \qquad K_k = |c|\, N_{k-1}\, D^{k/2-1/4}\, \frac{\sqrt{u_0^{(k)}(0)^2 + u_0^{(k)}(L)^2}}{(k-1)!},$$

for the first-order interaction (OPA-T2) and, with $\tilde{N}_k = kN_{k-1}$, for the full $E_{\mathrm{int}}^{\mathrm{mult}}$ (OPF5A-T5). Uniformly: $\|E(t)\| \le 1.301\,|c|D^{1/4}\|u_0'\|_\infty t^{5/4}$. Flat data give $O(t^p)$ for every p. **(P3-T3)** The first-order matrix elements are $V_{mn} = -4c\,mn/(L(m^2-n^2))$ for $m+n$ odd and 0 otherwise (a reflection symmetry).

Proof.

Step 1 A commutator identity

The first-order interaction is $\int_0^t S_1(t-s)[\,\cdot\,]S_1(s)\,ds$ applied to Vu_0; the commutator of $V = -c\partial$ with the Dirichlet heat flow is a boundary term involving u_0' at the ends (OPA-L2).

Step 2 Localisation at the ends

Near each end the Dirichlet heat kernel is that of the half-line, and $u_0(y) = \gamma y^k/k! + O(y^{k+1})$.

Step 3 Scaling

With $x = \sqrt{Dt}\,w$ the boundary term becomes $t^{k/2+3/4}$ times a universal profile, whose L^2 norm is N_{k-1}; the corrections are $O(t^{1/2})$ relative.

Step 4 Full interaction

A conjugation reduces $E_{\mathrm{int}}^{\mathrm{mult}}$ to the same boundary terms plus a remainder $O(t^{k+5/4})$ and an exponentially small image term (OPF5A-L5).

Why it works. the boundary interaction is born where the datum meets the wall, and the heat flow sees the wall at distance $\sqrt{Dt}$.

■

Uses: A23; research files `work/OP-ANA` §3, `work/OP-F5A` (L5, T5).

7.5 Where the interaction is continuous

Dim. B53 **OPA-T3**

The exact frontier of discontinuity [proved]

For an evolution $\tau - \sum_n a_n(\xi)$ on a bounded rectangle: (b) if the terms are at least two distinct non-constant monomials, $u_0 \mapsto E_{\mathrm{int}}^{\mathrm{add}}$ is unbounded from $(\mathfrak{B} \cap \ker Q, H^m)$ to L^2, for every m and every base time. **(OPA-P3)** If the terms are affinely proportional, $a_n = \theta_n a + \kappa_n$ with real θ_n, and $t_0 + \theta_n(t - t_0) \in \bar{I}$ (TC), then

$$E_{\mathrm{int}}^{\mathrm{add}}(u)(t) = u(t) + (N-1)u(t_0) - \sum_n e^{(t-t_0)\kappa_n} u\big(t_0 + \theta_n(t-t_0)\big),$$

which is continuous in H^1; without (TC) it is unbounded. The general non-proportional case (e) is proved modulo the Turán–Nazarov inequality [from cited source].

Proof.

Step 1 Exponential test data

On $u = e^{a(\xi)t + \xi x}$ with real ξ the interaction is $e_{\geq 2}(z)\, u$, $z_n = e^{s a_n(\xi)} - 1$ (A19). If some term grows faster than the full symbol along a ray of ξ (condition (S)), the ratio to any H^m norm grows exponentially, as in A21.

Step 2 Proportional terms

$e^{s a_n(\partial)} = e^{s\kappa_n} e^{s\theta_n a(\partial)}$, and $e^{\tau a(\partial)} u_0 = u(t_0 + \tau)$ for $t_0 + \tau \in \bar{I}$: the flows of the terms are time-rescaled copies of the true solution, controlled by $\|u\|_{H^1}$.

Why it works. the interaction is continuous only when every term is a rescaled copy of the whole equation.

■

Uses: A19, A21, B43.

Dim. B54 **P3-T6**

Balance decomposition [proof written, not audited]

For $u \in H^m(\Omega)$ with $(\partial_t - \sum_i a_i(\partial_x))u = 0$, $m \geq \max(\deg a_i, 1)$, put $u^{(i)} = \int_{t_0}^{t} a_i(\partial_x)u(s)\,\mathrm{d}s$. Then

$$u = u_0 \otimes 1 + \sum_i u^{(i)}, \qquad \|u^{(i)}\|_{H^{m-d_i}} \leq C\|u\|_{H^m}.$$

In $\mathfrak{B}$ the $u^{(i)}$ solve a coupled Faddeev system whose Neumann series converges on every bounded time interval, with $\|\mathbb{K}^n\| \leq (\kappa_0 |I|)^n / n!$.

Proof.

Step 1 Balance

Integrate the equation in time: $u(t) - u_0 = \sum_i \int_{t_0}^{t} a_i(\partial_x)u$. Each term loses d_i derivatives in x and gains one in t.

Step 2 Faddeev system

$\partial_t u^{(i)} = a_i(\partial)u = a_i(\partial)(u_0 + \sum_j u^{(j)})$, so $u^{(i)} = G_i a_i(\partial)(u_0 + \sum_{j\neq i} u^{(j)})$ with the Green operator G_i of $\partial_t - a_i$; the off-diagonal Volterra operators have the factorial bound.

Why it works. a split that uses the true solution, not the single-term flows, inherits its continuity.

■

Uses: B44.

7.6 Regularisation of the term split

Dim. B55 **P2-T1**

Every regulariser has the same limit residual [proved]

Let $S : W \to H$ be linear with closure of the range M, R any regulariser with $S(\operatorname{dom} R)$ dense in M, and ω_α near-minimisers of $\frac{1}{2}\|S\omega - u\|^2 + \alpha R(\omega)$ (errors $\varepsilon_\alpha \to 0$). Then $u - S\omega_\alpha \to P_M^{\perp} u$, so the limit residual is $\operatorname{dist}(u, M)$ for every regulariser. **(P2-L1)** If Q kills M, then $\|E^{\perp}(u)\|_{H^m} \geq \|Q(\partial)u\|_{H^{m-k}}/C_Q$.

Proof.

Step 1 Upper bound

For any $v \in \operatorname{dom} R$, $\frac{1}{2}\|S\omega_\alpha - u\|^2 \leq \frac{1}{2}\|Sv - u\|^2 + \alpha R(v) + \varepsilon_\alpha$; let $\alpha \to 0$, then choose Sv close to $P_M u$.

Step 2 Lower bound

$\|u - S\omega\| \geq \operatorname{dist}(u, M)$ always; Pythagoras in $u = P_M u + P_M^{\perp} u$ gives norm convergence.

Step 3 (L1)

$Q(\partial)u = Q(\partial)E^{\perp}(u)$ and $Q(\partial)$ is bounded $H^m \to H^{m-k}$.

Why it works. a regulariser only chooses among representations; it cannot reach what lies outside their closure.

■

Uses: none.

Dim. B56 **P2-P3**

The regulariser acts only on the ambiguity [proved]

The exact representations of u form $\omega^\dagger + \ker S$. For convex, lower semicontinuous, coercive R the regularised solutions converge (along subsequences) to an R-minimal exact representation, to $S^\dagger u$ for $R = \frac{1}{2}\|\cdot\|^2$. On a rectangle $\ker S$ has dimension $\delta(P)$ for factors and $\sum_{n<m} \deg(a_n - a_m)$ for terms. **(P2-P2)** For ODEs, the sum of the per-term kernels is $\ker \operatorname{lcm}(p_n)(D)$.

Proof.

Step 1 Bounded minimisers

$R(\omega_\alpha) \le R(\omega^\dagger) + \varepsilon_\alpha/\alpha$ and coercivity bound ω_α; weak limits are exact and R-minimal by weak lower semicontinuity (Mazur).

Step 2 Dimensions

A13(ii) for factors; for terms $\mathcal{R}/(\tau - a_n, \tau - a_m) \cong \mathbb{C}[\xi]/(a_n - a_m)$.

Step 3 (P2)

The roots of the lcm, with maximal multiplicities, give a basis contained in the union of the per-term kernels.

Why it works. all methods agree up to a finite-dimensional gauge, where the regulariser picks a representative.

■

Uses: A13.

Dim. B57 **P2-P4**

Linear encoders do not help [proved]

(a) A bounded, boundedly invertible linear encoder E preserves closedness of sums and distances up to the constants m_E, $\|E\|$. (b) An injective compact encoder can make a non-dense M have dense image, but then the encoded components diverge.

Proof.

Step 1 (a)

$m_E\|u - v\| \le \|Eu - Ev\| \le \|E\|\|u - v\|$; isomorphisms preserve closed subspaces.

Step 2 (b)

With $Ee_k = e_k/k$ and $M = w^\perp$, $w \propto \sum_k e_k/k$: $y \perp EM$ forces $Ey \in \mathbb{C}w$, impossible in ℓ^2 unless $y = 0$. If $Em_k \to Eu$ with $u \notin M$ and m_k bounded, a weak limit would give $u \in M$.

Why it works. stable encodings preserve geometry; unstable ones hide it by sending pieces to infinity.

■

Uses: none.

Dim. B58 P2-P5

Where the interaction is stable [proved]

(a) In $\mathfrak{B}_\rho$: $\sup_t \|E_{\mathrm{int}}(t)\|_{B_{\rho'}} \le e_{\ge 2}(\beta)\|u_0\|_{B_{\rho'}}$ with $\beta_n = e^{|I||a_n|(\rho')} - 1$. (b) On $\mathbb{R}$ with L^2 data: $\|E_{\mathrm{int}}(t)\|_{L^2} \le e_{\ge 2}(\beta)\|u_0\|$ with $\beta_n = \sup|e^{ta_n(i\kappa)} - 1|$; for advection–diffusion $\|E_{\mathrm{int}}\| \le 2\|u_0\|$.

Proof.

Step 1 (a)

A19(ii) and $\|e^{sa_n} - I\| \le e^{|s||a_n|(\rho')} - 1$; $e_{\ge 2}$ has nonnegative coefficients.

Step 2 (b)

Plancherel: $\widehat{E_{\mathrm{int}}} = e_{\ge 2}(z)\hat{u}_0$ and $|e_{\ge 2}(z)| \le e_{\ge 2}(|z|)$. For advection–diffusion $|z_1| \le 2$, $|z_2| \le 1$.

Why it works. stability of the interaction is a property of the data class, not of the method.

■

Uses: A19, B44.

Dim. B59 P3-C1

The hypotheses of Dyson–Phillips cannot be dropped [proved]

(a) Same order ($A_0 = \partial^2$, $V = -2\partial^2$): the Dyson terms have norm $(2n)^n e^{-n}/n! \sim 2^n/\sqrt{2\pi n}$ and $A_0 + V$ generates no semigroup. (b) Dispersive principal term ($A_0 = i\partial^2$, $V = i\partial$): $A_0 + V$ generates no semigroup. (c) $A_0 = i\partial^2$, $V = \partial$: the sum generates a group but the Dyson terms are unbounded.

Proof.

Step 1 Multipliers

All operators are Fourier multipliers; in (a) $\sup_y (2ty)^n e^{-ty}/n! = (2n)^n e^{-n}/n!$.

Step 2 Non-generation

The spectrum of a multiplier is the closure of its symbol's range; in (a), (b) the real part is unbounded above, so by Hille–Yosida there is no generator.

Why it works. without smoothing by the principal term, each perturbation step costs a full derivative.

■

Uses: A23.

Dim. B60 **OPF-T2**

Repeated factors: the multiplicity term [proof written, not audited]

For $Q = (\tau - a(\xi))^e$ and $u \in \mathfrak{B}$, $U = (u^{(0)}, \dots, u^{(e-1)})$ with $u^{(k)} = (\partial_t - a)^k u$ satisfies $U(t) = e^{s(a(\partial_x)I + N_e)}F$ with the nilpotent shift N_e; the multiplicity term commutes with the others, its flow is polynomial in s (secular growth), and $E_{\text{int}}^{\text{mult}} = 0$. **(OPF-T1)** For every P, every factor decomposition of $u \in \mathfrak{B}$ lies in $\mathfrak{B}$.

Proof.

Step 1 (T2)

$\partial_t U = (a(\partial_x)I + N_e)U$, a system with commuting constant-coefficient generator; B45(L3) for systems.

Step 2 (T1)

For each j, B26(c) gives $r(\partial_x)u_j$ and $s(\partial_t)u_j$ in $\mathfrak{B}$; B45(L4) inverts $r(\partial_x)$ inside $\mathfrak{B}$, and the remaining finite-dimensional part is controlled as in B47.

Why it works. a repeated factor adds a nilpotent term, which commutes with everything and grows polynomially.

■

Uses: B26, B45, B47.

7.7 Results used without proof here

TROP-3′, TROP-4	Uniform-in-time bound of the interaction on dissipative faces of the Newton polygon; no uniform bound on conservative faces.	[proved]

Proofs in `work/SCOUT-TROP/VERDETTO.md` (audited); not reproduced. TROP-5..8: [proof written, not audited].

R5-P6, R5-T4	Global Wei–Norman for Ornstein–Uhlenbeck; Duhamel with bound.	[proved]

Proofs in `work/CHECK/DECLASSAMENTI.md` §B; not reproduced.

P3-T2, P3-T4, P3-T5	Path formula for Dyson terms; closed form for an εx perturbation; Appell and Rosenbloom–Widder duals.	[proved] / [proof written, not audited]

Proofs in `work/P3`; not reproduced.

OPA-P4	Splitting error at full coupling equals the first-order term to relative $O(t^{1/2})$.	[proved]

Now contained in B52 (OPF5A-T5).

P1-Q6	No-go of continuity for every operator with at least two terms.	closed

By B53 (monomials) and B53(e) mod Turán–Nazarov.

P1-Q3, **P1-Q7**	No finite product of flows for ∂^2 and x^k, $k \geq 3$; A24 without the switching-off hypothesis.	[open]

CHAPTER 8

Phase 4: nonlinear equations

In this chapter	
Starting from	Part I, Chapter 4 (A25–A31), and its notation: Burgers flows S, S_B, S_A, the Lagrangian map X, J, the Hopf–Lax functional Φ_x, and the dictionary setting.
Why	to prove the tools behind A26–A28 (comparison principle, pre-shock bounds), the blow-up at the shock time, the jumps after the shock, the $\sqrt{\nu}$ splitting error, the KdV–Burgers remainder, and the remaining identification results (the negative theorem for complete dictionaries, closed forms, the degree bound for Burgers).
Role	Dim. B61–B72, followed by the list of nonlinear results whose long proofs are not reproduced here.

8.1 Tools before the shock

Dim. B61 — **S4T-L0**

Comparison principle [proved]

Let b, c be continuous and real on $[0,T]\times\mathbb{T}$, $|c|\le K$, $\nu\ge 0$, and $Lg = g_t + bg_x + cg - \nu g_{xx}$. (a) If $Lg \ge 0$ and $g(0)\ge 0$ then $g\ge 0$. (b) If $|Lg|\le F(t)$ then

$$\|g(t)\|_\infty \le e^{Kt}\Big(\|g(0)\|_\infty + \int_0^t F\Big).$$

Proof.

Step 1 (a) First touching point

For $\theta>0$ put $z = g + \theta e^{(K+1)t}$, so $Lz>0$ and $z(0)>0$. At a first point (t_1,x_1) with $z=0$: $z_t\le 0$, $z_x=0$, $z_{xx}\ge 0$, so $Lz\le 0$, a contradiction. Let $\theta\to 0$.

Step 2 (b)

$Y(t) = e^{Kt}(\|g(0)\| + \int_0^t F)$ satisfies $Y' \ge KY + F$; apply (a) to $Y \mp g$.

Why it works. a parabolic operator cannot create a new minimum.

■

Uses: none.

Dim. B62 **S4T-L1**

Derivative bounds before the shock [proved]

For real nonconstant u_0, $M_0 = \max u_0'$, $0 < T < t^*$, and w the viscous or inviscid Burgers solution: $-1/(t^* - t) \le \partial_x w \le M_0$, $\|w\|_\infty \le \|u_0\|_\infty$, and $\|\partial_x^j w(t)\|_\infty \le Y_j(t)$ with Y_j independent of ν.

Proof.

Step 1 The slope

$p = \partial_x w$ solves $p_t + wp_x + p^2 - \nu p_{xx} = 0$. Apply B61(a) to $M_0 - p$ (source $p^2 \ge 0$) and to $p - \vartheta$ with $\vartheta = -1/(t^* - t)$, which solves $\vartheta' = -\vartheta^2$.

Step 2 Higher derivatives

$p_j = \partial^j w$ solves $p_{j,t} + wp_{j,x} + (j+1)p_1 p_j - \nu p_{j,xx} = -\sum_{i=2}^{j-1} \binom{j}{i} p_i p_{j+1-i}$; B61(b) and induction give ν-independent bounds.

Why it works. before the shock every derivative obeys a transport equation with bounded coefficients.

■

Uses: B61.

Dim. B63 **S4T-P3**

The closed form blows up at the shock time [proof written, not audited]

If u_0' has a unique minimum a^* with $\kappa = u_0'''(a^*) > 0$ and $\eta = 1 - t/t^*$, the first-order coefficients of A26 satisfy, as $t \uparrow t^*$,

$$\eta^{3/2} \sup_x |E_1^{\mathrm{add}}(t)| \to \frac{3\sqrt{3}}{16}\sqrt{2t^*\kappa}.$$

Proof.

Step 1 Local form

With $y = a - a^*$: $u_0''(a) = \kappa y(1 + O(y))$ and $J = \eta + \frac{t\kappa}{2} y^2(1 + O(y))$.

Step 2 Scaling

With $y = z\sqrt{\eta/c_t}$, $c_t = t\kappa/2$, the quantity $|u_0''|/J^2$ equals $\frac{\kappa}{\sqrt{c_t}} \eta^{-3/2} z/(1+z^2)^2$ up to $1 + O(\eta^{1/4})$; the maximum of $z/(1+z^2)^2$ is $3\sqrt{3}/16$. Away from the window the function is smaller.

Why it works. the Jacobian vanishes quadratically at the cusp, and the coefficient is a ratio of a linear to a squared quadratic quantity.

■

Uses: A26.

8.2 After the shock

Dim. B64 **S4T-T3**

After the shock: jumps and the splitting order [proved mod. [F1]]

For smooth nonconstant u_0, $t > t^*$ and every $\nu > 0$: (i) $E_{\text{int}}^{\text{add}}(t)$ has a jump $+D$ at each shock, independent of ν, so $\|E_{\text{int}}^{\text{add}}\|_\infty \geq D/2$ and $E_{\text{int}}^{\text{add}} \notin H^{1/2}$; (ii) the same for the splitting error B then A; (iii) the splitting error A then B is smooth but $\liminf_\nu \nu^{-1/2}\|E_{AB}^{\text{mult}}\|_{L^1} \geq 2D\sqrt{t/\pi}$.

Proof.

Step 1 (i)

$E_{\text{int}}^{\text{add}} = S(t)u_0 - S_B(t)u_0 - S_A(t)u_0 + u_0$: the viscous, heat and initial terms are smooth, while $S_B(t)u_0$ jumps by $-D$ at the shock (Lax–Oleinik).

Step 2 (iii)

$E_{AB}^{\text{mult}} = S(t)u_0 - S_A(t)S_B(t)u_0$; heat-smoothing a jump of size D during time t leaves an error of L^1 size $\geq 2D\sqrt{\nu t/\pi}$, and the viscous solution is $O(\nu|\log\nu|)$-close to the shock profile.

Why it works. the inviscid term alone produces a discontinuity that no amount of viscosity in the other term can remove.

■

Uses: [F1], [F2].

Dim. B65 **OPF5D-T6**

The vanishing-viscosity L^1 bound [proved] (mod [F1] for the corollary)

For every $t, \nu > 0$,

$$\|S(t)u_0 - v(t)\|_{L^1(\mathbb{T})} \leq 8\pi\sqrt{\frac{2\nu\log(1 + t\|u_0'\|_\infty)}{t}}.$$

Consequently $\|E_{AB}^{\text{mult}}(t)\|_{L^1} = \Theta(\sqrt{\nu})$ for $t > t^*$, and $\|E_{\text{int}}^{\text{add}}(t)\|_{L^1} = O(\sqrt{\nu})$.

Stated in the paper as no. 9.2.2, p. 53

Proof.

Step 1 **Convex potentials**

$G_\nu(x) = 2\nu \log \int \exp\left((xy/t - h(y))/2\nu\right) \mathrm{d}y$ and $G_0(x) = \sup_y(xy/t - h(y))$, $h = U_0 + y^2/2t$, are convex, and $u - v = G_0' - G_\nu'$.

Step 2 **Laplace gap**

$G_\nu - G_0 = 2\nu \log I(x)$ with $I \geq c_0 = \sqrt{4\pi\nu/M}$, $M = \|u_0'\|_\infty + 1/t$; put $\varepsilon = G_\nu - c - G_0 \geq 0$.

Step 3 **Strong convexity and averaging**

G_0 is $(1/t^2M)$-strongly convex, so each $D_x(y)$ is quadratic-from-below in x; Fubini over a period and Jensen give $\int_{\mathbb{T}} \varepsilon \leq 4\pi\nu \log(tM)$.

Step 4 **Derivatives of close convex functions**

With $f = G_\nu - c$, $g = G_0$, $f - g = \varepsilon \geq 0$ and $h > 0$, convexity gives

$$|f' - g'|(x) \leq g'(x+h) - g'(x-h) + \frac{\varepsilon(x+h) + \varepsilon(x-h)}{h}.$$

Over one period $\int (g'(x+h) - g'(x-h))\,\mathrm{d}x = 4\pi h/t$, so

$$\|u - v\|_{L^1} \leq \frac{4\pi h}{t} + \frac{8\pi\nu \log(tM)}{h},$$

and $h = \sqrt{2\nu t \log(tM)}$ gives the bound.

Step 5 **The splitting error**

$E_{AB}^{\mathrm{mult}} = (u - v) - (S_A v - v)$ with $\|S_A v - v\|_{L^1} \leq 2\sqrt{\nu t/\pi}\,\mathrm{TV}(v(t))$; the lower bound is B64(iii).

Why it works. the viscous and inviscid solutions are derivatives of two convex potentials that differ by a Laplace correction of size ν log.

■

Uses: [F1], [F2].

Dim. B66 **OPR2-T5**

KdV–Burgers before the shock

[proved mod. local well-posedness]

For $u_t + uu_x = \nu u_{xx} - d u_{xxx}$, $k \geq 2$, $T < t^*$, and small $\nu + |d|$:

$$\sup_{t \leq T} \left\| u(t) - v(t) - \nu\delta(t) - d\,\delta_d(t) \right\|_{H^k} \leq C_k(\nu + |d|)^2,$$

so $E_{\mathrm{int}}^{\mathrm{add}}$ has an explicit first-order form in (ν, d).

Proof.

Step 1 **Energy estimate for $e = u - v$**

$e_t + (ve)_x + ee_x - \nu e_{xx} + de_{xxx} = \nu v_{xx} - dv_{xxx}$. In H^k the dispersive term is antisymmetric, the viscous term dissipates, and the transport terms are bounded by $C\|e\|_{H^k}^2$.

Step 2 **Bootstrap**

$(\sqrt{Y})' \le C\sqrt{Y} + G(\nu + |d|)$ keeps $\|e\|_{H^k} \le C(\nu + |d|)$ on $[0, T]$; the same for $e - \nu\delta - d\delta_d$ gives the square.

Why it works. before the shock both small terms are regular perturbations of the inviscid flow.

■

Uses: B62.

Dim. B67 **S4T-P5**

Fisher–KPP: the jet [proved]

For $u_t = \nu u_{xx} + ru(1-u)$ with terms $A = \nu u_{xx}$, $B = ru(1-u)$:

$$[s^2]\,\hat{E}^{\mathrm{add}} = \tfrac{1}{2}\big(A'[B] + B'[A]\big) = r\nu\big(u_0'' - u_0'^2 - 2u_0u_0''\big);$$

the estimate of the remainder is not uniform in time.

Proof.

Step 1 **The two derivatives**

$A'[w] = \nu w_{xx}$, $B'[w] = r(1-2u_0)w$. So $A'[B] = r\nu(u_0 - u_0^2)'' = r\nu(u_0'' - 2u_0'^2 - 2u_0u_0'')$ and $B'[A] = r\nu(1-2u_0)u_0''$.

Step 2 **Half the sum**

$\frac{1}{2}\big(A'[B] + B'[A]\big) = r\nu\big(u_0'' - u_0'^2 - 2u_0u_0''\big)$.

Why it works. A25 at order two.

■

Uses: A25.

Example 8.1 **The reference datum $u_0 = \varepsilon \sin x$** (OPA-C1)

Here $t^* = 1/\varepsilon$, there is one shock at $x = \pi$ for $t > t^*$, symmetric ($\rho = 0$), and A27 gives

$$\Lambda(t) = 2\log\frac{1}{\varepsilon t - 1} + 4\log 2 + o(1) \qquad (t \downarrow t^*),$$

consistent with the uniform rate of A28.

8.3 Identification in dictionaries

Dim. B68 **S4ID-P1**

The literal statement holds only in the minimal dictionary [proved]

For $F = y_t - \sigma y_n - N'$ with $\sigma \neq 0$ and $\deg N = 2$ (Burgers, Fisher–KPP, KdV, KdV–Burgers): (a) $F \in V_{d,k}$ iff $d \geq 2$ and $k \geq n$; (b) if $(d,k) \neq (2,n)$ then $\dim([F] \cap V_{d,k}) \geq 2$; (c) $[F] \cap V_{2,n} = \mathbb{C}F$.

Proof.

Step 1 (b) Two independent consequences

If $k \geq n+1$: $\partial_x F \in V_{d,k}$ contains y_{n+1}, which F does not. If $d \geq 3$: yF contains $y\,y_t$, which F does not.

Step 2 (c) Elimination

In $V_{2,n}$, the generators involving time derivatives other than y_t have normal forms of distinct orders $na+b \geq n+1$ with nonzero leading coefficients σ^a, so they must cancel; what remains is a multiple of F by degree count.

Why it works. a complete dictionary always contains derivatives or multiples of the equation.

■

Uses: A29.

Dim. B69 **S4ID-T4**

Closed forms do not identify; identifying solutions exist [proof written, not audited] ((iii) mod Hölder, Bona–Smith [from cited source])

(i) If u lies in a finite tower of field extensions of $\mathbb{C}(t,x)$ (closed forms), then $u, u_x, \dots, \partial_x^r u$ are algebraically dependent, so $\mathcal{J}_u \neq 0$ and u does not identify its equation in any dictionary containing that relation. (ii) For $u = P/Q$ in t, x and exponentials, an explicit degree $D^* = (n+1)c^n - n$ of such a relation exists. (iii) For a datum g that satisfies no algebraic ODE (for example Γ on an interval), the Burgers or KdV solution has $\mathcal{J}_u = 0$ and identifies its equation modulo consequences in every dictionary.

Proof.

Step 1 (i) Transcendence degree

Each step of the tower adds at most a bounded transcendence degree, so the field generated by u and its x-derivatives has finite transcendence degree r; $r+1$ derivatives are dependent. A29(c) concludes.

Step 2 (iii) A transcendental datum

At $t = 0$ any relation in the x-jet would be an algebraic ODE for g; by continuity a relation for u restricts to one for g. Hölder's theorem says Γ satisfies none [from cited source].

Why it works. closed forms live on finite-dimensional varieties of jets; differentially transcendental data do not.

■

Uses: A29.

Dim. B70 **OP-T7**

Burgers N-shocks: no relation of degree $\leq N$ [proved]

Under the hypotheses of A30, no nonzero polynomial of degree $\leq N$ in the jet of u (of any order) vanishes on u. Hence, for every order $K \geq N$, u identifies Burgers in the evolutionary dictionary iff $d \leq N$.

Stated in the paper as no. 9.3, p. 54

Proof.

Step 1 Cumulants

By A30, Steps 1–2, the jet of w is the sequence of cumulants κ_n of the moments r_j, polynomial in $r_1, \dots, r_N$, which are algebraically independent.

Step 2 Weights

Give r_j weight j. The top-weight part of κ_n is a truncated cumulant; a relation $P(\kappa) = 0$ of degree $\leq N$ has a nonzero top-weight part that must vanish identically.

Step 3 Confluent limit

The top-weight parts are cumulants of a measure with N atoms; in the confluent limit they become power sums of N variables, and power sums $p_1, \dots, p_N$ are algebraically independent (fundamental theorem of symmetric polynomials), which excludes any relation of degree $\leq N$.

Why it works. N atoms leave N independent power sums, so the first relation needs degree $N + 1$.

■

Uses: A30.

Dim. B71 **F4L-P3**

Analyticity radius of Burgers in the amplitude

[proof written, not audited]

For Burgers with datum $\varepsilon U_0'$, the solution is analytic in ε for $|\operatorname{Im}\varepsilon| < 2\pi\nu/\operatorname{osc}(U_0)$, uniformly in time, and this is optimal. **(KIEFFER-C)** The "infinite radius" claimed in a cited preprint is not proved there: its own formula has denominator zeros at finite ε.

Proof.

Step 1 A moment generating function

By Cole–Hopf, $\varphi_\varepsilon(t,x) = \mathbb{E}\, e^{sX}$ with $X = U_0(Y)$, Y distributed by the heat kernel, $s = -\varepsilon/2\nu$; X takes values in an interval of length $w = \operatorname{osc} U_0$.

Step 2 The strip lemma

For a random variable with values in an interval of length w, $M(s) = \mathbb{E}e^{sX}$ has no zeros for $|\operatorname{Im} s| < \pi/w$ (L-S), and $u = -2\nu\partial_x \log\varphi$ is analytic there. The symmetric Bernoulli law attains the bound (the tanh case).

Why it works. Cole–Hopf turns Burgers into the logarithm of a moment generating function, whose zeros are kept away by the bounded range.

■

Uses: [F2].

Dim. B72 **S4ID-L4**

Burgers in Hammerstein form

[proof written, not audited]

With $\mathcal{R}$ acting on differential polynomials, and $F_B = (\tau - \nu\xi^2)y + \frac{\xi}{2}y^2$: $[F_B] \cap (\mathcal{R}y \oplus \mathcal{R}y^2) = \mathcal{R}F_B$.

Proof.

Step 1 Degree splitting

Elements of $[F_B]$ are $\sum_j L_j\, \partial^j F_B$ with differential-polynomial coefficients; those of degree ≤ 2 in y with linear–quadratic structure must have constant-coefficient multipliers, because any y-dependent coefficient raises the degree.

Why it works. in the Hammerstein form the linear and quadratic parts cannot be mixed without leaving the space.

■

Uses: A29.

8.4 Nonlinear results whose proofs are not reproduced

The following results are proved in the research files named in the last column. Their proofs rely on long computations (scattering theory, Hirota bilinear forms, computer-assisted syzygies). They are listed with their labels so that the catalogue is complete as a list; the proofs are not reproduced in this edition.

INT-T1	KdV in the class $\mathcal{H}_M$: identification by the scattering data; (A) and unique continuation [proved]; (B), (C), INT-T2 [conditional] / [proved] at fixed time (OPF5D).	see label

`work/S4-INTEG` §1.1, `work/OP-F5D`.

INT-P5, INT-P6	Hirota N-solitons: the interaction is pairwise, integrates to 0, costs $O(N^3)$.	[proved]

`work/S4-INTEG`.

OP-P17, OP-P18	Lax and zero-curvature relations for all n (sign at $n = 0$ corrected).	[proved]

`work/OP-F5D` (OPF5D-T1, T2).

OPF5D-R2	NLS identification by scattering data without multiplicity.	[false] mod [DP]

`work/OP-F5D`.

OPF5D-T7	Lax–Oleinik equals Kružkov entropy solution ([F1]).	[proved mod. Kružkov]

`work/OP-F5D`.

OPF5A-T3, T4, C1	OPA-T1 with multiple points; genericity of (H_t); global uniform rate for generic data (several births [conditional]).	[proved] mod [F1]

`work/OP-F5A`.

OPF5C-T1, L1–L8, P7, P9, T6	Structure of Novikov's variety; upper bounds for all N; nine exact syzygy degrees $\delta_N(K)$ (computer-assisted).	[proved] mod (Nov′)

`work/OP-F5C`; the main line is in A31.

S4ID-L2, L5, C1, C2, P5–P7	Normal forms and elimination lemmas; evolutionary versus complete dictionaries.	[proved]

`work/S4-ID`.

OP-C11, **OP-T1**, **OP-E3**, **OP-C3′**, **OP-L2..P14**	Complete criteria for the Burgers family; KdV examples and obstructions.	[proved]

`work/OP-NL`.

ALG-T1, **MODREL-T1**, **F4LIN-T2**	Identification in the full ring for analytic u; moment relations; tropical interaction $4\nu \ln 2$.	[proved]

`work/S4-INTEG`.

OPR2-P4, **P6**, **O8**, **O9′**	Novikov relation, obstructions to INT-Q1.	[proved] / mod Novikov

`work/OP-R2`.

S4ID-P8, **OPR2-P7**, **OPF5D-P14**, **P15**	Nonlinear asymptotics with derivatives (Deift–Zhou).	[conditional]

`work/S4-ID`, `work/OP-F5D`.

INT-Q1, **OPF5C-Q1**	Analytic KdV data with reflection; general $\delta_N(K)$.	[open]

APPENDIX A

Index of results

Every result of this catalogue, sorted by its ledger identifier. “Dim.” is the catalogue number: A for Part I, B for Part II.

Identifier	Dim.	Result	Page
C3-C1	B12	The spectrum of an expansion is not intrinsic	65
C3-C2	B13	Exponential-polynomial data never identify	66
C3-L2	A1	Splitting off the gcd	3
C3-L3	A2	Exponential-polynomials are the finite-dimensional functions	4
C3-P1	A7	No finite certificate	14
C3-P3	B14	Translation symmetry	66
C3-P4	B15	Spectral criterion under uniqueness of the expansion	67
C3-P5	B16	Local data	67
C3-T1	A3	Identification criterion	7
C3-T2	A6	Bézout threshold and jet threshold	12
C3-T4	B10	Evolutions: Hermite interpolation	64
C3-T5	B11	Wave, Klein–Gordon, telegraph, beam	64
C3-T6	A4	The intrinsic operator	9
C12-C1	A15	A derivative can be lost	26
C12-L1	B22	One-dimensional lemmas on a rectangle	72
C12-L1′	A12	Twisted Poincaré lemma; no obstruction for finite modules	22
C12-L2	B23	Finite-dimensional modules on a rectangle	73
C12-L3	A11	The finite quotient of two coprime factors	21
C12-L5	B24	Functions constant along a direction on a convex set	73
C12-P1	B26	Constructive projectors	74
C12-P2	B28	Reducibility in second-order time; first order	75
C12-P4	B30	Two cases where the H^m split holds	77
C12-P5	B31	Variable coefficients: LCLM instead of the product	77
C12-P7	A17	Regularity criterion: closed and dense	29
C12-P10	A14	Holes break the decomposition	25
C12-T1	B25	Factor decomposition on rectangles and convex sets	74
C12-T1′	A13	Factor decomposition	23
C12-T2	B27	Vandermonde formula	75
C12-T3	A16	Chord criterion	27

Identifier	Dim.	Result	Page
C12-T4	B29	Interior regularity of the components	76
F4L-P3	B71	Analyticity radius of Burgers in the amplitude	105
II-C1	B9	The t-jet is a cyclic vector	63
II-L1	B2	First-order reduction as a module equivalence	58
II-P2	B4	The principal symbol sees only infinity	59
II-P5	B8	Normality of the companion matrix	62
II-T1	B3	When the companion form exists	58
II-T2	B5	Projectors of the companion matrix, and their poles	60
II-T3	B6	Polynomial projectors iff disjoint varieties	61
II-T4	B7	Only factors give single-valued projectors	62
OP-ALG-C1	B21	With variable coefficients the finite test fails	70
OP-ALG-T1	B18	Decidability in the holonomic model	68
OP-ALG-T3	B19	The minimal sufficient datum	69
OP-ALG-T4	B20	Saturated identification criterion	70
OP-T7	B70	Burgers N-shocks: no relation of degree $\leq N$	104
OPA-T1	A27	The interaction after the shock is $\Theta(\nu)$	48
OPA-T2	B52	Boundary interaction: sharp rate	91
OPA-T3	B53	The exact frontier of discontinuity	92
OPF5A-T2	A28	Uniform rate $\nu\lvert\log\nu\rvert$ across the shock time	50
OPF5B-P8	B40	The tangent model of a wedge is closed	82
OPF5B-T1	A18	Density on bounded convex regions	30
OPF5B-T2	B41	Saturated criterion for non-monic P	83
OPF5B-T6	B42	Repeated socle: no finite test	84
OPF5C-T2	A31	KdV N-solitons: degree $(N+2)2^{N-1}$	54
OPF5D-T6	B65	The vanishing-viscosity L^1 bound	100
OPF-P6	B36	Parallel families: pairwise closed iff totally closed	80
OPF-P7	B37	Separable families: the sum is always closed	81
OPF-T2	B60	Repeated factors: the multiplicity term	96
OPF-T4	B32	Cohomological criterion	78
OPF-T5	B35	Density via contracting symmetries	80
OPR2-T1	B39	Heat times transport: density	82
OPR2-T2	B38	Real transports in L^2	81
OPR2-T5	B66	KdV–Burgers before the shock	101
P1-L1	B43	Properties of $\mathfrak{B}$	85
P1-L2	B44	Flows of terms in $\mathfrak{B}$	86
P1-L3	B45	The Cauchy problem in the class	86
P1-P1	B46	Single flows are not determined without a class	87
P1-P1(e)	A21	The interaction is not continuous in H^m	37
P1-P2	B47	Factors in $\mathfrak{B}$	88
P1-P4	B48	Second-order factors: propagator and Gauss form	88

Identifier	Dim.	Result	Page
P1-P5	B50	Non-solvable: local only	90
P1-P6	B51	A boundary creates an interaction of order one	90
P1-T2	A19	Closed form of the interaction	34
P1-T3	A20	Cost: $O(N)$ versus $\Theta(2^N)$	35
P1-T4	B49	Solvable Lie algebra: exact global formula	89
P1-T5	A24	No finite product of flows for an infinite adjoint orbit	42
P2-C1	B34	Where regularising the factor split fails	79
P2-P3	B56	The regulariser acts only on the ambiguity	94
P2-P4	B57	Linear encoders do not help	94
P2-P5	B58	Where the interaction is stable	95
P2-T1	B55	Every regulariser has the same limit residual	93
P2-T2	A22	No regularisation removes the interaction	38
P2-T3	B33	Two subspaces: exact constants	79
P3-C1	B59	The hypotheses of Dyson–Phillips cannot be dropped	95
P3-T1	A23	Dyson–Phillips converges for every time	40
P3-T6	B54	Balance decomposition	92
P4-C1	B17	Restriction does not identify in the Weyl algebra	68
P4-T2	A5	Spectral theorem for the cyclic module	10
P4-T4	A8	Decidability with a bound	15
P4-T5	A9	Black-box impossibility	16
R2-T3	A10	Terms do not give components	19
R4-T1	B1	ODE identification	57
S4ID-L4	B72	Burgers in Hammerstein form	105
S4ID-P1	B68	The literal statement holds only in the minimal dictionary	103
S4ID-T1	A29	Dictionary criterion	52
S4ID-T2	A30	Burgers with N shocks	53
S4ID-T4	B69	Closed forms do not identify; identifying solutions exist	103
S4T-L0	B61	Comparison principle	98
S4T-L1	B62	Derivative bounds before the shock	99
S4T-P3	B63	The closed form blows up at the shock time	99
S4T-P5	B67	Fisher–KPP: the jet	102
S4T-T1	A25	Jet of the nonlinear interaction	45
S4T-T2	A26	Closed form before the shock	46
S4T-T3	B64	After the shock: jumps and the splitting order	100

Bibliography

[1] D. Cox, J. Little, D. O'Shea, *Ideals, Varieties, and Algorithms*, 4th ed., Springer, 2015.

[2] W. Fulton, *Algebraic Curves*, Benjamin, 1969; free edition 2008, §5.3.

[3] L. Ehrenpreis, Solution of some problems of division I, *Amer. J. Math.* 76 (1954).

[4] B. Malgrange, Existence et approximation des solutions des équations aux dérivées partielles et des équations de convolution, *Ann. Inst. Fourier* 6 (1955–56), Chap. I §2 Thm 2.

[5] V. P. Palamodov, *Linear Differential Operators with Constant Coefficients*, Springer, 1970.

[6] U. Oberst, Multidimensional constant linear systems, *Acta Appl. Math.* 20 (1990) 1–175.

[7] L. Hörmander, *The Analysis of Linear Partial Differential Operators II*, Springer, 1983.

[8] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton, 1970.

[9] K.-J. Engel, R. Nagel, *One-Parameter Semigroups for Linear Evolution Equations*, Springer, 2000.

[10] F. J. Dyson, The radiation theories of Tomonaga, Schwinger and Feynman, *Phys. Rev.* 75 (1949).

[11] J. Wei, E. Norman, Lie algebraic solution of linear differential equations, *J. Math. Phys.* 4 (1963).

[12] J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Springer, 1972, §4.1.

[13] E. Hopf, The partial differential equation $u_t + uu_x = \mu u_{xx}$, *Comm. Pure Appl. Math.* 3 (1950); J. D. Cole, *Quart. Appl. Math.* 9 (1951).

[14] S. N. Kružkov, First order quasilinear equations in several independent variables, *Math. USSR Sb.* 10 (1970), Thm 1 §3.

[15] S. P. Novikov, The periodic problem for the Korteweg–de Vries equation, *Funct. Anal. Appl.* 8 (1974).

[16] F. John, *Partial Differential Equations*, 4th ed., Springer, 1982, Chap. 7 §1.

[17] Project files: `claims/LEDGER.md`, `work/*/PROOF_PACKAGE.md`, `BIBLIOGRAFIA.md`.